UNITED STATES SECURITIES AND EXCHANGE COMMISSION

Washington, D.C. 20549

FORM 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the fiscal year ended September 30, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the transition period from _____ to _____
Commission file number 1-278

EMERSON ELECTRIC CO.

(Exact name of registrant as specified in its charter)

Missouri		43-0259330
(State or other jurisdiction of incorporation or organization)	logo_emersona10.jpg	(I.R.S. Employer Identification No.)
8000 W. Florissant Ave. P.O. Box 4100 St. Louis, Missouri		63136
(Address of principal executive offices)		(Zip Code)

Registrant's telephone number, including area code: (314) 553-2000

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock of $0.50 par value per share	EMR	New York Stock Exchange NYSE Chicago
0.375% Notes due 2024	EMR 24	New York Stock Exchange
1.250% Notes due 2025	EMR 25A	New York Stock Exchange
2.000% Notes due 2029	EMR 29	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: **None**

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issues its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

Aggregate market value of the voting stock held by nonaffiliates of the registrant as of close of business on March 31, 2023: $49.6 billion.

Common stock outstanding at October 31, 2023 : 570.1 million shares.

Documents Incorporated by Reference

1. Portions of Emerson Electric Co. Notice of 2024 Annual Meeting of Shareholders and Proxy Statement incorporated by reference into Part III hereof.

PART I

ITEM 1 - BUSINESS

Emerson (“the Company”) is a global technology and software company that provides innovative solutions for customers in a wide range of end markets around the world. Through its leading automation portfolio, Emerson helps process, hybrid and discrete manufacturers optimize operations, protect personnel, reduce emissions and achieve their sustainability goals. Sales by geographic destination in 2023 were: the Americas, 51 percent; Asia, Middle East & Africa, 30 percent (China, 12 percent); and Europe, 19 percent.

Portfolio management is an integral component of Emerson's growth and value creation strategy. Over the past two years, the Company has taken significant actions to accelerate the transformation of its portfolio through the completion of strategic acquisitions and divestitures of non-core businesses. These actions were undertaken to create a higher growth and cohesive industrial technology portfolio as a global automation leader serving a diversified set of end markets. The Company’s recent portfolio actions include the following transactions:

- On October 11, 2023, the Company completed the acquisition of National Instruments Corporation ("NI") at an equity value of $8.2 billion. NI, which provides software-connected automated test and measurement systems that enable enterprises to bring products to market faster and at a lower cost, had revenues of approximately $1.7 billion and pretax earnings of approximately $170 mllion for the 12 months ended September 30, 2023.

- On May 31, 2023, the Company completed the previously announced sale of a majority stake in its Climate Technologies business (which constitutes the former Climate Technologies segment, excluding Therm-O-Disc which was divested earlier in 2022) to private equity funds managed by Blackstone in a $14.0 billion transaction. Emerson received upfront, pre-tax cash proceeds of approximately $9.7 billion and a note receivable with a face value of $2.25 billion (which will accrue 5 percent interest payable in kind by capitalizing interest), while retaining a 40 percent non-controlling common equity interest in a new standalone joint venture between Emerson and Blackstone. The Climate Technologies business, which includes the Copeland compressor business and the entire portfolio of products and services across all residential and commercial HVAC and refrigeration end-markets, had 2022 net sales of approximately $5.0 billion. The new standalone business is named Copeland.

- On October 31, 2022, the Company completed the divestiture of its InSinkErator business, which manufactures food waste disposers, to Whirlpool Corporation for $3.0 billion. This business had net sales of $630 million in 2022.

- On May 31, 2022 the Company completed the divestiture of its Therm-O-Disc sensing and protection technologies business to an affiliate of One Rock Capital Partners, LLC.

- On May 16, 2022, the Company completed the transactions contemplated by its definitive agreement with Aspen Technology, Inc. ("Heritage AspenTech") to contribute two of Emerson's stand-alone industrial software businesses, Open Systems International, Inc. and the Geological Simulation Software business (collectively, the “Emerson Industrial Software Business”), along with approximately $6.0 billion in cash to Heritage AspenTech stockholders, to create "New AspenTech", a diversified, high-performance industrial software leader with greater scale, capabilities and technologies (defined as "AspenTech" herein). Upon closing of the transaction, Emerson owned 55 percent of the outstanding shares of AspenTech common stock (on a fully diluted basis). AspenTech had 2023 net sales of $1.04 billion.

Further information regarding acquisition and divestiture activity is set forth in Notes 4 and 5.

Certain prior year amounts have been reclassified to conform to the current year presentation. This includes reporting financial results for Climate Technologies, InSinkErator and Therm-O-Disc as discontinued operations for all periods presented, and the assets and liabilities of Climate Technologies and InSinkErator (prior to completion of the divestitures) as held-for-sale (see Note 5). In addition, as a result of its portfolio transformation, the Company now reports six segments and two business groups, which are highlighted in the table below (see Note 20 for further details). Beginning in 2024, the Company will report NI (which will be renamed Test & Measurement) as a new segment in the Software and Control business group.

INTELLIGENT DEVICES	SOFTWARE AND CONTROL
• Final Control • Measurement & Analytical • Discrete Automation • Safety & Productivity	• Control Systems & Software • AspenTech

The Company sells products and solutions that support customers in a variety of end markets, including process industries (such as chemical, power & renewables and energy), hybrid industries (life sciences, metals & mining, food & beverage, pulp & paper, and others), discrete industries (including automotive, medical, packaging and semiconductor) and more.

Emerson was incorporated in Missouri in 1890 and has evolved through internal growth and strategic acquisitions. Management has a well-established set of operating mechanisms to manage its business performance and set strategy. The Company also has processes undertaken by management with oversight from the Board of Directors to specifically focus on risks in areas such as cybersecurity, compliance, legal, sustainability, financial and reputational, among others. The Company periodically updates, assesses, and monitors its risk exposures, provides timely updates to the Board, and takes actions to mitigate these risks.

All Note references in this document refer to Notes to Consolidated Financial Statements set forth in Item 8 of this Annual Report on Form 10-K, which notes are hereby incorporated by reference. See also Item 1A - “Risk Factors” and Item 7 - “Management’s Discussion and Analysis of Financial Condition and Results of Operations.”

INTELLIGENT DEVICES

Final Control

The Final Control segment is a leading global provider of control valves, isolation valves, shutoff valves, pressure relief valves, pressure safety valves, actuators, and regulators for process and hybrid industries. These solutions respond to commands from a control system to continuously and precisely control and regulate the flow of liquids or gases to achieve safe operation along with reliability and optimized performance. Products within our Final Control segment are marketed under a variety of brands including: Anderson Greenwood, Bettis, Crosby, Fisher, Keystone, KTM and Vanessa.

Measurement & Analytical

The Measurement & Analytical segment is a leading supplier of intelligent instrumentation measuring the physical properties of liquids or gases, such as pressure, temperature, level, flow, acoustics, corrosion, pH, conductivity, water quality, toxic gases, and flame. The instrumentation transfers data to control systems and automation software, allowing process and hybrid industry operators to make educated decisions regarding production, reliability and safety. Products within our Measurement & Analytical segment are marketed under a variety of brands including: Flexim, Micro Motion and Rosemount.

Discrete Automation

The Discrete Automation segment includes solenoid valves, pneumatic valves, valve position indicators, pneumatic cylinders and actuators, air preparation equipment, pressure and temperature switches, electric linear motion solutions, programmable automation control systems and software, electrical distribution equipment, and materials joining solutions used primarily in discrete industries. Products within our Discrete Automation segment are marketed under a variety of brands including: Afag, Appleton, ASCO, Aventics, Branson, Movicon, PACSystems, SolaHD, TESCOM, and TopWorx.

Safety & Productivity

The Safety & Productivity segment offers tools for professionals and homeowners that promote safety and productivity. Pipe-working tools include pipe wrenches, pipe cutters, pipe threading and roll grooving equipment,

battery hydraulic tools for press connections, drain cleaners, tubing tools and diagnostic systems, including sewer inspection cameras and locating equipment. Electrical tools include conduit benders and cable pulling equipment, battery hydraulic tools for cutting and crimping electrical cable, and hole-making equipment. Other professional tools include water jetters, wet-dry vacuums, commercial vacuums and hand tools. Products within our Safety & Productivity segment are marketed under a variety of brands including: Greenlee, Klauke, ProTeam and RIDGID.

SOFTWARE AND CONTROL

Control Systems & Software

The Control Systems & Software segment provides control systems and software that control plant processes by collecting and analyzing information from measurement devices in the plant and using that information to adjust valves, pumps, motors, drives and other control hardware for maximum product quality, process efficiency and safety. These solutions include distributed control systems, safety instrumented systems, SCADA systems, application software, digital twins, asset performance management and cybersecurity. Control Systems & Software solutions are predominantly used by process and hybrid manufacturers. Products within our Control Systems & Software segment are marketed under a variety of brands including: AMS, DeltaV and Ovation.

AspenTech

AspenTech is a global leader in asset optimization software that enables industrial manufacturers to design, operate, and maintain their operations for maximum performance. AspenTech combines decades of modeling, simulation, and optimization capabilities with industrial operations expertise and applies advanced analytics to improve the profitability and sustainability of production assets. The purpose-built software drives value for customers by improving operational efficiency and maximizing productivity, reducing unplanned downtime and safety risks, and minimizing energy consumption and emissions.

Test & Measurement

As discussed above, Emerson completed the acquisition of NI on October 11, 2023. This business will be referred to as Test & Measurement and reported as a new segment in the Software and Control business group in 2024. Test & Measurement provides software-connected automated test and measurement systems that enable enterprises to bring products to market faster and at a lower cost. The Test & Measurement platform spans the full range of customer needs including modular instrumentation, data acquisition and control solutions, and general-purpose development software.

RESEARCH & DEVELOPMENT

Investing in innovation to accelerate organic growth is a critical component of Emerson's value creation strategy. The Company is focused on key growth initiatives across its software, control and intelligent devices portfolio. These initiatives include disruptive measurement technologies, software-defined automation systems, self-optimizing asset software and sustainability solutions. Total spending for R&D, engineering expense and customer-funded engineering and development was 6.9 percent of sales in 2023 compared to 6.3 percent in 2022.

DISTRIBUTION

The principal worldwide distribution channel for a majority of the Company's product offerings is through a direct sales force, while a network of independent sales representatives, and to a lesser extent independent distributors purchasing products for resale, are also utilized.

RAW MATERIALS

The Company's major requirements for basic raw materials include steel, cast iron, electronics, rare earth metals, aluminum and brass; and to a lesser extent, plastics and petroleum-based chemicals. The Company seeks to have many sources of supply for each of its major requirements in order to avoid significant dependence on any one or a few suppliers. However, the supply of materials or other items could be disrupted by natural disasters or other events. In 2023, freight costs and service levels returned to pre-pandemic levels, while the supply chain improved, including the availability of electronic components. Despite market price volatility for certain requirements, the raw materials and various purchased components needed for the Company’s products have generally been available in

sufficient quantities. See Item 7 - “Management’s Discussion and Analysis of Financial Condition and Results of Operations.”

PATENTS, TRADEMARKS AND LICENSES

The Company maintains an intellectual property portfolio it has developed or acquired over a number of years, including patents, trademarks and licenses. The Company also continues to develop or acquire new intellectual property. New patent applications are continuously filed to protect the Company’s ongoing research and development activities and the Company periodically reviews the continued utility of patent assets. The Company’s trademark registrations may be renewed and their duration is dependent upon national laws and trademark use. While this proprietary intellectual property portfolio is important to the Company in the aggregate, management does not regard any of its segments as being dependent on any single patent, trademark registration or license.

BACKLOG

The Company’s estimated consolidated order backlog was $7.8 billion and $7.0 billion at September 30, 2023 and 2022, respectively, of which approximately $1.2 billion and $1.1 billion related to AspenTech. Approximately 75 percent of the Company’s consolidated backlog is expected to be recognized as revenue over the next 12 months, with the remainder substantially over the subsequent two years thereafter. Backlog by business group at September 30, 2023 and 2022 follows (dollars in millions):

	2022	2023
Intelligent Devices	$ 3,930	4,471
Software and Control	3,058	3,302
Total Backlog	$ 6,988	7,773

COMPETITION

The Company's businesses operate in highly competitive markets. The Company competes based on product performance, quality, branding, service and/or price across the industries and markets served. A significant element of the Company's competitive strategy is to deliver solutions to our customers by manufacturing high-quality products at the best relevant global cost. Although no single company competes directly with Emerson in all of the Company's product lines, various companies compete in one or more product lines with the number of competitors varying by product line. Some competitors have substantially greater sales, assets and financial resources than Emerson and the Company also competes with many smaller companies. Management believes Emerson has a market leadership position in many of its product lines.

REGULATIONS

The Company's operations, products and services are subject to various government regulations, including environmental regulations. Our manufacturing locations generate waste, of which treatment, storage, transportation and disposal are subject to U.S. federal, state, foreign and/or local laws and regulations relating to protection of the environment. The Company continually works to minimize the environmental impact of its operations through safe technologies, facility design and operating procedures. Compliance with government regulations, including environmental regulations, has not had, and based on current information and the applicable laws and regulations currently in effect, is not expected to have a material effect on the Company's capital expenditures (including expenditures for environmental control facilities), earnings or competitive position. However, laws and regulations may be changed, accelerated or adopted that impose significant operational restrictions and compliance requirements upon the Company and which could negatively impact our operating results. See Item 1A - "Risk Factors."

HUMAN CAPITAL RESOURCES

Emerson is dedicated to modernizing our workplace culture to meet the needs and expectations of today's workers and attract talent that will help us thrive. In 2023, the Company launched its first-ever employee value proposition, *Let's Go,* which serves as an invitation to employees and potential employees to join the Company in our bold aspiration to create a healthier, safer, smarter and more sustainable world. We believe the Company’s success depends on our ability to attract, develop and retain key personnel. The skills, experience and industry knowledge of

key employees significantly benefit our operations and performance. The Company's Board of Directors and management oversee various employee initiatives.

The Company supports and develops its employees through global training and development programs that build and strengthen employees' leadership and professional skills. Leadership development programs include intensive learning programs for new leaders as well as more established leaders. The Company also partners with educational institutions and nonprofit organizations to help prepare current and future workers with the knowledge and skills they need to succeed. To assess and improve employee retention and engagement, the Company surveys employees with the assistance of third-party consultants, and takes actions to address areas of employee concern. In 2023, we initiated a continuous listening strategy, with more than 85 percent of employees participating in the survey and an overall engagement score of 78 percent. The categories of safety, well-being, growth and development, and access to resources and support earned the most favorable scores.

Employee health and safety in the workplace is also one of the Company's core values. The Corporate Safety Council is led by our Chief Operating Officer and oversees our safety efforts, supported by health and safety committees and leaders that operate at the local site level. Hazards in the workplace are actively identified and management tracks incidents so remedial actions can be taken to improve workplace safety. In 2023, the Company's total recordable rate of injuries was 0.30, and its lost or restricted workday case rate was 0.22 (both measured as the number of incidents per 100 employees).

We have identified other human capital priorities, including, among other things, providing competitive wages and benefits and promoting an inclusive work environment. The Company is committed to efforts to elevate the representation of women and U.S. minorities and foster an inclusive work environment that supports our large global workforce and helps us innovate for our customers. Employee Resource Groups have been created to support our diverse workforce and have grown to over 13,000 members. We also have taken actions to enhance diversity, including setting diversity targets for interview slates and targeted recruiting to increase the representation of women, minorities, U.S. military veterans, individuals with a disability and LGBTQ+ talent within Emerson. In 2021, the Company introduced diversity goals at the leadership level. We continue to make progress on our goals and were named a "Best Employer for Diversity" by Forbes. Overall, women represent 33 percent of our global workforce and 23 percent of leadership positions are held by women. In the U.S., minorities represent 35 percent of our workforce and 21 percent of our leadership positions.

Employee levels are managed to align with the pace of business and management believes it has sufficient human capital to operate its business successfully. The Company and its subsidiaries had approximately 67,000 employees at September 30, 2023. Management believes that the Company's employee relations are favorable.

A small portion of the Company's U.S. employees are unionized, while outside the U.S., we have employees in certain countries, particularly in Europe, that are represented by an employee representative organization, such as a union, works council or employee association.

ENVIRONMENTAL SUSTAINABILITY

Emerson's global purpose is to drive innovation that makes the world healthier, safer, smarter and more sustainable. Our environmental sustainability strategy is focused on driving progress within our facilities and helping our customers achieve their ESG objectives. In 2021, we appointed Mike Train as Chief Sustainability Officer. This role, part of our Office of the Chief Executive, reflects our focus on sustainability across our company. Under his leadership, Emerson has made significant strides, and we are strengthening our leadership position as our customers and suppliers work to deliver their environmental targets.

In 2022, we set an ambitious target to achieve net zero greenhouse gas (GHG) emissions across our value chain by 2045 compared to a 2021 baseline. To set us on the right pathway, we will target net zero operations and a 25 percent reduction of our value chain emissions by 2030, also compared to a 2021 baseline. The Company also added ESG targets, including GHG reduction targets, as a component in the determination of annual bonuses for leadership beginning in 2022. In 2023, we established a goal to achieve zero waste to landfill in our manufacturing facilities by 2032, from a 2022 fiscal year baseline, wherever this is compatible with local conditions and regulations. We also introduced a new Technology and Environmental Sustainability Board committee, which is tasked with overseeing strategy related to technology and R&D, the Company's product cybersecurity practices and Emerson's environmental sustainability goals and programs.

Our environmental sustainability strategy is summarized by our “Greening Of, Greening By, Greening With” framework. **Greening Of Emerson** demonstrates our efforts to improve our internal environmental sustainability performance, including reducing our GHG emissions and energy and water consumption. **Greening By Emerson** is our approach to delivering technology, solutions and expertise (including through our software offerings) that support and enable our customers’ decarbonization and environmental sustainability efforts. **Greening With Emerson** reflects how we foster collaboration among stakeholders by participating in environmental sustainability industry forums, partnering to develop innovative solutions, and engaging with governments globally to support sustainability-related policies and regulations.

Emerson’s environmental sustainability initiatives and strategy are discussed further in our 2022 Environmental, Social and Governance Report, which can be found on our website at www.Emerson.com; this report is not incorporated by reference and should not be considered part of this Form 10-K.

INTERNET ACCESS

Emerson's reports on Forms 10-K, 10-Q, 8-K and all amendments to those reports, as well as proxy statements, are available without charge through the Company’s website on the internet as soon as reasonably practicable after they are electronically filed with, or furnished to, the U.S. Securities and Exchange Commission (SEC). They may be accessed as follows: www.Emerson.com, Investors, SEC Filings. Information on the Company’s website does not constitute part of this Form 10-K.

The information set forth under Item 1A - “Risk Factors” is hereby incorporated by reference.

ITEM 1A - RISK FACTORS

Investing in our securities involves risks. You should carefully consider, among other matters, the factors set forth below and the other information in this report. The Company’s risk factors set forth below are not the only risks facing the Company. Additional risks and uncertainties not currently known to management or that management currently deems immaterial also may materially, adversely affect the Company’s business, financial condition or operating results. We may amend or supplement the risk factors set forth below from time to time by other reports we file with the SEC.

Business and Operational Risks

We Operate in Businesses That Are Subject to Competitive Pressures That Could Affect Prices or Demand for Our Products

Our businesses operate in markets that are highly competitive and potentially volatile, and we compete on the basis of product performance, quality, service and/or price across the industries and markets served. Our businesses are largely dependent on the current and future business environment, including capital and consumer spending. A significant element of our competitive strategy is to deliver solutions to our customers by manufacturing high-quality products at the best relevant global cost. Various companies compete with us in one or more product lines and the number of competitors varies by product line. Some of our competitors have substantially greater sales, assets and financial resources than our Company and we also compete with many smaller companies. Competitive pressures could adversely affect prices or customer demand for our products, impacting our sales or profit margins, and/or resulting in a loss of market share. In addition, certain of our businesses rely, in part, on independent sales representatives and distributors. Any disruption or adverse change in our relationships with these independent sales representatives could weaken our competitive position and adversely affect our results of operations, cash flows and financial condition. A disruption or adverse change could result from the sale or financial instability of an independent sales representative or distributor, changes to our relationship including favoring competing products for any reason, or other events.

Our Operating Results Depend in Part on Continued Successful Research, Development and Marketing of New and/or Improved Products and Services, and There Can Be No Assurance That We Will Continue to Successfully Introduce New Products and Services

The success of new and improved products and services depends on their initial and continued acceptance by our customers. Our businesses are affected by varying degrees of technological change and corresponding shifts in

customer demand, which result in unpredictable product transitions, shortened life cycles and increased importance of bein g first to market with new products and services. We may experience difficulties or delays in the research, development, production and/or marketing of new products and services which may negatively impact our operating results and prevent us from recouping or realizing a return on the investments required to continue to bring new products and services to market.

We must anticipate and respond to market and technological changes driven by broader trends such as decarbonization and electrification efforts in response to climate change. Market growth from the use of cleaner energy sources, as well as emissions management, energy efficiency and decarbonization efforts are likely to depend in part on technologies not yet deployed or widely adopted today. We may not adequately innovate or position our businesses for the adoption of technologies such as battery storage solutions, hydrogen use cases in industry, mobility, and power generation, enhanced electrical grid demand management, carbon capture and sequestration or advanced nuclear power.

These trends and the relative competitiveness of our product and service offerings will continue to be impacted by uncertain factors such as the pace of technological developments and related cost considerations, the levels of economic growth in different markets around the world and the adoption of climate change-related policies such as carbon taxes, greenhouse gas emission reductions, incentives or mandates for particular types of energy, or policies that impact the availability of financing for certain types of projects.

If We Are Unable to Defend or Protect Our Intellectual Property Rights, the Company's Competitive Position Could Be Adversely Affected

The Company's intellectual property rights are important to its business and include numerous patents, trademarks, copyrights, trade secrets and other confidential information. This intellectual property may be subject to challenge, infringement, invalidation or circumvention by third parties. Despite extensive security measures, our intellectual property may be subject to misappropriation through unauthorized access of our information technology systems, employee theft, or other acts of industrial espionage. Should the Company be unable to adequately defend or protect its intellectual property, it may suffer competitive harm.

We Engage in Acquisitions and Divestitures, Which Are Subject to Domestic and Foreign Regulatory Requirements, and May Encounter Difficulties in Integrating and Separating These Businesses and Therefore We May Not Realize the Anticipated Benefits

We regularly seek growth through strategic acquisitions as well as evaluate our portfolio for potential divestitures. These activities require favorable environments to execute these transactions, and we may encounter difficulties in obtaining the necessary regulatory approvals in both domestic and foreign jurisdictions. In 2023 and in past years, we have made various acquisitions and divestitures, including our acquisition of National Instruments which closed after year-end, our divestiture of a majority stake in the Climate Technologies business (now renamed Copeland), and our majority stake in Aspen Technology, Inc., and entered into joint venture arrangements intended to complement or expand our business, and may continue to do so in the future. As a result of these transactions, the Company has a narrower business which is focused on higher growth markets including software, innovation and disruptive technologies, and may encounter more volatility and be more vulnerable to changing market conditions. The success of these transactions will depend on our ability to achieve higher rates of growth, integrate assets and personnel acquired in these transactions and to cooperate with our strategic partners. We may encounter difficulties in integrating acquisitions with our operations as well as separating divested businesses, and in managing strategic investments. Furthermore, we may not realize the degree, or timing, of anticipated benefits including, among others, increasing rates of profitability and growth. Any of the foregoing could adversely affect our business and results of operations.

As part of the Copeland transaction, the Company received a note receivable and retained a 40 percent non-controlling common equity interest. As the Company no longer has a controlling interest in this business, the future value or proceeds from the note receivable and common equity interest will depend on the business performance of Copeland and how the controlling owner manages the business. Therefore, the Company can make no assurance regarding the amount or timing of any future proceeds or value to be derived from the note receivable and common equity interest.

We Use a Variety of Raw Materials and Components in Our Businesses, and Significant Shortages or Price Increases Could Increase Our Operating Costs and Adversely Impact the Competitive Positions of Our Products

Our major requirements for raw materials include steel, cast iron, electronics, rare earth metals, aluminum, brass and, to a lesser extent, plastics and petroleum-based chemicals. The Company seeks multiple sources of supply for each of its major requirements in order to avoid significant dependence on any one or a few suppliers. However, the supply of materials or other items could be disrupted by natural disasters, a health epidemic or pandemic, or other events. Significant shortages or price increases could impact the prices our affected businesses charge, their operating costs and the competitive position of their products and services, which could adversely affect our results of operations. While we monitor market prices of the commodities we require and attempt to mitigate price exposure through hedging activities, this risk could adversely affect our operating results.

Our Operations Depend on Production Facilities Throughout the World, a Majority of Which Are Located Outside the United States and Subject to Increased Risks of Disrupted Production, Causing Delays in Shipments and Loss of Customers and Revenue

We manage businesses with manufacturing facilities worldwide, a majority of which are located outside the United States, and also source certain materials globally. Emerging market sales represent over one-third of total sales and serving a global customer base requires that we place more materials sourcing and production in emerging markets to capitalize on market opportunities and maintain a best-cost position. Our and our suppliers' non-U.S. production facilities and operations could be disrupted by weather and natural disaster (including the potential effects of climate change), labor strife, war (including the Russia-Ukraine and other global conflicts), political unrest, terrorist activity or public health concerns such as an epidemic or pandemic, particularly in emerging countries that are not well-equipped to handle such occurrences.

Our manufacturing facilities abroad are dependent on the stability of governments and business conditions and may be more susceptible to changes in laws, policies and regulations in host countries, as well as economic and political upheaval, than our domestic facilities. These facilities face increased risks of nationalization as well as operational disruptions which could cause delays in shipments of products and the loss of sales and customers, and insurance proceeds may not adequately compensate us.

Access to Funding Through the Capital Markets is Essential to the Execution of Our Business Plan, and if We Are Unable to Maintain Such Access We Could Experience a Material Adverse Effect on Our Business and Financial Results

Our ability to invest in our businesses, make strategic acquisitions and refinance maturing debt obligations requires access to the capital markets and sufficient bank credit lines to support short-term borrowings. Volatility in the capital markets may increase costs associated with issuing commercial paper or other debt instruments, or affect the Company's ability to access those markets. If we are unable to continue to access the capital markets, we could experience a material adverse effect on our business and financial results. Additionally, if our customers, suppliers or financial institutions are unable to access the capital markets to meet their commitments to the Company, our business could be adversely impacted.

Our Business Success Depends on the Ability to Attract, Develop and Retain Key Personnel

Our success depends in part on the efforts and abilities of our management and key employees. Their skills, experience and industry knowledge significantly benefit our operations and performance. The failure to attract, develop and retain highly qualified personnel could adversely affect our ability to succeed in our human capital goals and priorities as well as negatively impact our business and operating results.

Security and/or Data Privacy Breaches, or Disruptions of Our Information Technology Systems Could Adversely Affect Our Business

The Company relies on information technology networks and systems, including the internet, to process, transmit and store electronic information, and to manage or support a variety of business processes and activities. These technology networks and systems may be susceptible to damage, disruptions or shutdowns due to failures during the process of upgrading or replacing software, databases or components; power outages; telecommunications or system failures; terrorist attacks; natural disasters; employee error or malfeasance; server or cloud provider breaches; and computer viruses or cyberattacks. Cybersecurity threats and incidents can range from uncoordinated individual attempts to gain unauthorized access to information technology networks and systems to more sophisticated and targeted measures, known as advanced persistent threats, directed at the Company, its products,

its customers and/or its third-party service providers. Despite the implementation of cybersecurity measures (including access controls, data encryption, vulnerability assessments, continuous monitoring, and maintenance of backup and protective systems), the Company's information technology systems may still be vulnerable to cybersecurity threats and other electronic security breaches. It is possible for such vulnerabilities to remain undetected for an extended period. In addition, it is possible a security breach could result in theft of trade secrets or other intellectual property or disclosure of confidential customer, supplier or employee information. Should the Company be unable to prevent security breaches or other damage to our information technology systems, disruptions could have an adverse effect on our operations, as well as expose the Company to litigation, liability or penalties under privacy laws, increased cybersecurity protection costs, reputational damage and product failure. In addition, we must comply with increasingly complex and rigorous regulatory standards enacted to protect business and personal data in the U.S. and elsewhere. Compliance with privacy and localization laws and regulations increases operational complexity. Failure to comply with these regulatory standards could subject us to fines and penalties, as well as legal and reputational risks, including proceedings against the Company by governmental entities or others.

Our Products and Services are Highly Sophisticated and Specialized, and a Major Product Failure or Similar Event Caused by Defects, Cybersecurity Incidents or Other Failures, Could Adversely Affect Our Business, Reputation, Financial Position and Results of Operations

We produce highly sophisticated products and provide specialized services that incorporate or use complex or leading-edge technology, including both hardware and software. Many of our products and services, including measurement and analytical instrumentation, industrial valves and equipment, and process control systems, are integrated and used in complex process, hybrid and discrete manufacturing environments. As a result, the impact of a catastrophic product failure or similar event could be significant. While we have built operational processes to ensure that our product design, manufacture, performance and servicing meet rigorous quality standards, there can be no assurance that we or our customers or other third parties will not experience operational process or product failures and other problems, including through manufacturing or design defects, process or other failures of contractors or third-party suppliers, cybersecurity incidents or other intentional acts, that could result in potential product, safety, regulatory or environmental risks. Cybersecurity incidents aimed at the software embedded in our products could lead to third-party claims resulting from damages caused by our product failures, and this risk is enhanced by the increasingly connected nature of our products. The potential consequences of a material cybersecurity incident include financial loss, reputational damage, litigation with third parties, diminution in the value of our investment in research, development and engineering, and increased cybersecurity protection and remediation costs due to the increasing sophistication and proliferation of threats, which in turn could adversely affect our competitiveness and results of operations.

Industry and General Economic Risks

The Coronavirus (COVID-19) Outbreak Adversely Impacted our Business and a Resurgence or Development of New Strains or Variants of COVID-19, or Other Public Health Emergencies, Could in the Future Have a Material Adverse Impact on our Business, Results of Operation, Financial Condition and Liquidity, the Nature and Extent of Which is Highly Uncertain

The global outbreak of the coronavirus (COVID-19) significantly increased economic, demand and operational uncertainty. Our operations have generally stabilized since the peak of the COVID-19 pandemic and in May 2023, the World Health Organization declared an end to COVID-19 as a public health emergency. However, a resurgence or development of new strains of COVID-19, or other public health emergencies, could result in unpredictable responses by authorities around the world which could negatively impact our global operations, customers and suppliers. Any future pandemics or public health emergencies could result in disruptions to our manufacturing operations, including higher rates of employee absenteeism, and supply chain, which could negatively impact our ability to meet customer demand. The extent to which new strains or variants of COVID-19, or other public health emergencies, could impact our business, results of operations, financial condition or liquidity is highly uncertain and would depend on future developments, including the spread and duration of any such virus and the variants, potential actions taken by governmental authorities, and how quickly economic conditions stabilize and recover.

Our Substantial Sales Both in the U.S. and Abroad Subject Us to Economic Risk as Our Results of Operations May Be Adversely Affected by Changes in Government Regulations and Policies and Currency Fluctuations

We sell, manufacture, engineer and purchase products globally, with significant sales in both mature and emerging markets. We expect sales in non-U.S. markets to continue to represent a significant portion of our total sales. Our U.S. and international operations subject the Company to changes in government regulations and policies in a large number of jurisdictions around the world, including those related to trade, investments, taxation, exchange controls and repatriation of earnings. Changes in laws or policies governing the terms of foreign trade, trade restrictions or barriers, tariffs or taxes, trade protection measures, and retaliatory countermeasures, including on imports from countries where we manufacture products, could adversely impact our business and financial results. In addition, changes in the relative values of currencies occur from time to time and have affected our operating results and could do so in the future. While we monitor our exchange rate exposures and attempt to mitigate this exposure through hedging activities, this risk could adversely affect our operating results.

Recessions, Adverse Market Conditions or Downturns in End Markets We Serve May Negatively Affect Our Operations

In the past, our operations have been exposed to significant volatility due to changes in general economic conditions or consumer preferences, recessions or adverse conditions in the end markets we serve. In the future, similar changes could adversely impact overall sales, operating results (including potential impairment charges for goodwill or other long-lived assets) and cash flows. Moreover, during economic downturns we may undertake more extensive restructuring actions, including workforce reductions, global facility consolidations, centralization of certain business support activities, and other cost reduction initiatives, and incur higher costs. As these plans and actions can be complex, the anticipated operational improvements, efficiencies and other benefits might be delayed or not realized.

Legal and Regulatory Risks

Changes in Tax Rates, Laws or Regulations and the Resolution of Tax Disputes Could Adversely Impact Our Financial Results

As a global company, we are subject to taxation in the U.S. and numerous non-U.S. jurisdictions. Significant judgment is required to determine our consolidated income tax provision and related liabilities. The Company's effective tax rate, cash flows and operating results could be affected by changes in the mix of earnings in countries with different statutory tax rates, as well as by changes in the local tax laws and regulations, or the interpretations thereof. In addition, the Company's tax returns are subject to regular review and audit by U.S. and non-U.S. tax authorities. While we believe our tax provisions are appropriate, the final outcome of tax audits or disputes could result in adjustments to the Company's tax liabilities, which could adversely affect our financial results.

Our Reputation, Ability To Do Business and Results of Operations Could Be Impaired By Improper Conduct By Any of Our Employees, Agents or Business Partners

We are subject to regulation under a wide variety of U.S. federal and state and non-U.S. laws, regulations and policies, including laws related to anti-corruption, anti-bribery, export and import compliance, anti-trust and money laundering, due to our global operations. In particular, the U.S. Foreign Corrupt Practices Act, the U.K. Bribery Act and similar anti-bribery laws in other jurisdictions generally prohibit companies and their intermediaries from making improper payments to government officials for the purpose of obtaining or retaining business, and we operate in many parts of the world that have experienced government corruption to some degree. We cannot provide assurance our internal controls will always protect us from the improper conduct of our employees, agents and business partners. Any such violation of law or improper actions could subject us to civil or criminal investigations in the U.S. and other jurisdictions, could lead to substantial civil or criminal, monetary and non-monetary penalties and related shareholder lawsuits, could lead to increased costs of compliance and could damage our reputation, our business and results of operations.

We Are Subject to Litigation and Environmental Regulations That Could Adversely Impact Our Operating Results

We are, and may in the future be, a party to a number of legal proceedings and claims, including those involving intellectual property, product liability (including asbestos) and environmental matters, several of which claim, or may in the future claim, significant damages. Given the inherent uncertainty of litigation, we can offer no assurance that existing litigation or a future adverse development will not have a material adverse impact. We also are subject to various laws and regulations relating to environmental protection and the discharge of materials into the environment, and we could incur substantial costs as a result of the noncompliance with or liability for cleanup or

other costs or damages under environmental laws. In addition, increased public awareness and concern regarding global climate change may result in more international, federal, and/or state or other stakeholder requirements or expectations that could result in more restrictive or expansive standards, such as stricter limits on greenhouse gas emissions or more prescriptive reporting of environmental, social, and governance metrics. There continues to be a lack of consistent climate change legislation and standards, which creates economic and regulatory uncertainty. While the Company has adopted certain voluntary targets, environmental laws, regulations or standards may be changed, accelerated or adopted and impose significant operational restrictions and compliance requirements upon the Company, its products or customers, which could negatively impact the Company's business, capital expenditures, results of operations, financial condition and competitive position.

Increasing Interest and Expectations with Respect to Environmental, Social, and Governance (ESG) Matters by Our Various Stakeholders Could Adversely Affect Our Business and Operating Results

In response to growing customer, investor, employee, governmental, and other stakeholder interest in our ESG practices, we have increased reporting of our ESG programs and performance and have established and announced our aspirational purpose, causes, values, and related commitments, goals or targets, including those regarding sustainability, greenhouse gas emissions, our net zero ambition, and diversity, equity and inclusion. Our ability to achieve such goals and aspirations is subject to numerous risks and uncertainties, many of which rely on the collective efforts of others or may be outside of our control. Such risks include, among others, the availability and adoption of new or additional technologies that reduce carbon or eliminate energy sources on a commercially reasonable basis, competing and evolving economic, policy and regulatory factors, the ability of suppliers and others to meet our sustainability, diversity and other goals, the availability of qualified candidates in our labor markets and our ability to recruit and retain diverse talent, and customer engagement in our goals. There may be times where actual outcomes vary from those aimed for or expected and sometimes challenges may delay or block progress. As a result, we cannot offer assurances that the results reflected or implied by any such statements will be realized or achieved. Moreover, standards and expectations for ESG matters continue to evolve and may be subject to varying interpretations, which may result in significant revisions to our goals or progress. In addition, certain of our product offerings may become less attractive as standards evolve. A failure or perceived failure to meet our aspirational purpose, causes, values, and related commitments, goals or targets within the timelines we announce, or at all, or a failure or perceived failure to meet evolving stakeholders expectations and standards, could damage our reputation, adversely affect employee retention or engagement or support from our various stakeholders and could subject us to government enforcement actions or penalties and private litigation. Such outcomes could negatively impact the Company's business, capital expenditures, results of operations, financial condition and competitive position.

ITEM 1B - UNRESOLVED STAFF COMMENTS

None.

ITEM 2 - PROPERTIES

At September 30, 2023, the Company had approximately 130 manufacturing locations worldwide, of which approximately 40 were located in the United States and 90 were located outside the United States, primarily in Europe and Asia, and to a lesser extent in Canada and Latin America. Manufacturing locations by business are: Intelligent Devices, 115, including 35 in the Final Control segment, 30 in the Measurement & Analytical segment, 40 in the Discrete Automation segment, and 10 in the Safety & Productivity segment; and Software and Control, 10, all in the Control Systems & Software segment. Additionally, there are 5 locations that support multiple segments. The majority of the locations are owned, with the remainder occupied under lease. The Company considers its facilities suitable and adequate for the purposes for which they are used. The Company also maintains a smaller number of administrative, sales, research and development, and distribution facilities.

ITEM 3 - LEGAL PROCEEDINGS

The Company and its subsidiaries are party to various legal proceedings, some of which claim substantial amounts of damages. It is not possible to predict the outcome of these matters, but historically the Company has been largely successful in both prosecuting and defending claims and lawsuits.

Given the uncertainties of litigation, a remote possibility exists that litigation could have a material adverse impact on the Company; however, the Company believes a material adverse impact of any pending litigation is unlikely.

Information regarding legal proceedings is set forth in Note 15.

ITEM 4 - MINE SAFETY DISCLOSURES

Not applicable.

INFORMATION ABOUT OUR EXECUTIVE OFFICERS

The following sets forth certain information as of November 13, 2023, with respect to the Company's executive officers. T hese officers have been elected or appointed to terms which expire February 6, 2024:

Name	Position	Age	Year First Appointed an Executive Officer
S. L. Karsanbhai	President and Chief Executive Officer	54	2018
R. R. Krishnan	Executive Vice President and Chief Operating Officer	52	2021
M. J. Baughman	Executive Vice President, Chief Financial Officer and Chief Accounting Officer	58	2018
S. Y. Bosco	Senior Vice President, Chief Legal Officer	65	2016
M. H. Train	Senior Vice President and Chief Sustainability Officer	61	2016
L. A. Flavin	Senior Vice President, Chief Transformation and Chief Compliance Officer	58	2021
P. Zornio	Senior Vice President and Chief Technology Officer	60	2022
V. Ramnath	Senior Vice President and Chief Marketing Officer	56	2023
N. Piazza	Senior Vice President and Chief People Officer	45	2023

There are no family relationships among any of the executive officers and directors.

Lal Karsanbhai has been Chief Executive Officer since February 2021 and President since March 2021. Prior to his current position, Mr. Karsanbhai was Executive President - Automation Solutions from October 2018 through January 2021, President - Measurement & Analytical from 2016 through September 2018, and President Emerson Network Power Europe, Middle East & Africa from 2014 through 2016.

Ram R. Krishnan was appointed Executive Vice President and Chief Operating Officer in February 2021. Prior to his current position, Mr. Krishnan was President Final Control from November 2017 to February 2021, Chief Operating Officer Final Control from January 2017 to November 2017, and President Flow Solutions from 2016 through January 2017.

Michael J. Baughman was appointed Executive Vice President and Chief Financial Officer in May 2023, and Chief Accounting Officer in February 2018. Prior to his current position, Mr. Baughman was named Vice President and Controller in October 2017. Prior to that Mr. Baughman was Vice President, Finance, Global Operations, Quality, and Research and Development of Baxter International Inc., a global healthcare products company, from 2015 through September 2017, and Vice President, Finance, Medical Products of Baxter from 2013 to 2015.

Sara Y. Bosco was appointed to Senior Vice President, Secretary and Chief Legal Officer in February 2023. Prior to her current position, Ms. Bosco was Senior Vice President, Secretary and General Counsel from May 2016 through February 2023, and President, Emerson Asia-Pacific from 2008 through May 2016.

Michael H. Train was appointed Senior Vice President and Chief Sustainability Officer in March 2021. Prior to his current position, Mr. Train was President from October 2018 to March 2021 and Executive President - Automation Solutions from October 2016 through October 2018, Executive Vice President - Automation Solutions from May 2016 through October 2016 and President of Global Sales for Emerson Process Management from 2010 through May 2016.

Lisa A. Flavin was appointed Senior Vice President and Chief Compliance Officer in March 2021, and assumed the additional role of Chief Transformation Officer in 2023. Prior to her current position, Ms. Flavin was Vice President and Chief Compliance Officer from February 2019 through March 2021 and Vice President, Audit and Chief Compliance Officer from February 2015 through February 2019.

Peter Zornio was appointed Senior Vice President and Chief Technology Officer in December 2022. Prior to his current position, Mr. Zornio was the Chief Technology Officer for the Automation Solutions Group from June 2017 to December 2022 and Chief Strategy Officer for Automation Solutions – Systems and Solutions from June 2006 to June 2017.

Vidya Ramnath was appointed to Senior Vice President and Chief Marketing Office r in June 2023. Prior to her current position, Ms. Ramnath was President of Middle East & Africa from 2019 through June 2023 and Vice President of Asia Pacific for Measurement & Analytical from 2017 through 2019.

Nick Piazza was appointed Senior Vice President and Chief People Officer in August 2023. Prior to his current position, Mr. Piazza was Vice President of Global Talent and Human Resource Operations from August 2021 through July 2023, and Vice President of Human Resources in Asia-Pacific for the company’s Automation Solutions business from July 2017 through July 2021.

PART II

ITEM 5 - MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Information regarding the market for the Company's common stock and dividend payments is set forth in Note 22 and is hereby incorporated by reference. There were approximately 15,200 stockholders of record at September 30, 2023.

Neither the Company nor any "affiliated purchaser" repurchased any shares of Company common stock during the three-month period ended September 30, 2023. In March 2020, the Board of Directors authorized the purchase of 60 million shares and a total of approximately 33.3 million shares remain available under the authorization.

ITEM 6 [RESERVED]

ITEM 7 - MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

Safe Harbor Statement

This Annual Report on Form 10-K contains various forward-looking statements and includes assumptions concerning Emerson's operations, future results and prospects. These forward-looking statements are based on current expectations and are subject to risks and uncertainties. Emerson undertakes no obligation to update any such statements to reflect later developments. In connection with the “safe harbor” provisions of the Private Securities Litigation Reform Act of 1995, Emerson provides the cautionary statements set forth under Item 1A - “Risk Factors,” which are hereby incorporated by reference and identify important economic, political and technological factors, among others, changes in which could cause the actual results or events to differ materially from those set forth in or implied by the forward-looking statements and related assumptions.

Non-GAAP Financial Measures

To supplement the Company's financial information presented in accordance with U.S. generally accepted accounting principles (U.S. GAAP), management periodically uses certain "non-GAAP financial measures," as such term is defined in Regulation G under SEC rules, to clarify and enhance understanding of past performance and prospects for the future. Generally, a non-GAAP financial measure is a numerical measure of a company's operating performance, financial position or cash flows that excludes or includes amounts that are included in or excluded from the most directly comparable measure calculated and presented in accordance with U.S. GAAP. For example, non-GAAP measures may exclude the impact of certain items such as acquisitions or divestitures, amortization of intangibles, restructuring costs, discrete taxes, changes in reporting segments, gains, losses and impairments, or items outside of management's control, such as foreign currency exchange rate fluctuations. Management believes that the following non-GAAP financial measures provide investors and analysts useful insight into the Company's financial position and operating performance. Any non-GAAP measure provided should be viewed in addition to, and not as an alternative to, the most directly comparable measure determined in accordance with U.S. GAAP, as identified in italics below. Further, the calculation of these non-GAAP financial measures may differ from the calculation of similarly titled financial measures presented by other companies and therefore may not be comparable among companies.

Underlying sales, which exclude the impact of significant acquisitions, divestitures and fluctuations in foreign currency exchange rates during the periods presented, are provided to facilitate relevant period-to-period comparisons of sales growth by excluding those items that impact overall comparability (U.S. GAAP measure: *net sales*).

Operating profit (defined as net sales less cost of sales and selling, general and administrative expenses) and operating profit margin (defined as operating profit divided by net sales) are indicative of short-term operational performance and ongoing profitability. Management closely monitors operating profit and operating profit margin of each business to evaluate past performance and actions required to improve profitability. EBIT (defined as earnings before deductions for interest expense, net, related party interest income, and income taxes) and total segment EBIT, and EBIT margin (defined as EBIT divided by net sales) and total segment EBIT margin, are financial measures that exclude the impact of financing on the capital structure and income taxes. Adjusted EBITA and adjusted segment EBITA (defined as earnings excluding interest expense, net, related party interest income, income taxes, intangibles amortization expense, restructuring expense, first year purchase accounting related items and transaction fees, and certain gains, losses or impairments) and adjusted EBITA margin and adjusted segment EBITA margin (defined as adjusted EBITA divided by net sales) are measures used by management to evaluate the Company's operational performance, as they exclude the impact of acquisition-related investments and non-operational items. EBITDA (defined as EBIT excluding depreciation and amortization) and EBITDA margin (defined as EBITDA divided by net sales) are also used as measures of the Company's current operating performance, as they exclude the impact of capital and acquisition-related investments. Adjusted EBITDA (defined as EBITDA excluding restructuring expense, first year purchase accounting related items and transaction fees, and certain gains, losses or impairments) and adjusted EBITDA margin (defined as Adjusted EBITDA divided by net sales) are also used to exclude the impact of non-operational items. All of these are commonly used financial measures utilized by management to evaluate performance (U.S. GAAP measures: *pretax earnings or pretax profit margin, segment earnings or segment margin*).

Earnings, earnings per share, return on common stockholders' equity and return on total capital excluding certain gains and losses, impairments, restructuring costs, impacts of acquisitions or divestitures, amortization of intangibles, discrete taxes, or other items provide additional insight into the underlying, ongoing operating performance of the Company and facilitate period-to-period comparisons by excluding the earnings impact of these items. Management believes that presenting earnings, earnings per share, return on common stockholders' equity and return on total capital excluding these items is more representative of the Company's operational performance and may be more useful for investors (U.S. GAAP measures: *earnings, earnings per share, return on common stockholders' equity, return on total capital*).

Free cash flow (operating cash flow less capital expenditures) and free cash flow as a percent of net sales are indicators of the Company’s cash generating capabilities, dividends as a percent of free cash flow is an indicator of the Company's ability to support its dividend, and free cash flow conversion of adjusted net earnings (free cash flow divided by net earnings adjusted for intangibles amortization expense, restructuring expense, first year purchase accounting related items and transaction fees, and certain gains, losses or impairments) is an indicator of the quality of the Company's earnings, after considering investments in capital assets which are necessary to maintain and enhance existing operations. The determination of operating cash flow adds back noncash depreciation expense to earnings and thereby does not reflect a charge for necessary capital expenditures. Management believes that free cash flow, free cash flow as a percent of net sales and dividends as a percent of free cash flow are useful to both management and investors as measures of the Company’s ability to generate cash and support its dividend (U.S. GAAP measures: *operating cash flow, operating cash flow as a percent of net sales, dividends as a percent of operating cash flow*).

FINANCIAL REVIEW

Report of Management

The Company's management is responsible for the integrity and accuracy of the financial statements. Management believes that the financial statements for each of the years in the three-year period ended September 30, 2023 have been prepared in conformity with U.S. generally accepted accounting principles appropriate in the circumstances. In preparing the financial statements, management makes informed judgments and estimates where necessary to reflect the expected effects of events and transactions that have not been completed. The Company's disclosure controls and procedures ensure that material information required to be disclosed is recorded, processed, summarized and communicated to management and reported within the required time periods.

In meeting its responsibility for the reliability of the financial statements, management relies on a system of internal accounting controls. This system is designed to provide reasonable assurance that assets are safeguarded and transactions are executed in accordance with management's authorization and recorded properly to permit the preparation of financial statements in accordance with U.S. generally accepted accounting principles. Although the design of this system recognizes that errors or irregularities may occur, management believes that the Company's internal accounting controls provide reasonable assurance that errors or irregularities that could be material to the financial statements are prevented or would be detected within a timely period.

The Audit Committee of the Board of Directors, which is composed solely of independent directors, is responsible for overseeing the Company's financial reporting process. The Audit Committee meets with management and the Company's internal auditors periodically to review the work of each and to monitor the discharge by each of its responsibilities. The Audit Committee also meets periodically with the independent auditors, who have free access to the Audit Committee and the Board of Directors, to discuss the quality and acceptability of the Company's financial reporting and internal controls, as well as nonaudit-related services.

The independent auditors are engaged to express an opinion on the Company's consolidated financial statements and on the Company's internal control over financial reporting. Their opinions are based on procedures that they believe to be sufficient to provide reasonable assurance that the financial statements contain no material errors and that the Company's internal controls are effective.

Management's Report on Internal Control Over Financial Reporting

The Company's management is responsible for establishing and maintaining adequate internal control over financial reporting for the Company. With the participation of the Chief Executive Officer and the Chief Financial Officer, management conducted an evaluation of the effectiveness of internal control over financial reporting based on the framework and the criteria established in *Internal Control - Integrated Framework* (2013), issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on this evaluation, management has concluded that internal control over financial reporting was effective as of September 30, 2023.

The Company's auditor, KPMG LLP, an independent registered public accounting firm, has issued an audit report on the effectiveness of the Company's internal control over financial reporting.

/s/ S. L. Karsanbhai
S. L. Karsanbhai
President
and Chief Executive Officer

/s/ Michael J. Baughman
Michael J. Baughman
Executive Vice President
and Chief Financial Officer

Results of Operations

Years ended September 30
(Dollars in Item 7 are in millions, except per share amounts or where noted)

		2021	2022	2023	22 vs. 21	23 vs. 22
Net sales	$	12,932	13,804	15,165	7 %	10 %
Gross profit	$	5,730	6,306	7,427	10 %	18 %
Percent of sales		*44.3 %*	*45.7 %*	*49.0 %*	1.4 pts	3.3 pts
SG&A	$	3,494	3,614	4,186		
Percent of sales		*27.0 %*	*26.2 %*	*27.6 %*	(0.8) pts	1.4 pts
Gain on subordinated interest	$	—	(453)	(161)		
Other deductions, net	$	319	519	683		
Amortization of intangibles	*$*	*277*	*336*	*482*		
Restructuring costs	*$*	*132*	*75*	*72*		
Interest expense, net	$	155	194	34		
Interest income from related party	$	—	—	(41)		
Earnings from continuing operations before income taxes	$	1,762	2,432	2,726	38 %	12 %
Percent of sales		*13.6 %*	*17.6 %*	*18.0 %*	4.0 pts	0.4 pts
Earnings from continuing operations common stockholders	$	1,414	1,886	2,152	33 %	14 %
Percent of sales		*10.9 %*	*13.7 %*	*14.2 %*	2.8 pts	0.5 pts
Net earnings common stockholders	$	2,303	3,231	13,219	40 %	309 %
Percent of sales		*17.8 %*	*23.4 %*	*87.2 %*	5.6 pts	63.8 pts
Diluted EPS – Earnings from continuing operations	$	2.35	3.16	3.72	34 %	18 %
Diluted EPS – Net earnings	$	3.82	5.41	22.88	42 %	323 %
Adjusted Diluted EPS – Earnings from continuing operations	$	3.01	3.64	4.44	21 %	22 %
Return on common stockholders' equity		25.2 %	31.9 %	85.1 %	6.7 pts	53.2 pts
Return on total capital		18.1 %	20.4 %	66.5 %	2.3 pts	46.1 pts

OVERVIEW

Overall, sales for 2023 were $15.2 billion, up 10 percent compared with the prior year, reflecting strong growth across the majority of the Company's business segments and all geographies.

Net earnings from continuing operations attributable to common stockholders were $2,152 in 2023, up 14 percent compared with prior year earnings of $1,886, and diluted earnings per share from continuing operations were $3.72, up 18 percent versus $3.16 in 2022. Adjusted diluted earnings per share from continuing operations were $4.44 compared with $3.64 in the prior year, reflecting strong sales growth and operating performance.

The Company generated operating cash flow from continuing operations of $2.7 billion in 2023, an increase of $678, or 33 percent, reflecting higher earnings (excluding the impacts in both years from the Vertiv subordinated interest gains and higher Heritage AspenTech intangibles amortization in the current year).

The table below presents the Company's diluted earnings per share from continuing operations on an adjusted basis to facilitate period-to-period comparisons and provide additional insight into the underlying, ongoing operating performance of the Company. Adjusted diluted earnings per share from continuing operations excludes intangibles

amortization expense, restructuring expense, first year purchase accounting related items and transaction-related costs, interest income on undeployed proceeds related to the Copeland transaction, gains or losses on the Copeland equity method investment, and certain gains, losses or impairments.

	2021	2022	2023
Diluted earnings from continuing operations per share	$ 2.35	3.16	3.72
Amortization of intangibles	0.38	0.45	0.62
Restructuring and related costs	0.21	0.14	0.14
Acquisition/divestiture costs and pre-acquisition interest on AspenTech debt	—	0.15	0.13
Gain on subordinated interest	—	(0.60)	(0.21)
National Instruments investment gain	—	—	(0.07)
Other investment-related gains	—	(0.02)	—
AspenTech Micromine purchase price hedge	—	0.04	(0.02)
Interest income on undeployed proceeds from Copeland transaction	—	—	(0.19)
Loss on Copeland equity method investment	—	—	0.24
Russia business exit charge	—	0.32	0.08
OSI first year acquisition accounting charges and fees	0.07	—	—
Adjusted diluted earnings from continuing operations per share	$ 3.01	3.64	4.44

The table below summarizes the changes in adjusted diluted earnings per share from continuing operations. The items identified below are discussed throughout MD&A, see further discussion above and in the Business Segments and Financial Position sections below.

	2022	2023
Adjusted diluted earnings from continuing operations per share - prior year	$ 3.01	3.64
Operations, including impact of AspenTech acquisition	0.58	0.77
Corporate and other	—	0.07
Stock compensation	0.12	(0.16)
Foreign currency	(0.02)	(0.12)
Pensions	0.03	0.07
Gains on sales of capital assets in 2022	0.02	(0.02)
Gains on sales of investments in 2021	(0.03)	—
Effective tax rate	(0.09)	0.01
Interest income on Copeland note receivable	—	0.05
Other	(0.01)	(0.01)
Share repurchases	0.03	0.14
Adjusted diluted earnings from continuing operations per share - current year	$ 3.64	4.44

NET SALES

Net sales for 2023 were $15.2 billion, an increase of $1.4 billion, or 10 percent compared with 2022. Intelligent Devices sales increased 7 percent, while Software and Control sales increased 20 percent, which included the impact of the Heritage AspenTech acquisition. Underlying sales were up 10 percent on 6 percent higher volume and 4 percent higher price. Foreign currency translation subtracted 2 percent, the Heritage AspenTech acquisition added 3 percent and the divestiture of Metran, Emerson's Russia-based manufacturing subsidiary, deducted 1 percent. Underlying sales were up 11 percent in the U.S. and up 9 percent internationally.

Net sales for 2022 were $13.8 billion, an increase of $0.9 billion, or 7 percent compared with 2021. Intelligent Devices sales increased 5 percent, while Software and Control sales increased 16 percent. Underlying sales increased 7 percent on 4 percent higher volume and 3 percent higher price. The Heritage AspenTech acquisition

added 3 percent and foreign currency translation deducted 3 percent. Underlying sales were up 12 percent in the U.S. and up 5 percent internationally.

INTERNATIONAL SALES

Emerson is a global business with international sales representing 58 percent of total sales in 2023, including U.S. exports . The Company generally expects faster economic growth in emerging markets in Asia, Latin America, Eastern Europe and Middle East/Africa.

International destination sales, including U.S. exports, increased 9 percent, to $8.9 billion in 2023, reflecting the Company's overall increase in sales and the impact of the Heritage AspenTech acquisition. U.S. exports of $1.0 billion were up 6 percent compared with 2022. Underlying international destination sales were up 9 percent, as foreign currency translation had a 3 percent unfavorable impact on the comparison, the Heritage AspenTech acquisition added 3 percent and the divestiture of Metran, Emerson's Russia-based manufacturing subsidiary, deducted 1 percent. Underlying sales increased 10 percent in Europe, 9 percent in Asia, Middle East & Africa (China up 4 percent), 14 percent in Latin America and 1 percent in Canada. Origin sales by international subsidiaries, including shipments to the U.S., totaled $7.7 billion in 2023, up 5 percent compared with 2022.

International destination sales, including U.S. exports, increased 2 percent, to $8.2 billion in 2022, reflecting the impact of the Heritage AspenTech acquisition. U.S. exports of $1.0 billion were up 51 percent compared with 2021, including an increase of approximately $200 due to the Heritage AspenTech acquisition. Underlying international destination sales were up 5 percent, as foreign currency translation had a 5 percent unfavorable impact on the comparison and the Heritage AspenTech acquisition added 2 percent. Underlying sales increased 5 percent in Asia, Middle East & Africa (China up 11 percent), 18 percent in Latin America and 14 percent in Canada, while Europe was down slightly. Origin sales by international subsidiaries, including shipments to the U.S., totaled $7.4 billion in 2022, down 2 percent compared with 2021.

ACQUISITIONS AND DIVESTITURES

Portfolio management is an integral component of Emerson's growth and value creation strategy. Over the past two years, the Company has taken significant actions to accelerate the transformation of its portfolio through the completion of strategic acquisitions and divestitures of non-core businesses. These actions were undertaken to create a higher growth and cohesive industrial technology portfolio as a global automation leader serving a diversified set of end markets. The Company’s recent portfolio actions include the following transactions:

On October 11, 2023, subsequent to Emerson's fiscal year-end, the Company completed the acquisition of National Instruments Corporation ("NI") at an equity value of $8.2 billion. NI, which provides software-connected automated test and measurement systems that enable enterprises to bring products to market faster and at a lower cost, had revenues of approximately $1.7 billion and pretax earnings of approximately $170 for the 12 months ended September 30, 2023.

In 2023, the Company acquired two businesses, Flexim, which will be reported in the Measurement & Analytical segment, and Afag, which will be reported in the Discrete Automation segment, for $705, net of cash acquired.

On March 31, 2023, Emerson completed the divestiture of Metran, its Russia-based manufacturing subsidiary. In 2023, the Company recognized a pretax loss of $47 in Other deductions ($47 after-tax, in total $0.08 per share) related to its exit of business operations in Russia. The Company had previously announced its intention to exit business operations in 2022 and recognized a pretax loss of $181 ($190 after-tax, in total $0.32 per share). This charge included a loss of $36 in operations and $145 reported in Other deductions ($10 of which is reported in restructuring costs) and was primarily non-cash. Emerson's historical net sales in Russia represented approximately 2.0 percent of consolidated annual sales.

On May 31, 2023, the Company completed the previously announced sale of a majority stake in its Climate Technologies business (which constitutes the former Climate Technologies segment, excluding Therm-O-Disc which was divested earlier in 2022) to private equity funds managed by Blackstone in a $14.0 billion transaction. The Company recognized a pretax gain of approximately $10.6 billion (approximately $8.4 billion after-tax including tax expense recognized in prior quarters related to subsidiary restructurings). The new standalone business is named Copeland.

On October 31, 2022, the Company completed the divestiture of its InSinkErator business, which manufactures food waste disposers, to Whirlpool Corporation for $3.0 billion, and the Company recognized a pretax gain of approximately $2.8 billion (approximately $2.1 billion after-tax) in 2023.

On May 31, 2022 the Company completed the divestiture of its Therm-O-Disc sensing and protection technologies business to an affiliate of One Rock Capital Partners, LLC. The Company recognized a pretax gain of $486 ($429 after-tax) in 2022.

Climate Technologies, Therm-O-Disc and InSinkErator are reported within discontinued operations for all periods presented.

On May 16, 2022, the Company completed the transactions contemplated by its definitive agreement with Aspen Technology, Inc. ("Heritage AspenTech") to contribute two of Emerson's stand-alone industrial software businesses, Open Systems International, Inc. and the Geological Simulation Software business (collectively, the “Emerson Industrial Software Business”), along with approximately $6.0 billion in cash to Heritage AspenTech stockholders, to create "New AspenTech" (defined as "AspenTech" herein). Upon closing of the transaction, Emerson owned 55 percent of the outstanding shares of AspenTech common stock (on a fully diluted basis). AspenTech and its subsidiaries now operate under Heritage AspenTech’s previous name “Aspen Technology, Inc.” and AspenTech common stock is traded on NASDAQ under AspenTech’s previous stock ticker symbol “AZPN.” Due to the timing of the acquisition, the results for the first half of fiscal 2022 do not include the results of Heritage AspenTech.

On October 1, 2020, the Company completed the acquisition of Open Systems International, Inc. (OSI), a leading operations technology software provider in the global power industry, for approximately $1.6 billion, net of cash acquired. This business had net sales of $191 in 2021 and is now reported in the AspenTech segment.

See Notes 4, 5, 8 and 23 and Item 1A - "Risk Factors" for further information on acquisitions and divestitures.

COST OF SALES

Cost of sales for 2023 were $7,738, an increase of $240 compared with $7,498 in 2022. Gross profit was $7,427 in 2023 compared to $6,306 in 2022, while gross margin increased 3.3 percentage points to 49.0 percent due to favorable price less net material inflation, the impact of the Heritage AspenTech acquisition which benefited margins by 0.6 percentage points, and favorable mix.

Cost of sales for 2022 were $7,498, an increase of $296 compared with $7,202 in 2021, primarily due to higher sales volume and higher materials costs. Gross profit was $6,306 in 2022 compared to $5,730 in 2021, while gross margin increased 1.4 percentage points to 45.7 percent. The Heritage AspenTech acquisition benefited gross margin 0.9 percentage points and favorable mix also contributed to the increase. Price less net material inflation was favorable but had a slightly dilutive impact on margins, while higher freight and other inflation also negatively impacted margins.

SELLING, GENERAL AND ADMINISTRATIVE EXPENSES

SG&A expenses of $4,186 in 2023 increased $572 compared with 2022 and SG&A as a percent of sales increased 1.4 percentage points to 27.6 percent, reflecting the Heritage AspenTech acquisition and higher stock compensation expense of $125, of which $75 related to Emerson stock plans due to a higher share price and $50 was attributable to AspenTech stock plans. These items were partially offset by strong operating leverage on higher sales.

SG&A expenses of $3,614 in 2022 increased $120 compared with 2021, reflecting the impact of higher sales and higher wage and other inflation. SG&A as a percent of sales decreased 0.8 percentage points to 26.2 percent, reflecting lower stock compensation expense of $72 due to a lower share price in 2022 (0.6 percentage points) and leverage on higher sales.

GAIN ON SUBORDINATED INTEREST

In the first quarter of 2022, the Company received a distribution of $438 related to its subordinated interest in Vertiv (in total, a pretax gain of $453 was recognized in the first quarter of 2022, $358 after-tax, $0.60 per share) and received the remaining $15 related to the pretax gain in the first quarter of 2023. In 2023, the Company received additional distributions totaling $161 ($122 after-tax, $0.21 per share). Based on the terms of the agreement and the current calculation, the Company could receive additional distributions of approximately $40. The remaining distributions are contingent on the timing and price at which Vertiv shares are sold by the equity holders and therefore, there can be no assurance as to the amount or timing of the remaining distributions to the Company.

OTHER DEDUCTIONS, NET

Other deductions, net were $683 in 2023, an increase of $164 compared with 2022, reflecting a loss of $177 on the Company's equity method investment in Copeland, higher intangibles amortization of $146 primarily related to the Heritage AspenTech acquisition, and an unfavorable impact from foreign currency transactions of $112 reflecting losses in the current year compared to gains in the prior year. The prior year included a charge of $145 related to the Company exiting its business in Russia compared to a charge of $47 in the current year. The current year also included a mark-to-market gain of $56 on the Company's equity investment in NI, and a mark-to-market gain of $24 related to foreign currency forward contracts entered into by AspenTech to mitigate the impact of foreign currency exchange associated with the Micromine purchase price compared to a loss of $50 in the prior year. On June 21, 2023, AspenTech terminated all outstanding foreign currency forward contracts.

Other deductions, net were $519 in 2022, an increase of $200 compared with 2021, reflecting a charge of $145 related to the Company exiting its business in Russia ($10 of which is reported in restructuring costs), acquisition/divestiture costs of $91, higher intangibles amortization of $59, primarily related to the Heritage AspenTech acquisition, and a mark-to-market loss of $50 related to foreign currency forward contracts entered into by AspenTech to mitigate the impact of foreign currency exchange associated with the Micromine purchase price. These items were partially offset by lower restructuring costs of $57. See Notes 6 and 7.

INTEREST EXPENSE, NET

Interest expense, net was $34, $194 and $155 in 2023, 2022 and 2021, respectively. The decrease in 2023 reflects interest income on undeployed proceeds from the Copeland transaction of $141 ($108 after-tax, $0.19 per share). The increase in 2022 compared to 2021 reflects the issuance of $3 billion of long-term debt in December 2021 to support the AspenTech transaction, partially offset by $500 of notes that matured in the first quarter of 2022.

Interest income from related party was $41 in 2023 and reflects non-cash interest income on the Copeland note receivable, which is capitalized to the carrying value of the note.

EARNINGS BEFORE INCOME TAXES

Pretax earnings from continuing operations of $2,726 increased $294 in 2023, up 12 percent compared with 2022, reflecting strong operating results in the current year. Earnings increased $447 in Intelligent Devices and decreased $27 in Software and Control (reflecting the impact of higher intangibles amortization due to the Heritage AspenTech acquisition).

Pretax earnings from continuing operations of $2,432 increased $670 in 2022, up 38 percent compared with 2021, reflecting the impact of the Vertiv gain discussed above and strong earnings, which increased $340 in Intelligent Devices and increased $74 in Software and Control. See the Business Segments discussion that follows and Note 20.

INCOME TAXES

Income taxes were $599, $549 and $346 for 2023, 2022 and 2021, respectively, resulting in effective tax rates of 22 percent, 23 percent and 20 percent in 2023, 2022 and 2021, respectively. The rate in 2022 reflected the impact of the Russia business exit which was essentially offset by a benefit related to the completion of tax examinations. See Note 16.

NET EARNINGS AND EARNINGS PER SHARE

Net earnings from continuing operations attributable to common stockholders in 2023 were $2,152, up 14 percent compared with 2022, and diluted earnings per share from continuing operations were $3.72, up 18 percent compared with $3.16 in 2022, reflecting strong operating results. Adjusted diluted earnings per share from continuing operations were $4.44 compared with $3.64 in the prior year. See the analysis of adjusted earnings per share in the Overview section for further details. Earnings from discontinued operations attributable to common stockholders in 2023 were $11,067 ($19.16 per share) which included the $8.4 billion after-tax gain on the Copeland transaction and the $2.1 billion after-tax gain on the divestiture of InSinkErator, compared to $1,345 ($2.25 per share) in 2022. See Note 5. Net earnings attributable to common stockholders were $13,219 ($22.88 per share) compared with $3,231 ($5.41 per share) in 2022.

Net earnings from continuing operations attributable to common stockholders in 2022 were $1,886, up 33 percent compared with 2021, and diluted earnings per share from continuing operations were $3.16, up 34 percent compared with $2.35 in 2021. Results reflected strong operating results and included a gain of $0.60 per share related to the Company's subordinated interest in Vertiv. Adjusted diluted earnings per share from continuing

operations were $3.64 compared with $3.01 in the prior year. See the analysis of adjusted earnings per share in the Overview section for further details. Earnings from discontinued operations attributable to common stockholders in 2022 were $1,345 ($2.25 per share) compared to $889 ($1.47 per share) in 2021, reflecting an after-tax gain of $429 ($0.72 per share) related to the Therm-O-Disc divestiture in 2022. See Note 5. Net earnings common stockholders were $3,231 ($5.41 per share) in 2022 compared with $2,303 ($3.82 per share) in 2021 .

The table below, which shows results on an adjusted EBITA basis, is intended to supplement the Company's discussion of its results of operations herein.

Twelve Months Ended September 30	2021	2022	2023	22 vs. 21	23 vs. 22
Earnings from continuing operations before income taxes	$ 1,762	2,432	2,726	38 %	12 %
Percent of sales	13.6 %	17.6 %	18.0 %	*4.0 pts*	*0.4 pts*
Interest expense, net	155	194	34		
Interest income from related party	—	—	(41)		
Amortization of intangibles	304	430	678		
Restructuring and related costs	166	105	92		
Acquisition/divestiture and related costs	—	91	84		
Gain on subordinated interest	—	(453)	(161)		
National Instruments investment gain	—	—	(56)		
Other investment-related gains		(14)	—		
AspenTech Micromine purchase price hedge	—	50	(24)		
Loss on Copeland equity method investment	—	—	177		
Russia business exit charge	—	181	47		
OSI first year acquisition accounting charges	50	—	—		
Adjusted EBITA from continuing operations	$ 2,437	3,016	3,556	24 %	18 %
Percent of sales	18.8 %	21.8 %	23.4 %	*3.0 pts*	*1.6 pts*

RETURNS ON EQUITY AND TOTAL CAPITAL

Return on common stockholders' equity (net earnings attributable to common stockholders divided by average common stockholders' equity) was 85.1 percent in 2023 compared with 31.9 percent in 2022 and 25.2 percent in 2021. Return on total capital (computed as net earnings attributable to common stockholders excluding after-tax net interest expense, divided by average common stockholders' equity plus short- and long-term debt less cash and short-term investments) was 66.5 percent in 2023 compared with 20.4 percent in 2022 and 18.1 percent in 2021. The higher returns in 2023 included the impact of the after-tax gain from the Copeland transaction (approximately $8.4 billion), the after-tax gain on the InSinkErator divestiture (approximately $2.1 billion), the Vertiv subordinated interest after-tax gain of $122, the National Instruments investment after-tax gain of $43, the after-tax loss on the Copeland equity method investment of $134, after-tax acquisition/divestiture costs of $78, and the Russia business exit after-tax loss of $47. The higher returns in 2022 included the impact of the Vertiv subordinated interest after-tax gain of $358, the after-tax gain on the Therm-O-Disc divestiture of $429, after-tax acquisition/divestiture costs of $93 (including amounts reported in discontinued operations), and the Russia business exit after-tax loss of $190. Excluding these items in both years, return on common stockholders' equity was 17.9 percent and 26.9 percent in 2023 and 2022, respectively, and return on total capital was 14.0 percent and 17.4 percent, respectively. The decrease in 2023 reflects the increase to equity from the after-tax gains on the Copeland transaction and InSinkErator divestiture.

Business Segments

Following is an analysis of segment results for 2023 compared with 2022, and 2022 compared with 2021. The Company defines segment earnings as earnings before interest and income taxes.

INTELLIGENT DEVICES

	2022	2023	Change	FX	Acq/Div	U/L
Sales:						
Final Control	$ 3,607	3,970	10 %	2 %	1 %	13 %
Measurement & Analytical	3,215	3,595	12 %	2 %	2 %	16 %
Discrete Automation	2,612	2,635	1 %	2 %	— %	3 %
Safety & Productivity	1,402	1,388	(1) %	— %	— %	(1)%
Total	$ 10,836	11,588	7 %	2 %	1 %	10 %
Earnings:						
Final Control	$ 592	865	46 %			
Measurement & Analytical	785	936	19 %			
Discrete Automation	542	509	(6) %			
Safety & Productivity	250	306	22 %			
Total	$ 2,169	2,616	21 %			
Margin	*20.0 %*	*22.6 %*	2.6 pts			
Amortization of intangibles:						
Final Control	$ 94	88				
Measurement & Analytical	21	27				
Discrete Automation	30	29				
Safety & Productivity	26	26				
Total	$ 171	170				
Restructuring and related costs:						
Final Control	$ 75	28				
Measurement & Analytical	3	13				
Discrete Automation	—	27				
Safety & Productivity	10	—				
Total	$ 88	68				
Adjusted EBITA	$ 2,428	2,854	18 %			
Adjusted EBITA Margin	*22.4 %*	*24.6 %*	2.2 pts			

2023 vs. 2022 - Intelligent Devices sales were $11.6 billion in 2023, an increase of $752, or 7 percent. Underlying sales increased 10 percent on 5 percent higher volume and 5 percent higher price. Underlying sales increased 11 percent in the Americas (U.S. up 12 percent), increased 9 percent in Europe and increased 8 percent in Asia, Middle East & Africa (China up 2 percent). Sales for Final Control increased $363, or 10 percent. Underlying sales increased 13 percent, reflecting strength in chemical and energy end markets and across all geographies, particularly in the U.S. Sales for Measurement & Analytical increased $380, or 12 percent. Underlying sales increased 16 percent, reflecting robust growth in the Americas and Europe due to strong demand, while Asia, Middle East & Africa was up moderately due to softness in China. Discrete Automation sales increased $23, or 1 percent, while underlying sales increased 3 percent, reflecting softening demand in the second half of the year, with all geographies up low-to-mid single digits for the full year. Safety & Productivity sales decreased $14, or 1 percent, and underlying sales decreased 1 percent, reflecting softness in the Americas and Europe, while Asia, Middle East & Africa was up slightly. Earnings for Intelligent Devices were $2,616, an increase of $447, or 21 percent, and margin increased 2.6 percentage points to 22.6 percent, reflecting favorable price less net material inflation, leverage on higher sales and favorable mix, partially offset by wage and other inflation. Adjusted EBITA margin was 24.6 percent, an increase of 2.2 percentage points.

INTELLIGENT DEVICES

	2021	2022	Change	FX	Acq/Div	U/L
Sales:						
Final Control	$ 3,488	3,607	4 %	3 %	— %	7 %
Measurement & Analytical	3,078	3,215	4 %	3 %	— %	7 %
Discrete Automation	2,474	2,612	6 %	4 %	— %	10 %
Safety & Productivity	1,340	1,402	5 %	2 %	— %	7 %
Total	$ 10,380	10,836	5 %	3 %	— %	8 %
Earnings:						
Final Control	$ 432	592	37 %			
Measurement & Analytical	684	785	15 %			
Discrete Automation	457	542	18 %			
Safety & Productivity	256	250	(2) %			
Total	$ 1,829	2,169	19 %			
Margin	*17.6 %*	*20.0 %*	2.4 pts			
Amortization of intangibles:						
Final Control	$ 107	94				
Measurement & Analytical	25	21				
Discrete Automation	34	30				
Safety & Productivity	28	26				
Total	$ 194	171				
Restructuring and related costs:						
Final Control	$ 66	75				
Measurement & Analytical	58	3				
Discrete Automation	11	—				
Safety & Productivity	4	10				
Total	$ 139	88				
Adjusted EBITA	$ 2,162	2,428	12 %			
Adjusted EBITA Margin	*20.8 %*	*22.4 %*	1.6 pts			

2022 vs. 2021 - Intelligent Devices sales were $10.8 billion in 2022, an increase of $456, or 5 percent. Underlying sales increased 8 percent on 5 percent higher volume and 3 percent higher price. Foreign currency translation had a 3 percent unfavorable impact. Underlying sales increased 12 percent in the Americas (U.S. up 12 percent), increased 1 percent in Europe and increased 5 percent in Asia, Middle East & Africa (China up 11 percent). Sales for Final Control increased $119, or 4 percent, and underlying sales increased 7 percent, reflecting strong demand in the Americas and China, partially offset by softness in the rest of Asia, Middle East & Africa. Sales for Measurement & Analytical increased $137, or 4 percent, and underlying sales increased 7 percent. Sales were strong in China and North America, while sales were down moderately in Europe due to supply chain constraints. Discrete Automation sales increased $138, or 6 percent, and underlying sales increased 10 percent, reflecting strong demand across all geographies. Safety & Productivity sales increased $62, or 5 percent, and underlying sales increased 7 percent. Sales of professional tools were strong, while wet/dry vacuums decreased moderately due to difficult comparisons. Earnings for Intelligent Devices were $2,169, an increase of $340, or 19 percent, and margin increased 2.4 percentage points to 20.0 percent, reflecting leverage on higher volume, favorable mix, lower restructuring expenses which benefited margins 0.5 percentage points, savings from cost reduction actions and favorable price less net material inflation, partially offset by higher freight and other inflation. Adjusted EBITA margin was 22.4 percent, an increase of 1.6 percentage points.

SOFTWARE AND CONTROL

	2022	2023	Change	FX	Acq/Div	U/L
Sales:						
Control Systems & Software	$ 2,398	2,606	9 %	1 %	1 %	11 %
AspenTech	656	1,042	59 %	— %	(60) %	(1)%
Total	$ 3,054	3,648	20 %	1 %	(11) %	10 %
Earnings:						
Control Systems & Software	$ 437	529	21 %			
AspenTech	12	(107)	(967) %			
Total	$ 449	422	(6) %			
Margin	*14.7 %*	*11.6 %*	(3.1) pts			
Amortization of intangibles:						
Control Systems & Software	$ 22	22				
AspenTech	237	486				
Total	$ 259	508				
Restructuring and related costs:						
Control Systems & Software	$ 11	9				
AspenTech	—	1				
Total	$ 11	10				
Adjusted EBITA	$ 719	940	31 %			
Adjusted EBITA Margin	*23.5 %*	*25.8 %*	2.3 pts			

2023 vs. 2022 - Software and Control sales were $3.6 billion in 2023, an increase of $594, or 20 percent compared to the prior year, reflecting the impact of the Heritage AspenTech acquisition and strong growth in Control Systems & Software. Underlying sales increased 10 percent on 8 percent higher volume and 2 percent higher price. Underlying sales increased 7 percent in the Americas (U.S. up 6 percent), increased 11 percent in Europe and increased 13 percent in Asia, Middle East & Africa (China up 16 percent). Sales for Control Systems & Software increased $208, or 9 percent, and underlying sales increased 11 percent, reflecting global strength in process end markets while power end markets were up modestly. Sales for AspenTech increased $386, or 59 percent, due to the acquisition of Heritage AspenTech. Earnings for Software and Control were $422, a decrease of $27, or 6 percent, and margin decreased 3.1 percentage points to 11.6 percent, reflecting the impact from $249 of incremental intangibles amortization related to the Heritage AspenTech acquisition. Adjusted EBITA margin was 25.8 percent, an increase of 2.3 percentage points, reflecting leverage on higher sales, higher price and favorable mix, partially offset by inflation and unfavorable foreign currency transactions.

SOFTWARE AND CONTROL

	2021	2022	Change	FX	Acq/Div	U/L
Sales:						
Control Systems & Software	$ 2,321	2,398	3 %	4 %	— %	7 %
AspenTech	319	656	106 %	— %	(106)%	— %
Total	$ 2,640	3,054	16 %	4 %	(13)%	7 %
Earnings:						
Control Systems & Software	$ 382	437	14 %			
AspenTech	(7)	12	269 %			
Total	$ 375	449	20 %			
Margin	*14.2 %*	*14.7 %*	0.5 pts			
Amortization of intangibles:						
Control Systems & Software	$ 20	22				
AspenTech	89	237				
Total	$ 109	259				
Restructuring and related costs:						
Control Systems & Software	$ 11	11				
AspenTech	2	—				
Total	$ 13	11				
Adjusted EBITA	$ 497	719	45 %			
Adjusted EBITA Margin	*18.8 %*	*23.5 %*	4.7 pts			

2022 vs. 2021 - Software and Control sales were $3.1 billion in 2022, an increase of $414, or 16 percent compared to 2021, reflecting the impact of the Heritage AspenTech acquisition and growth in Control Systems & Software. Underlying sales increased 7 percent on higher volume. Underlying sales increased 13 percent in the Americas (U.S. up 12 percent), decreased 4 percent in Europe and increased 6 percent in Asia, Middle East & Africa (China up 11 percent). Sales for Control Systems & Software increased $77, or 3 percent, and underlying sales increased 7 percent, reflecting strength in process end markets in North America and China, partially offset by weakness in Europe, while power end markets were strong in North America and Europe. Sales for AspenTech increased $337, or 106 percent, due to the acquisition of Heritage AspenTech. Earnings for Software and Control were $449, an increase of $74, or 20 percent, and margin increased 0.5 percentage points to 14.7 percent. Results for 2022 included intangibles amortization of $148 related to the Heritage AspenTech acquisition. Adjusted EBITA margin was 23.5 percent, an increase of 4.7 percentage points, reflecting the impact of the Heritage AspenTech acquisition and leverage on higher volume in Control Systems & Software.

Financial Position, Liquidity and Capital Resources

Emerson maintains a conservative financial structure to provide the strength and flexibility necessary to achieve our strategic objectives and efficiently deploy cash where needed worldwide to fund operations, complete acquisitions and sustain long-term growth. Emerson is in a strong financial position, with total assets of $43 billion and stockholders' equity of $21 billion, and has the resources available for reinvestment in existing businesses, strategic acquisitions and managing its capital structure on a short- and long-term basis.

The Company continues to generate substantial operating cash flow, including over $2.7 billion from continuing operations in 2023. Cash flows have been and are expected to be sufficient for at least the next 12 months to meet the Company’s operating requirements, including those related to salaries and wages, working capital, capital expenditures, and other liquidity requirements associated with operations. The Company also has certain contractual obligations, primarily long-term debt and operating leases (see Notes 9, 12 and 13). The Company has been able to readily meet all its funding requirements and currently believes that sufficient funds will be available to meet its needs for the foreseeable future through operating cash flow, existing resources, short- and long-term debt capacity, or its $3.5 billion revolving backup credit facility under which it has not incurred any borrowings.

CASH FLOW

		2021	2022	2023
Operating Cash Flow	$	2,458	2,048	2,726
Percent of sales		*19.0 %*	*14.8 %*	*18.0 %*
Capital Expenditures	$	404	299	363
Percent of sales		*3.1 %*	*2.2 %*	*2.4 %*
Free Cash Flow (Operating Cash Flow less Capital Expenditures)	$	2,054	1,749	2,363
Percent of sales		*15.9 %*	*12.7 %*	*15.6 %*
Operating Working Capital	$	457	990	1,283
Percent of sales		*3.5 %*	*7.2 %*	*8.5 %*

Operating cash flow from continuing operations for 2023 was $2.7 billion, an increase of $678, or 33 percent compared with 2022, reflecting higher earnings (excluding the impacts in both years from the Vertiv subordinated interest gains and higher Heritage AspenTech intangibles amortization in the current year). Operating cash flow included approximately $310 generated by AspenTech. Operating cash flow from continuing operations of $2.0 billion in 2022 decreased 17 percent compared to $2.5 billion in 2021, reflecting higher working capital due to increased sales and ongoing supply chain constraints.

At September 30, 2023, operating working capital as a percent of sales was 8.5 percent compared with 7.2 percent in 2022 and 3.5 percent in 2021. Operating working capital remained elevated in 2023 due to higher inventory levels to support sales growth and higher receivables. The increase for 2022 compared to 2021 was due to higher inventory levels to support sales growth and reflecting supply chain constraints. In addition, the Heritage AspenTech acquisition increased operating working capital by approximately $250 in 2022. As of September 30, 2023, Emerson's cash and equivalents totaled $8.1 billion, reflecting the after-tax proceeds related to the Copeland transaction, which were used along with other available liquidity to fund the National Instruments transaction subsequent to year-end (see the Leverage/Capitalization section for further discussion of Emerson's post-close financial position). Going forward, Copeland is not expected to issue dividends to the Company but will distribute cash for the Company to pay its share of U.S. taxes. The Company's cash also includes approximately $120 attributable to AspenTech which is intended to be used for its own purposes and is not available to return to Emerson shareholders.

Free cash flow from continuing operations (operating cash flow less capital expenditures) was $2,363 in 2023, up 35 percent, reflecting the increase in operating cash flow. Free cash flow from continuing operations was $1,749 in 2022, compared with $2,054 in 2021. Net cash paid in connection with acquisitions was $705, $5,702 and $1,592 in 2023, 2022 and 2021, respectively.

Total cash provided by operating activities including the impact of discontinued operations was $637, $2,922 and $3,575 in 2023, 2022 and 2021, respectively. The decrease in 2023 was due to approximately $2.3 billion of income taxes paid related to the gains on the Copeland transaction and InSinkErator divestiture and subsidiary restructurings related to the Copeland transaction. Investing cash flow from discontinued operations of $12.5 billion in 2023 reflects the proceeds from the Copeland transaction and InSinkErator divestiture.

On March 27, 2020, the CARES Act was enacted in response to the COVID-19 pandemic, and among other things, provides tax relief to businesses. Tax provisions of the CARES Act included the deferral of certain payroll taxes, relief for retaining employees, and other provisions. The Company deferred $73 of certain payroll taxes through the end of calendar year 2020, of which approximately $37 was paid in December 2021 and the remainder paid in December 2022.

Dividends were $1,198 ($2.08 per share) in 2023, compared with $1,223 ($2.06 per share) in 2022 and $1,210 ($2.02 per share) in 2021. In November 2023, the Board of Directors voted to increase the quarterly cash dividend 1 percent, to an annualized rate of $2.10 per share.

Purchases of Emerson common stock totaled $2,000, $500 and $500 in 2023, 2022 and 2021, respectively, at average per share prices of $94.09, $87.64 and $94.65. AspenTech repurchases were $214 in 2023, which increased the Company's common ownership percentage to approximately 57 percent.

In November 2015, the Board of Directors authorized the purchase of up to 70 million shares, and during 2022, the remaining shares available under this authorization were purchased. In March 2020, the Board of Directors authorized the purchase of an a dditional 60 million shares and a total of approximately 33.3 million shares remain available. The Company purchased 21.3 million shares in 2023, 5.7 million shares in 2022 and 5.3 million shares in 2021 under the authorizations.

LEVERAGE/CAPITALIZATION

	2021	2022	2023
Total Assets	$ 24,715	35,672	42,746
Long-term Debt	$ 5,793	8,259	7,610
Common Stockholders' Equity	$ 9,883	10,364	20,689
Total Debt-to-Total Capital Ratio	40.3 %	50.0 %	28.3 %
Net Debt-to-Net Capital Ratio	30.4 %	45.3 %	0.5 %
Operating Cash Flow-to-Debt Ratio	36.9 %	19.7 %	33.4 %
Interest Coverage Ratio	11.6X	11.7X	11.5X

Total debt, which includes long-term debt, current maturities of long-term debt, commercial paper and other short-term borrowings, was $8,157, $10,374 and $6,665 as of September 30, 2023, 2022 and 2021, respectively. The decrease in 2023 included a net reduction in short-term borrowings of approximately $1.6 billion and repayments of long-term debt of $741 (including $264 related to AspenTech's repayment of the outstanding balance on its existing term loan facility plus accrued interest). The increase in 2022 was due to the issuance of $3 billion of long-term debt and increased commercial paper borrowings of approximately $1.3 billion compared to September 30, 2021. The Company used the net proceeds from the sale of the notes and the increased commercial paper borrowings to fund the majority of its contribution of approximately $6.0 billion to existing stockholders of Heritage AspenTech as part of the transaction. Long-term debt was issued in December 2021 as follows: $1 billion of 2.0% notes due December 2028, $1 billion of 2.2% notes due December 2031, and $1 billion of 2.8% notes due December 2051. Additionally, the Company repaid $500 of 2.625% notes that matured in 2022, and in 2021 repaid $300 of 4.25% notes that matured. See Note 4 and Note 13.

The total debt-to-total capital ratio and net debt-to-net capital ratio (less cash and short-term investments) decreased in 2023 due to the proceeds and after-tax gains (which increased common stockholder's equity) on the Copeland transaction and InSinkErator divestiture. Considering the cash paid to complete the National Instruments transaction in October 2023, the Company's net debt-to-net capital ratio was approximately 29.0 percent, reflecting moderate levels of debt consistent with prior years. These ratios increased in 2022 due to the increased borrowings to support the AspenTech transaction discussed above. The interest coverage ratio is computed as earnings before income taxes plus interest expense, divided by interest expense. The Company's earnings increased in 2023 and 2022 which offset higher interest expense due to the increased long-term debt and commercial paper borrowings to fund the Heritage AspenTech acquisition.

In February 2023, the Company entered into a $3.5 billion five-year revolving backup credit facility with various banks, which replaced the May 2018 $3.5 billion facility. The credit facility is maintained to support general corporate purposes, including commercial paper borrowings. The Company has not incurred any borrowings under this or previous facilities. The credit facility contains no financial covenants and is not subject to termination based on a change of credit rating or material adverse changes. The facility is unsecured and may be accessed under various interest rate alternatives at the Company’s option. Fees to maintain the facility are immaterial. The Company also maintains a universal shelf registration statement on file with the SEC under which it can issue debt securities, preferred stock, common stock, warrants, share purchase contracts or share purchase units without a predetermined limit. Securities can be sold in one or more separate offerings with the size, price and terms to be determined at the time of sale.

FINANCIAL INSTRUMENTS

In the normal course of business, the Company is exposed to changes in interest rates and foreign currency exchange rates due to its worldwide presence and diverse business pr ofile and selectively uses derivative financial instruments, including forwards, swaps and purchased options to manage these risks. The Company does not hold

derivatives for trading or speculative purposes. The value of derivatives and other financial instruments is subject to change as a result of market movements in rates and prices. Sensitivity analysis is one technique used to forecast the impact of these movements. Based on a hypothetical 10 percent increase in interest rates or a 10 percent weakening in the U.S. dollar across all currencies, the potential losses in future earnings, fair value or cash flows are not material. Sensitivity analysis has limitations; for example, a weaker U.S. dollar would benefit future earnings through favorable translation of non-U.S. operating results. See Notes 1, and 11 through 13.

Critical Accounting Policies

Preparation of the Company's financial statements requires management to make judgments, assumptions and estimates regarding uncertainties that could affect reported revenue, expenses, assets, liabilities and equity. Note 1 describes the significant accounting policies used in preparation of the consolidated financial statements. The most significant areas where management judgments and estimates impact the primary financial statements are described below. Actual results in these areas could differ materially from management's estimates under different assumptions or conditions.

REVENUE RECOGNITION

The Company evaluates its contracts with customers to identify the promised goods or services and recognizes revenue for the identified performance obligations at the amount the Company expects to be entitled to in exchange for those goods or services. A performance obligation is a promise in a contract to transfer a distinct good or service to a customer. Revenue is recognized when, or as, performance obligations are satisfied and control has transferred to the customer, typically when products are shipped or delivered, title and risk of loss pass to the customer, and the Company has a present right to payment. The majority of the Company's revenues relate to a broad offering of manufactured products and software which are recognized at the point in time when control transfers, generally in accordance with shipping terms, or the first day of the contractual term for software. A portion of the Company's revenues relate to the sale of post-contract customer support, parts and labor for repairs, and engineering services.

In some circumstances, contracts include multiple performance obligations, where revenue is recognized separately for each good or service, as well as contracts where revenue is recognized over time as control transfers to the customer. Tangible products represent a large majority of the delivered items in contracts with multiple performance obligations or where revenue is recognized over time, while a smaller portion is attributable to installation, service and maintenance. In sales arrangements that involve multiple performance obligations, revenue is allocated based on the relative standalone selling price for each performance obligation. Observable selling prices from actual transactions are used whenever possible. In other instances, the Company determines the standalone selling price based on third-party pricing or management's best estimate. For projects where revenue is recognized over time, the Company typically uses an input method to determine progress and recognize revenue, based on costs incurred. The Company believes costs incurred closely correspond with its performance under the contract and the transfer of control to the customer. The Company also has software maintenance contracts where revenue is recognized ratably over the maintenance term.

VALUATION OF ASSETS AND LIABILITIES

Assets and liabilities acquired in business combinations, including intangible assets, are accounted for using the acquisition method and recorded at their respective fair values. In 2022, the Company completed the acquisition of Aspen Technology, Inc. and engaged an independent third-party valuation specialist to assist in the determination of the fair value of intangible assets. This included the use of certain assumptions and estimates, including the projected revenue for the customer relationship and developed technology intangible asset and the obsolescence rate for the developed technology intangible asset. Although we believe the assumptions and estimates to be reasonable and appropriate, they require judgement and are based on experience and historical information obtained from Aspen Technology, Inc.

In 2023, the consideration received from the divestiture of a majority stake in Copeland included a note receivable with a face value of $2.25 billion and the Company also retained a 40 percent non-controlling common equity interest in Copeland. The note receivable and common equity interest were required to be initially valued at fair value as part of the overall consideration received for the transaction. The fair value of the common equity investment was determined using a discounted cash flow model, which included estimating financial projections for Copeland and applying an appropriate discount rate, and an option pricing model based on various assumptions. Fair value for the note receivable was determined using a market approach primarily based on interest rates for companies with similar credit quality and the expected duration of the note.

LONG-LIVED ASSETS

Long-lived assets, which include property, plant and equipment, goodwill and identifiable intangible assets, are reviewed for impairment whenever events or changes in business circumstances indicate impairment may exist. If the Company determines that the carrying value of a long-lived asset may not be recoverable, a permanent impairment charge is recorded for the amount by which the carrying value of the long-lived asset exceeds its estimated fair value. Reporting units are also reviewed for possible goodwill impairment at least annually, in the fourth quarter. If an initial assessment indicates it is more likely than not an impairment may exist, it is evaluated by comparing the reporting unit's estimated fair value to its carrying value. Fair value is generally estimated using an income approach that discounts estimated future cash flows using discount rates judged by management to be commensurate with the applicable risk. Estimates of future sales, operating results, cash flows and discount rates are subject to changes in the economic environment, including such factors as the general level of market interest rates, expected equity market returns and the volatility of markets served, particularly when recessionary economic circumstances continue for an extended period of time.

RETIREMENT PLANS

The Company maintains a prudent long-term investment strategy consistent with the duration of pension obligations. The determination of defined benefit plan expense and liabilities is dependent on various assumptions, including the expected annual rate of return on plan assets, the discount rate and the rate of annual compensation increases. In accordance with U.S. generally accepted accounting principles, actual results that differ from the Company's assumptions are accumulated as deferred actuarial gains or losses and amortized to expense in future periods. The Company's principal U.S. defined benefit plan is closed to employees hired after January 1, 2016 while shorter-tenured employees ceased accruing benefits effective October 1, 2016.

As of September 30, 2023, the U.S. pension plans were overfunded by $656 in total (approximately 22 percent in excess of the projected benefit obligation), including unfunded plans totaling $159. The non-U.S. plans were underfunded by $62, including unfunded plans totaling $213. The Company contributed a total of $46 to defined benefit plans in 2023 and expects to contribute approximately $45 in 2024. At year-end 2023, the discount rate for U.S. plans was 6.03 percent, and was 5.64 percent in 2022. The assumed investment return on plan assets was 6.00 percent in 2023, 6.00 percent in 2022 and 6.50 percent in 2021, and will be 6.50 percent for 2024. While management believes its assumptions used are appropriate, actual experience may differ. A 0.25 percentage point decrease in the U.S. and non-U.S. discount rates would have increased the total projected benefit obligation at September 30, 2023 by $100 and increased 2024 pension expense by $15. A 0.25 percentage point decrease in the expected return on plan assets would increase 2024 pension expense by $15. See Note 14.

CONTINGENT LIABILITIES

The Company is a party to a number of pending legal proceedings and claims, including those involving general and product liability (including asbestos) and other matters, several of which claim substantial amounts of damages. The Company accrues for such liabilities when it is probable that future costs (including legal fees and expenses) will be incurred and such costs can be reasonably estimated. Accruals are based on developments to date; management's estimates of the outcomes of these matters; and the Company's experience in contesting, litigating and settling similar matters. The Company engages an outside expert to develop an actuarial estimate of its expected costs to resolve all pending and future asbestos claims, including defense costs, as well as its related insurance receivables. The reserve for asbestos litigation, which is recorded on an undiscounted basis, is based on projected claims through 2065.

Although it is not possible to predict the ultimate outcome of these matters, the Company historically has been largely successful in defending itself against claims and suits that have been brought against it, and will continue to defend itself vigorously in all such matters. While the Company believes a material adverse impact is unlikely, given the inherent uncertainty of litigation, a remote possibility exists that a future development could have a material adverse impact on the Company. See Note 15.

INCOME TAXES

Income tax expense and tax assets and liabilities reflect management's assessment of taxes paid or expected to be paid (received) on items included in the financial statements. Deferred tax assets and liabilities arise from temporary differences between the consolidated financial statement carrying amounts of existing assets and liabilities and their respective tax bases, and consideration of operating loss and tax credit carryforwards. Deferred income taxes are measured using enacted tax rates in effect for the year in which the temporary differences are expected to be recovered or settled. The impact on deferred tax assets and liabilities of a change in tax rates is recognized in the period that includes the enactment date. Valuation allowances are provided to reduce deferred tax assets to the

amount that will more likely than not be realized. This requires management to make judgments and estimates regarding the amount and timing of the reversal of taxable temporary differences, expected future taxable income, and the impact of tax planning strategies.

Uncertainty exists regarding tax positions taken in previously filed tax returns which remain subject to examination, along with positions expected to be taken in future returns. The Company provides for unrecognized tax benefits, based on the technical merits, when it is more likely than not that an uncertain tax position will not be sustained upon examination. Adjustments are made to the uncertain tax positions when facts and circumstances change, such as the closing of a tax audit; changes in applicable tax laws, including tax case rulings and legislative guidance; or expiration of the applicable statute of limitations.

Cash repatriated to the U.S. is generally not subject to U.S. federal income taxes. No provision is made for withholding taxes and any applicable U.S. income taxes on the undistributed earnings of non-U.S. subsidiaries where these earnings are considered indefinitely invested or otherwise retained for continuing international operations. Determination of the amount of taxes that might be paid on these undistributed earnings if eventually remitted is not practicable. See Notes 1 and 16.

Other Items

LEGAL MATTERS

At September 30, 2023, there were no known contingent liabilities (including guarantees, pending litigation, taxes and other claims) that management believes will be material in relation to the Company's financial statements, nor were there any material commitments outside the normal course of business.

NEW ACCOUNTING PRONOUNCEMENTS

In 2023, the Company adopted ASU No. 2021-10 (Topic 832), *Government Assistance,* which requires annual disclosures about certain types of government assistance received. This standard has no impact on the accounting for government assistance and did not materially impact the Company's disclosures.

In 2022, the Company adopted three accounting standard updates, and in 2021 adopted two accounting standard updates and one new accounting standard, each of which had an immaterial or no impact on the Company's financial statements. These included:

- Updates to Accounting Standards Codification ("ASC") 805, *Business Combinations*, which clarify the accounting for contract assets and liabilities assumed in a business combination. In general, this will result in contract liabilities being recognized at their historical amounts under ASC 606, rather than at fair value in accordance with the general requirements of ASC 805.

- Updates to ASC 740, *Income Taxes*, which require the recognition of a franchise tax that is partially based on income as an income-based tax with any incremental amount as a non-income based tax. These updates also make certain changes to intra-period tax allocation principles and interim tax calculations.

- Updates to ASC 321, *Equity Securities*, ASC 323 *Investments - Equity Method and Joint Ventures*, and ASC 815, *Derivatives and Hedging*, which clarify how to account for the transition into and out of the equity method of accounting when evaluating observable transactions.

- Updates to ASC 350, *Intangibles - Goodwill and Other*, which eliminate the requirement to measure impairment based on the implied fair value of goodwill compared to the carrying amount of a reporting unit’s goodwill. Instead, goodwill impairment will be measured as the excess of a reporting unit’s carrying amount over its estimated fair value.

- Updates to ASC 350, *Intangibles - Goodwill and Other*, which align the requirements for capitalizing implementation costs incurred in a software hosting arrangement with the requirements for costs incurred to develop or obtain internal-use software.

- Adoption of ASC 326, *Financial Instruments - Credit Losses*, which amends the impairment model by requiring entities to use a forward-looking approach to estimate lifetime expected credit losses on certain types of financial instruments, including trade receivables.

FISCAL 2024 OUTLOOK

For fiscal year 2024, consolidated net sales from continuing operations are expected to be up 13 to 15.5 percent, with underlying sales up 4 to 6 percent excluding a 1 percent unfavorable impact from foreign currency translation and a 10 to 10.5 percent impact from the NI acquisition. Earnings per share, which incorporate the NI acquisition other than as set forth below, are expected to be $3.82 to $4.02, while adjusted earnings per share are expected to be $5.15 to $5.35 (see the following reconciliation).

Outlook for Fiscal 2024 Earnings Per Share	2024
Diluted earnings per share	$3.82 - $4.02
Amortization of intangibles	~ 0.67
Restructuring and related costs	~ 0.22
Acquisition/divestiture fees and related costs	~ 0.22
Copeland equity loss	~ 0.22
Adjusted diluted earnings per share	$5.15 - $5.35

Operating cash flow is expected to be $3.0 to $3.1 billion and free cash flow, which excludes projected capital spending of approximately $0.4 billion, is expected to be $2.6 to $2.7 billion. The fiscal 2024 outlook assumes approximately $500 million returned to shareholders through share repurchases and approximately $1.2 billion of dividend payments.

GAAP earnings per share guidance for fiscal 2024 does not include the impact of intangibles amortization and other purchase accounting-related costs related to the NI transaction. The initial accounting for this transaction is not yet complete and therefore Emerson is unable to estimate these amounts. Although these items may have a significant impact on GAAP earnings per share, they will be excluded from adjusted earnings per share and will have no impact on cash flows.

ITEM 7A - QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

The information from this Annual Report on Form 10-K set forth in Item 7 under "Financial Instruments" is hereby incorporated by reference.

ITEM 8 - FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

See the Company's consolidated financial statements and accompanying notes and the report thereon of KPMG LLP (PCAOB ID 185) that follow.

Consolidated Statements of Earnings
EMERSON ELECTRIC CO. & SUBSIDIARIES

Years ended September 30
(Dollars and shares in millions, except per share amounts)

	2021	2022	2023
Net sales	$ 12,932	13,804	15,165
Cost of sales	7,202	7,498	7,738
Selling, general and administrative expenses	3,494	3,614	4,186
Gain on subordinated interest	—	(453)	(161)
Other deductions, net	319	519	683
Interest expense, net of interest income of: 2021, $ 10; 2022, $34; 2023, $227	155	194	34
Interest income from related party	—	—	(41)
Earnings from continuing operations before income taxes	1,762	2,432	2,726
Income taxes	346	549	599
Earnings from continuing operations	1,416	1,883	2,127
Discontinued operations, net of tax of $ 239, $306 and $3,012, respectively	911	1,347	11,073
Net earnings	2,327	3,230	13,200
Less: Noncontrolling interests in earnings of subsidiaries	24	(1)	(19)
Net earnings common stockholders	$ 2,303	3,231	13,219
Earnings common stockholders:			
Earnings from continuing operations	$ 1,414	1,886	2,152
Discontinued operations	889	1,345	11,067
Net earnings common stockholders	$ 2,303	3,231	13,219
Basic earnings per share common stockholders:			
Earnings from continuing operations	$ 2.36	3.17	3.74
Discontinued operations	1.49	2.27	19.26
Basic earnings per common share	$ 3.85	5.44	23.00
Diluted earnings per share common stockholders:			
Earnings from continuing operations	$ 2.35	3.16	3.72
Discontinued operations	1.47	2.25	19.16
Diluted earnings per common share	$ 3.82	5.41	22.88
Weighted average outstanding shares:			
Basic	598.1	592.9	574.2
Diluted	601.8	596.3	577.3

See accompanying Notes to Consolidated Financial Statements.

Consolidated Statements of Comprehensive Income
EMERSON ELECTRIC CO. & SUBSIDIARIES

Years ended September 30
(Dollars in millions)

	2021	2022	2023
Net earnings	$ 2,327	3,230	13,200
Other comprehensive income (loss), net of tax:			
Foreign currency translation	81	(644)	254
Pension and postretirement	605	37	(25)
Cash flow hedges	18	(14)	4
Total other comprehensive income (loss)	704	(621)	233
Comprehensive income	3,031	2,609	13,433
Less: Noncontrolling interests in comprehensive income of subsidiaries	23	(9)	(18)
Comprehensive income common stockholders	$ 3,008	2,618	13,451

See accompanying Notes to Consolidated Financial Statements.

Consolidated Balance Sheets

EMERSON ELECTRIC CO. & SUBSIDIARIES

Years ended September 30 (Dollars and shares in millions, except per share amounts)

	2022	2023
ASSETS		
Current assets		
Cash and equivalents	$ 1,804	8,051
Receivables, less allowances of $100 in 2022 and $100 in 2023	2,261	2,518
Inventories	1,742	2,006
Other current assets	1,301	1,244
Current assets held-for-sale	1,398	—
Total current assets	8,506	13,819
Property, plant and equipment, net	2,239	2,363
Other assets		
Goodwill	13,946	14,480
Other intangible assets	6,572	6,263
Copeland note receivable and equity investment	—	3,255
Other	2,151	2,566
Noncurrent assets held-for-sale	2,258	—
Total other assets	24,927	26,564
Total assets	$ 35,672	42,746
LIABILITIES AND EQUITY		
Current liabilities		
Short-term borrowings and current maturities of long-term debt	$ 2,115	547
Accounts payable	1,276	1,275
Accrued expenses	3,038	3,210
Current liabilities held-for-sale	1,348	—
Total current liabilities	7,777	5,032
Long-term debt	8,259	7,610
Other liabilities	3,153	3,506
Noncurrent liabilities held-for-sale	167	—
Equity		
Common stock, $0.50 par value; authorized, 1,200.0 shares; issued, 953.4 shares; outstanding, 591.4 shares in 2022; 572.0 shares in 2023	477	477
Additional paid-in-capital	57	62
Retained earnings	28,053	40,070
Accumulated other comprehensive income (loss)	(1,485)	(1,253)
Cost of common stock in treasury, 362.0 shares in 2022; 381.4 shares in 2023	(16,738)	(18,667)
Common stockholders' equity	10,364	20,689
Noncontrolling interests in subsidiaries	5,952	5,909
Total equity	16,316	26,598
Total liabilities and equity	$ 35,672	42,746

See accompanying Notes to Consolidated Financial Statements.

Consolidated Statements of Equity
EMERSON ELECTRIC CO. & SUBSIDIARIES

Years ended September 30
(Dollars in millions, except per share amounts)

	2021	2022	2023
Common stock	$ 477	477	477
Additional paid-in-capital			
Beginning balance	470	522	57
Stock plans	52	85	127
AspenTech purchases of common stock	—	—	(122)
AspenTech acquisition	—	(550)	—
Ending balance	522	57	62
Retained earnings			
Beginning balance	24,955	26,047	28,053
Net earnings common stockholders	2,303	3,231	13,219
Dividends paid (per share: 2021, $2.02; 2022, $2.06; 2023, $2.08)	(1,210)	(1,225)	(1,202)
Adoption of accounting standard updates	(1)	—	—
Ending balance	26,047	28,053	40,070
Accumulated other comprehensive income (loss)			
Beginning balance	(1,577)	(872)	(1,485)
Foreign currency translation	82	(636)	253
Pension and postretirement	605	37	(25)
Cash flow hedges	18	(14)	4
Ending balance	(872)	(1,485)	(1,253)
Treasury stock			
Beginning balance	(15,920)	(16,291)	(16,738)
Purchases	(500)	(500)	(2,000)
Issued under Emerson stock plans	129	53	71
Ending balance	(16,291)	(16,738)	(18,667)
Common stockholders' equity	9,883	10,364	20,689
Noncontrolling interests in subsidiaries			
Beginning balance	42	40	5,952
Net earnings	24	(1)	(19)
Stock plans	—	35	94
AspenTech purchases of common stock	—	—	(92)
Other comprehensive income	(1)	(8)	1
Dividends paid	(25)	(4)	(1)
AspenTech acquisition	—	5,890	—
Purchase of noncontrolling interests	—	—	3
Climate Technologies divestiture	—	—	(29)
Ending balance	40	5,952	5,909
Total equity	$ 9,923	16,316	26,598

See accompanying Notes to Consolidated Financial Statements.

Consolidated Statements of Cash Flows
EMERSON ELECTRIC CO. & SUBSIDIARIES

Years ended September 30 (Dollars in millions)

	2021	2022	2023
Operating activities			
Net earnings	$ 2,327	3,230	13,200
Earnings from discontinued operations, net of tax	(911)	(1,347)	(11,073)
Adjustments to reconcile net earnings to net cash provided by operating activities:			
Depreciation and amortization	762	842	1,051
Stock compensation	197	125	250
Pension expense (income)	28	2	(71)
Pension funding	(41)	(43)	(43)
Changes in operating working capital	167	(312)	(190)
Gain on subordinated interest	—	(453)	(161)
Other, net	(71)	4	(237)
Cash from continuing operations	2,458	2,048	2,726
Cash from discontinued operations	1,117	874	(2,089)
Cash provided by operating activities	3,575	2,922	637
Investing activities			
Capital expenditures	(404)	(299)	(363)
Purchases of businesses, net of cash and equivalents acquired	(1,592)	(5,702)	(705)
Divestitures of businesses	30	17	—
Proceeds from subordinated interest	—	438	176
Proceeds from related party note receivable	—	—	918
Other, net	(25)	(138)	(141)
Cash from continuing operations	(1,991)	(5,684)	(115)
Cash from discontinued operations	(129)	350	12,530
Cash provided by (used in) investing activities	(2,120)	(5,334)	12,415
Financing activities			
Net increase (decrease) in short-term borrowings	(504)	1,241	(1,578)
Proceeds from short-term borrowings greater than three months	71	1,162	395
Payments of short-term borrowings greater than three months	(71)	(1,165)	(400)
Proceeds from long-term debt	—	2,975	—
Payments of long-term debt	(308)	(522)	(741)
Dividends paid	(1,210)	(1,223)	(1,198)
Purchases of common stock	(500)	(500)	(2,000)
AspenTech purchases of common stock	—	—	(214)
Payment of related party note payable	—	—	(918)
Other, net	100	80	(169)
Cash provided by (used in) financing activities	(2,422)	2,048	(6,823)
Effect of exchange rate changes on cash and equivalents	6	(186)	18
Increase (Decrease) in cash and equivalents	(961)	(550)	6,247
Beginning cash and equivalents	3,315	2,354	1,804
Ending cash and equivalents	$ 2,354	1,804	8,051
Changes in operating working capital			
Receivables	$ (18)	(143)	(191)
Inventories	(11)	(334)	(160)
Other current assets	(91)	(56)	(1)
Accounts payable	107	147	(17)
Accrued expenses	180	74	179
Total changes in operating working capital	$ 167	(312)	(190)

See accompanying Notes to Consolidated Financial Statements.

Years ended September 30
(Dollars in millions, except per share amounts or where noted)

(1) SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES

Financial Statement Presentation

The preparation of the financial statements in conformity with U.S. generally accepted accounting principles (U.S. GAAP) requires management to make estimates and assumptions that affect reported amounts and related disclosures. Actual results could differ from these estimates. Certain prior year amounts have been reclassified to conform to the current year presentation. This includes reporting financial results for Climate Technologies, InSinkErator and Therm-O-Disc as discontinued operations for all periods presented, and the assets and liabilities of Climate Technologies and InSinkErator (prior to completion of the divestitures) as held-for-sale (see Note 5). In addition, as a result of its portfolio transformation, the Company now reports six segments and two business groups (see Note 20).

In 2023, the Company adopted ASU No. 2021-10 (Topic 832), *Government Assistance,* which requires annual disclosures about certain types of government assistance received. This standard has no impact on the accounting for government assistance and did not materially impact the Company's disclosures.

In 2022, the Company adopted three accounting standard updates, and in 2021 adopted two accounting standard updates and one new accounting standard, each of which had an immaterial or no impact on the Company's financial statements. These included:

- Updates to Accounting Standards Codification ("ASC") 805, *Business Combinations*, which clarify the accounting for contract assets and liabilities assumed in a business combination. In general, this will result in contract liabilities being recognized at their historical amounts under ASC 606, rather than at fair value in accordance with the general requirements of ASC 805.

- Updates to ASC 740, *Income Taxes*, which require the recognition of a franchise tax that is partially based on income as an income-based tax with any incremental amount as a non-income based tax. These updates also make certain changes to intra-period tax allocation principles and interim tax calculations.

- Updates to ASC 321, *Equity Securities*, ASC 323 *Investments - Equity Method and Joint Ventures*, and ASC 815, *Derivatives and Hedging*, which clarify how to account for the transition into and out of the equity method of accounting when evaluating observable transactions.

- Updates to ASC 350, *Intangibles - Goodwill and Other*, which eliminate the requirement to measure impairment based on the implied fair value of goodwill compared to the carrying amount of a reporting unit’s goodwill. Instead, goodwill impairment will be measured as the excess of a reporting unit’s carrying amount over its estimated fair value.

- Updates to ASC 350, *Intangibles - Goodwill and Other*, which align the requirements for capitalizing implementation costs incurred in a software hosting arrangement with the requirements for costs incurred to develop or obtain internal-use software.

- Adoption of ASC 326, *Financial Instruments - Credit Losses*, which amends the impairment model by requiring entities to use a forward-looking approach to estimate lifetime expected credit losses on certain types of financial instruments, including trade receivables.

Principles of Consolidation

The consolidated financial statements include the accounts of the Company and its controlled affiliates. Intercompany transactions, profits and balances are eliminated in consolidation. Investments of 20 percent to 50 percent of the voting shares of other entities are accounted for by the equity method. Investments in publicly traded companies of less than 20 percent are carried at fair value, with changes in fair value reflected in earnings.

Investments in nonpublicly traded companies of less than 20 percent are carried at cost, minus impairment, and adjusted for observable price changes in orderly transactions.

Foreign Currency Translation

The functional currency for most of the Company's non-U.S. subsidiaries is the local currency. Adjustments resulting from translating local currency financial statements into U.S. dollars are reflected in accumulated other comprehensive income.

Cash Equivalents

Cash equivalents consist of highly liquid investments with original maturities of three months or less.

Inventories

Inventories are stated at the lower of cost and net realizable value. The majority of inventory is valued based on standard costs, which are revised at the beginning of each year and approximate average costs, while the remainder is principally valued on a first-in, first-out basis. Following are the components of inventory as of September 30:

	2022	2023
Finished products	$ 417	446
Raw materials and work in process	1,325	1,560
Total inventories	$ 1,742	2,006

Fair Value Measurement

ASC 820, *Fair Value Measurement*, establishes a formal hierarchy and framework for measuring certain financial statement items at fair value, and requires disclosures about fair value measurements and the reliability of valuation inputs. Under ASC 820, measurement assumes the transaction to sell an asset or transfer a liability occurs in the principal or at least the most advantageous market for that asset or liability. Within the hierarchy, Level 1 instruments use observable market prices for an identical item in active markets and have the most reliable valuations. Level 2 instruments are valued through broker/dealer quotation or other approaches using market-observable inputs for similar items in active markets, including forward and spot prices, interest rates and volatilities. Level 3 instruments are valued using inputs not observable in an active market, such as company-developed future cash flow estimates, and are considered the least reliable. Valuations for all of the Company's financial instruments fall within Level 2. The fair value of the Company's long-term debt and note receivable from Copeland are Level 2, estimated using current interest rates and pricing from financial institutions and other market sources for debt with similar maturities and characteristics.

Property, Plant and Equipment

The Company records investments in land, buildings, and machinery and equipment at cost. Depreciation is computed principally using the straight-line method over estimated service lives, which for principal assets are 30 to 40 years for buildings and 8 to 12 years for machinery and equipment. Long-lived tangible assets are reviewed for impairment whenever events or changes in business circumstances indicate the carrying value of the assets may not be recoverable. Impairment losses are recognized based on estimated fair values if the sum of estimated future undiscounted cash flows of the related assets is less than the carrying values.

The components of property, plant and equipment as of September 30 follow:

	2022	2023
Land	$ 200	255
Buildings	1,500	1,758
Machinery and equipment	3,300	3,228
Construction in progress	390	283
Property, plant and equipment, at cost	5,390	5,524
Less: Accumulated depreciation	3,151	3,161
Property, plant and equipment, net	$ 2,239	2,363

Goodwill and Other Intangible Assets

Assets and liabilities acquired in business combinations are accounted for using the acquisition method and recorded at their respective fair values. Substantially all goodwill is assigned to the reporting unit that acquires a business. A reporting unit is an operating segment as defined in ASC 280, *Segment Reporting*, or a business one level below an operating segment if discrete financial information for that business unit is prepared and regularly reviewed by the segment manager. The Company conducts annual impairment tests of goodwill in the fourth quarter. If an initial assessment indicates it is more likely than not goodwill might be impaired, it is evaluated by comparing the reporting unit's estimated fair value to its carrying value. An impairment charge would be recorded for the amount by which the carrying value of the reporting unit exceeds the estimated fair value. Goodwill is also tested for impairment between annual tests if events or circumstances indicate the fair value of a unit may be less than its carrying value. Estimated fair values of reporting units are Level 3 measures and are developed generally under an income approach that discounts estimated future cash flows using risk-adjusted interest rates, as well as earnings multiples or other techniques as warranted. Fair values are subject to changes in underlying economic conditions.

With the exception of certain trade names, all of the Company's identifiable intangible assets are subject to amortization on a straight-line basis over their estimated useful lives. Identifiable intangibles consist of intellectual property such as technology, patents and trademarks, customer relationships and capitalized software. Identifiable intangibles are also subject to evaluation for potential impairment if events or circumstances indicate the carrying amount may not be recoverable. See Note 10.

Leases

The Company leases offices; manufacturing facilities and equipment; and transportation, information technology and office equipment under operating lease arrangements. Finance lease arrangements are immaterial. The Company determines whether an arrangement is, or contains, a lease at contract inception. An arrangement contains a lease if the Company has the right to direct the use of and obtain substantially all of the economic benefits of an identified asset. Right-of-use assets and lease liabilities are recognized at lease commencement based on the present value of lease payments over the lease term. Leases with an initial term of 12 months or less are not recognized on the balance sheet and are recorded as short-term lease expense. The discount rate used to calculate present value is the Company's incremental borrowing rate based on the lease term and the economic environment of the applicable country or region.

Certain leases contain renewal options or options to terminate prior to lease expiration, which are included in the measurement of right-of-use assets and lease liabilities when it is reasonably certain they will be exercised. The Company has elected to account for lease and non-lease components as a single lease component for its offices and manufacturing facilities. Some lease arrangements include payments that are adjusted periodically based on actual charges incurred for common area maintenance, utilities, taxes and insurance, or changes in an index or rate referenced in the lease. The fixed portion of these payments is included in the measurement of right-of-use assets and lease liabilities at lease commencement, while the variable portion is recorded as variable lease expense. The Company's leases typically do not contain material residual value guarantees or restrictive covenants.

Product Warranty

Warranties vary by product line and are competitive for the markets in which the Company operates. Warranties are largely offered to provide assurance that the product will function as intended and generally extend for a period of one to two years from the date of sale or installation. Provisions for warranty expense are estimated at the time of sale based on historical experience and adjusted quarterly for any known issues that may arise. Product warranty expense is less than one-half of one percent of sales.

Revenue Recognition

Emerson is a global manufacturer that designs and manufactures products and delivers services that bring technology and engineering together to provide innovative solutions for its customers. The Company evaluates its contracts with customers to identify the promised goods or services and recognizes revenue for the identified performance obligations at the amount the Company expects to be entitled to in exchange for those goods or services. A performance obligation is a promise in a contract to transfer a distinct good or service to a customer. Revenue is recognized when, or as, performance obligations are satisfied and control has transferred to the customer, typically when products are shipped or delivered, title and risk of loss pass to the customer, and the Company has a present right to payment. The majority of the Company's revenues relate to a broad offering of manufactured products and software which are recognized at the point in time when control transfers, generally in accordance with shipping terms, or the first day of the contractual term for software. A portion of the Company's

revenues relate to the sale of post-contract customer support, parts and labor for repairs, and engineering services. In some circumstances, contracts include multiple performance obligations, where revenue is recognized separately for each good or service, as well as contracts where revenue is recognized over time as control transfers to the customer.

Revenue is recognized over time for approximately 10 percent of the Company's revenues. These revenues primarily relate to projects in the Control Systems & Software segment where revenue is recognized using the percentage-of-completion method to reflect the transfer of control over time, and software maintenance contracts in the AspenTech and Control Systems & Software segments where revenue is typically recognized on a straight-line basis. Approximately 10 percent of revenues relate to sales arrangements with multiple performance obligations, principally in the AspenTech and Control Systems & Software segments. Tangible products represent a large majority of the delivered items in contracts with multiple performance obligations or where revenue is recognized over time, while a smaller portion is attributable to installation, service and maintenance.

For projects where revenue is recognized over time, the Company typically uses an input method to determine progress and recognize revenue, based on costs incurred. The Company believes costs incurred closely correspond with its performance under the contract and the transfer of control to the customer. For software maintenance contracts, revenue is recognized ratably over the maintenance term.

In sales arrangements that involve multiple performance obligations, revenue is allocated based on the relative standalone selling price for each performance obligation. Observable selling prices from actual transactions are used whenever possible. In other instances, the Company determines the standalone selling price based on third-party pricing or management's best estimate. Generally, contract duration is short-term, and cancellation, termination or refund provisions apply only in the event of contract breach and are rarely invoked.

Payment terms vary but are generally short-term in nature. The Company's long-term contracts, where revenue is generally recognized over time, are typically billed as work progresses in accordance with the contract terms and conditions, either at periodic intervals or upon achievement of certain milestones. The timing of revenue recognition and billings under these contracts results in either unbilled receivables (contract assets) when revenue recognized exceeds billings, or customer advances (contract liabilities) when billings exceed revenue recognized. Unbilled receivables are reclassified to accounts receivable when an unconditional right to consideration exists, typically when a milestone in the contract is achieved. The Company does not evaluate whether the transaction price includes a significant financing component for contracts where the time between cash collection and performance is less than one year.

Certain arrangements with customers include variable consideration, typically in the form of rebates, cash discounts or penalties. In limited circumstances, the Company sells products with a general right of return. In most instances, returns are limited to product quality issues. The Company records a reduction to revenue at the time of sale to reflect the ultimate amount of consideration it expects to receive. The Company's estimates are updated quarterly based on historical experience, trend analysis, and expected market conditions. Variable consideration is typically not constrained at the time revenue is recognized. See Notes 2 and 20 for additional information about the Company's revenues.

Derivatives and Hedging

In the normal course of business, the Company is exposed to changes in interest rates and foreign currency exchange rates due to its worldwide presence and diverse business pr ofile. The Company's foreign currency exposures relate to transactions denominated in currencies that differ from the functional currencies of its business units, primarily in euros, Mexican pesos, and Singapore dollars. As part of the Company's risk management strategy, derivative instruments are selectively used in an effort to minimize the impact of these exposures. Foreign exchange forwards and options are utilized to hedge foreign currency exposures impacting sales or cost of sales transactions, firm commitments and the fair value of assets and liabilities. Non-U.S. dollar obligations are utilized to reduce foreign currency risk associated with the Company's net investments in foreign operations. All derivatives are associated with specific underlying exposures and the Company does not hold derivatives for trading or speculative purposes. The duration of hedge positions is generally two years or less, except for the Company's net investment hedges.

All derivatives are accounted for under ASC 815, *Derivatives and Hedging*, and recognized at fair value. For derivatives hedging variability in future cash flows, any gain or loss is deferred in stockholders' equity and recognized when the underlying hedged transaction impacts earnings. The majority of the Company's derivatives

that are designated as hedges and qualify for hedge accounting are cash flow hedges. For derivatives hedging the fair value of existing assets or liabilities, both the gain or loss on the derivative and the offsetting loss or gain on the hedged item are recognized in earnings each period. Currency fluctuations on non-U.S. dollar obligations that have been designated as hedges of net investments in foreign operations are recognized in accumulated other comprehensive income (loss) and reclassified to income in the same period when a foreign operation is sold or substantially liquidated and the gain or loss related to the sale is included in income. To the extent that any hedge is not fully effective at offsetting changes in the underlying hedged item, there could be a net earnings impact.

The Company also uses derivatives to hedge economic exposures that do not receive hedge accounting under ASC 815. The underlying exposures for these hedges relate primarily to the revaluation of certain foreign-currency-denominated assets and liabilities. In addition, in 2022 AspenTech entered into foreign currency forward contracts to mitigate the impact of foreign currency exchange associated with the Micromine purchase price. On June 21, 2023, AspenTech terminated all outstanding foreign currency forward contracts and on August 1, 2023, announced the termination of the agreement to purchase Micromine. Gains or losses on derivative instruments not designated as hedges are recognized in the income statement immediately.

Counterparties to derivative arrangements are companies with investment-grade credit ratings. The Company has bilateral collateral arrangements with counterparties with credit rating-based posting thresholds that vary depending on the arrangement. If credit ratings on the Company's debt fall below pre-established levels, counterparties can require immediate full collateralization on all derivatives in net liability positions. The maximum amount that could potentially have been required was immaterial. The Company also can demand full collateralization of derivatives in net asset positions should any counterparty credit ratings fall below certain thresholds. No collateral was posted with counterparties and none was held by the Company at year end. Risk from credit loss when derivatives are in asset positions is not considered material. The Company has master netting arrangements in place with its counterparties that allow the offsetting of certain derivative-related amounts receivable and payable when settlement occurs in the same period. Accordingly, counterparty balances are netted in the consolidated balance sheet and are reported in other current assets or accrued expenses as appropriate, depending on positions with counterparties as of the balance sheet date. See Note 11.

Income Taxes

The provision for income taxes is based on pretax income reported in the consolidated statements of earnings and tax rates currently enacted in each jurisdiction. Certain income and expense items are recognized in different time periods for financial reporting and income tax filing purposes, and deferred income taxes are provided for the effect of temporary differences. The Tax Cuts and Jobs Act subjects the Company to U.S. tax on global intangible low-taxed income earned by certain of its non-U.S. subsidiaries. The Company has elected to recognize this tax as a period expense when it is incurred. The Company also provides for withholding taxes and any applicable U.S. income taxes on earnings intended to be repatriated from non-U.S. locations. No provision has been made for these taxes on approximately $6.5 billion of undistributed earnings of non-U.S. subsidiaries as of September 30, 2023, as these earnings are considered indefinitely invested or otherwise retained for continuing international operations. Recognition of withholding taxes and any applicable U.S. income taxes on undistributed non-U.S. earnings would be triggered by a management decision to repatriate those earnings. Determination of the amount of taxes that might be paid on these undistributed earnings if eventually remitted is not practicable. See Note 16.

(2) REVENUE RECOGNITION

The following table summarizes the balances of the Company's unbilled receivables (contract assets), which are reported in Other assets (current and noncurrent), and its customer advances (contract liabilities), which are reported in Accrued expenses and Other liabilities.

	2022	2023
Unbilled receivables (contract assets)	$ 1,390	1,453
Customer advances (contract liabilities)	(776)	(897)
Net contract assets	$ 614	556

The majority of the Company's contract balances relate to (1) arrangements where revenue is recognized over time and payments from customers are made according to a contractual billing schedule, and (2) revenue from term software license arrangements sold by AspenTech where the license revenue is recognized upfront upon delivery. The decrease in net contract assets was due to customer billings exceeding revenue recognized for performance

completed during the period. Revenue recognized for 2023 included approximately $534 that was included in the beginning contract liability balance. Other factors that impacted the change in net contract liabilities were immaterial.

Revenue recognized for 2023 for performance obligations that were satisfied in previous periods, including cumulative catchup adjustments on the Company's long-term contracts, was not material. Capitalized amounts related to incremental costs to obtain customer contracts and costs to fulfill contracts are immaterial.

As of September 30, 2023, the Company's backlog relating to unsatisfied (or partially unsatisfied) performance obligations in contracts with its customers was approximately $7.8 billion (of which approximately $1.2 billion related to AspenTech). AspenTech's remaining performance obligations primarily relate to software maintenance in long-term contracts for unspecified future software updates provided on a when-and-if available basis. The Company expects to recognize approximately 75 percent of its remaining performance obligations as revenue over the next 12 months, with the remainder substantially over the subsequent two years thereafter.

See Note 20 for additional information about the Company's revenues.

(3) WEIGHTED-AVERAGE COMMON SHARES

Basic earnings per common share consider only the weighted-average of common shares outstanding while diluted earnings per common share also consider the dilutive effects of stock options and incentive shares. An inconsequential number of shares of common stock were excluded from the computation of dilutive earnings per share in 2023, 2022 and 2021 as the effect would have been antidilutive. Earnings allocated to participating securities were inconsequential for all years presented.

Reconciliations of weighted-average shares for basic and diluted earnings per common share follow (shares in millions):

	2021	2022	2023
Basic shares outstanding	598.1	592.9	574.2
Dilutive shares	3.7	3.4	3.1
Diluted shares outstanding	601.8	596.3	577.3

(4) ACQUISITIONS AND DIVESTITURES

Aspen Technology

On May 16, 2022, the Company completed the transactions contemplated by its definitive agreement with Aspen Technology, Inc. ("Heritage AspenTech") to contribute two of Emerson's stand-alone industrial software businesses, Open Systems International, Inc. and the Geological Simulation Software business (collectively, the “Emerson Industrial Software Business”), along with approximately $6.0 billion in cash to Heritage AspenTech stockholders, to create "New AspenTech", a diversified, high-performance industrial software leader with greater scale, capabilities and technologies (defined as "AspenTech" herein). Upon closing of the transaction, Emerson beneficially owned 55 percent of the outstanding shares of AspenTech common stock (on a fully diluted basis) and former Heritage AspenTech stockholders owned the remaining outstanding shares of AspenTech common stock. AspenTech and its subsidiaries now operate under Heritage AspenTech’s previous name “Aspen Technology, Inc.” and AspenTech common stock is traded on NASDAQ under AspenTech’s previous stock ticker symbol “AZPN.”

The business combination has been accounted for using the acquisition method of accounting with Emerson considered the accounting acquirer of Heritage AspenTech. The net assets of Heritage AspenTech were recorded at their estimated fair value and the Emerson Industrial Software Business continues at its historical basis. The Company recorded a noncontrolling interest of $5.9 billion for the 45 percent ownership interest of former Heritage AspenTech stockholders in AspenTech. The noncontrolling interest associated with the Heritage AspenTech acquired net assets was recorded at fair value determined using the closing market price per share of Heritage AspenTech as of May 16, 2022, while the portion attributable to the Emerson Industrial Software business was recorded at its historical carrying amount. The impact of recognizing the noncontrolling interest in the Emerson Industrial Software Business resulted in a decrease to additional paid-in-capital of $550.

The following table summarizes the components of the purchase consideration reflected in the acquisition accounting using Heritage AspenTech's shares outstanding and closing market price per share as of May 16, 2022 (in millions except share and per share data):

Heritage AspenTech shares outstanding		66,662,482
Heritage AspenTech share price	$	166.30
Purchase price	$	11,086
Value of stock-based compensation awards attributable to pre-combination service		102
Total purchase consideration	$	11,188

The total purchase consideration for Heritage AspenTech was allocated to assets and liabilities as follows.

Cash and equivalents	$	274
Receivables		43
Other current assets		280
Property, plant equipment		4
Goodwill ($34 expected to be tax-deductible)		7,225
Other intangible assets		4,390
Other assets		513
Total assets		12,729
Short-term borrowings		27
Accounts payable		8
Accrued expenses		115
Long-term debt		255
Deferred taxes and other liabilities		1,136
Total purchase consideration	$	11,188

Emerson's cash contribution of approximately $6.0 billion was paid out at approximately $ 87.69 per share (on a fully diluted basis) to holders of issued and outstanding shares of Heritage AspenTech common stock as of the closing of the transactions, with $168 of cash remaining on AspenTech's balance sheet as of the closing which is not included in the allocation of purchase consideration above.

The estimated intangible assets attributable to the transaction are comprised of the following (in millions):

	Amount		Estimated Useful Life (Years)
Developed technology	$	1,350	10
Customer relationships		2,300	15
Trade names		430	Indefinite-lived
Backlog		310	3
Total	$	4,390	

Results of operations for 2023 attributable to the Heritage AspenTech acquisition include sales of $ 752 compared to $356 for 2022, while the impact to GAAP net earnings was not material in both years.

Pro Forma Financial Information

The following unaudited proforma consolidated condensed financial results of operations are presented as if the acquisition of Heritage AspenTech occurred on October 1, 2020. The pro forma information is presented for

informational purposes only and is not indicative of the results of operations that would have been achieved had the acquisition occurred as of that time ($ in millions, except per share amounts).

		2021	2022
Net Sales	$	13,662	14,218
Net earnings from continuing operations common stockholders	$	1,217	1,916
Diluted earnings per share from continuing operations	$	2.02	3.21

The pro forma results for 2021 include $159 of transaction costs which were assumed to be incurred in the first quarter of 2021. Of these transaction costs, $91 were included in the Company's reported results for 2022, but have been excluded from the 2022 pro forma results above. In addition, Heritage AspenTech incurred $68 of transaction costs prior to the completion of the acquisition that were not included in Emerson's reported results. The pro forma results for 2021 include estimated interest expense of $147 related to the issuance of $3.0 billion of term debt and increased commercial paper borrowings to fund the acquisition, while results for 2022 include additional interest expense of $56 to reflect the increased borrowings as if they were outstanding for the entire year.

Other Transactions

In 2023, the Company acquired two businesses, Flexim, which will be reported in the Measurement & Analytical segment, and Afag, which will be reported in the Discrete Automation segment, for $705, net of cash acquired. The Company recognized goodwill of $ 429 (none of which is expected to be tax deductible) and other identifiable intangible assets of $314, primarily customer relationships and intellectual property with a weighted-average useful life of approximately 9 years.

On July 27, 2022, AspenTech entered into an agreement to acquire Micromine, a global leader in design and operational solutions for the mining industry, for AU $900 (approximately $623 USD based on exchange rates when the transaction was announced). On August 1, 2023, AspenTech announced the termination of the agreement to purchase Micromine. AspenTech, along with the sellers of Micromine, had been waiting to secure a final Russian regulatory approval as a condition to the closing of the transaction. As this process continued, the timing and requirements necessary to get this approval became increasingly unclear. This lack of clarity on the potential for, and timing of, a successful review led AspenTech and the sellers of Micromine to this mutual course of action. AspenTech did not pay any termination fee as part of this arrangement.

On March 31, 2023, Emerson completed the divestiture of Metran, its Russia-based manufacturing subsidiary. In 2023, the Company recognized a pretax loss of $47 in Other deductions ($47 after-tax, in total $0.08 per share) related to its exit of business operations in Russia. The Company had previously announced its intention to exit business operations in 2022 and recognized a pretax loss of $181 ($190 after-tax, in total $0.32 per share). This charge included a loss of $36 in operations and $145 reported in Other deductions ($10 of which is reported in restructuring costs) and was primarily non-cash. Emerson's historical net sales in Russia represented approximately 2.0 percent of consolidated annual sales.

In 2022, the Company acquired three other businesses, two in the Control Systems & Software segment and one in the AspenTech segment, for $130, net of cash acquired. The three businesses had combined annual sales of approximately $ 40.

In the first quarter of 2022, the Company received a distribution of $ 438 related to its subordinated interest in Vertiv (in total, a pretax gain of $ 453 was recognized in the first quarter of 2022, $358 after-tax, $0.60 per share) and received the remaining $ 15 related to the pretax gain in the first quarter of 2023. In 2023, the Company received additional distributions totaling $161 ($122 after-tax, $0.21 per share). Based on the terms of the agreement and the current calculation, the Company could receive additional distributions of approximately $40. The remaining distributions are contingent on the timing and price at which Vertiv shares are sold by the equity holders and therefore, there can be no assurance as to the amount or timing of the remaining distributions to the Company.

(5) DISCONTINUED OPERATIONS

On May 31, 2023, the Company completed the previously announced sale of a majority stake in its Climate Technologies business (which constitutes the former Climate Technologies segment, excluding Therm-O-Disc which was divested earlier in 2022) to private equity funds managed by Blackstone in a $14.0 billion transaction. Emerson received upfront, pre-tax cash proceeds of approximately $ 9.7 billion (an increase of $0.2 billion from when the

transaction was announced due to Blackstone's decision to purchase an additional 5 percent of the common equity) and a note receivable with a face value of $2.25 billion (which will accrue 5 percent interest payable in kind by capitalizing interest), while retaining a 40 percent non-controlling common equity interest (down from 45 percent when the transaction was announced) in a new standalone joint venture between Emerson and Blackstone. The Climate Technologies business, which includes the Copeland compressor business and the entire portfolio of products and services across all residential and commercial HVAC and refrigeration end-markets, had 2022 net sales of approximately $5.0 billion and pretax earnings of $ 1.0 billion. The Company recognized a pretax gain of approximately $10.6 billion (approximately $8.4 billion after-tax including tax expense recognized in prior quarters related to subsidiary restructurings). The new standalone business is named Copeland. See Note 8 for further details.

On October 31, 2022, the Company completed the divestiture of its InSinkErator business, which manufactures food waste disposers, to Whirlpool Corporation for $3.0 billion. This business had net sales of $ 630 and pretax earnings of $ 152 in 2022. The Company recognized a pretax gain of approximately $2.8 billion (approximately $2.1 billion after-tax) in the first quarter of 2023.

On May 31, 2022 the Company com pleted the divestiture of its Therm-O-Disc sensing and protection technologies business to an affiliate of One Rock Capital Partners, LLC. The Company recognized a pretax gain of $486 ($429 after-tax) in the third quarter of 2022.

The financial results of Climate Technologies, InSinkErator ("ISE") and Therm-O-Disc ("TOD") (through the completion of the divestitures), are reported as discontinued operations for all years presented and were as follows:

	Climate Technologies			ISE and TOD			Total		
	2021	2022	2023	2021	2022	2023	2021	2022	2023
Net sales	$ 4,401	4,976	3,156	903	848	49	5,304	5,824	3,205
Cost of sales	2,899	3,405	2,000	572	538	29	3,471	3,943	2,029
SG&A	560	514	390	125	119	7	685	633	397
Gain on sale of business	—	—	(10,610)	—	(486)	(2,783)	—	(486)	(13,393)
Other deductions, net	(10)	55	75	8	26	12	(2)	81	87
Earnings before income taxes	952	1,002	11,301	198	651	2,784	1,150	1,653	14,085
Income taxes	196	209	2,358	43	97	654	239	306	3,012
Earnings, net of tax	$ 756	793	8,943	155	554	2,130	911	1,347	11,073

Climate Technologies' results for 2023 include lower expense of $ 96 due to ceasing depreciation and amortization upon the held-for-sale classification and $57 of transaction-related costs reported in Other deductions, net. Income taxes for 2023 included approximately $ 2.2 billion for the gain on the Copeland transaction and subsidiary restructurings, and approximately $660 related to the gain on the InSinkErator divestiture.

The aggregate carrying amounts of the major classes of assets and liabilities classified as held-for-sale as of September 30, 2023 and 2022 are summarized as follows:

	Climate Technologies		ISE		Total	
	September 30,		September 30,		September 30,	
Assets	2022	2023	2022	2023	2022	2023
Receivables	$ 747	—	68	—	815	—
Inventories	449	—	81	—	530	—
Other current assets	49	—	4	—	53	—
Property, plant & equipment, net	1,122	—	141	—	1,263	—
Goodwill	716	—	2	—	718	—
Other noncurrent assets	265	—	12	—	277	—
Total assets held-for-sale	$ 3,348	—	308	—	3,656	—
Liabilities						
Accounts payable	$ 752	—	60	—	812	—
Other current liabilities	475	—	61	—	536	—
Deferred taxes and other noncurrent liabilities	154	—	13	—	167	—
Total liabilities held-for-sale	$ 1,381	—	134	—	1,515	—

Net cash from operating and investing activities for Climate Technologies, InSinkErator and Therm-O-Disc were as follows:

	Climate Technologies			ISE and TOD			Total		
	2021	2022	2023	2021	2022	2023	2021	2022	2023
Cash from operating activities	$ 906	881	(1,330)	211	(7)	(759)	1,117	874	(2,089)
Cash from investing activities	$ (80)	(202)	9,475	(49)	552	3,055	(129)	350	12,530

Cash from operating activities for 2023 reflects approximately $2.3 billion of income taxes paid related to the gains on the Copeland transaction and InSinkErator divestiture and subsidiary restructurings related to the Copeland transaction. Cash from investing activities for 2023 reflects the proceeds of approximately $9.7 billion related to the Copeland transaction and approximately $ 3.0 billion related to the InSinkErator divestiture.

(6) OTHER DEDUCTIONS, NET

Other deductions, net are summarized below:

	2021	2022	2023
Amortization of intangibles (intellectual property and customer relationships)	$ 277	336	482
Restructuring costs	132	75	72
Acquisition/divestiture costs	—	91	69
Foreign currency transaction (gains) losses	2	12	50
Investment-related gains & gains from sales of capital assets	(21)	(30)	(69)
Loss on Copeland equity method investment	—	—	177
Russia business exit	—	135	47
Other	(71)	(100)	(145)
Total	$ 319	519	683

In 2023, intangibles amortization included $258 related to the Heritage AspenTech acquisition compared to $ 97 in 2022, while 2021 included backlog amortization related to the OSI acquisition of $30. Foreign currency transaction losses included a mark-to-market gain of $ 24 in 2023 related to foreign currency forward contracts entered into by AspenTech to mitigate the impact of foreign currency exchange associated with the Micromine purchase price compared to a mark-to-market loss of $50 in 2022. On June 21, 2023, AspenTech terminated all outstanding foreign currency forward contracts. The Company recognized a mark-to-market gain of $56 in 2023 related to its equity investment in National Instruments Corporation (see Note 11 for further information). Other is composed of several items, including pension expense, litigation costs, provision for bad debt and other items, none of which is individually significant.

(7) RESTRUCTURING COSTS

Each year the Company incurs costs to size its businesses to levels appropriate for current economic conditions and to continually improve its cost structure and operational efficiency, deploy assets globally, and remain competitive on a worldwide basis. Costs result from numerous individual actions implemented across the Company's various operating units on an ongoing basis and can include costs for moving facilities to best-cost locations, restarting plants after relocation or geographic expansion to better serve local markets, reducing headcount or the number of facilities, exiting certain product lines, and other costs resulting from asset deployment decisions (such as contract termination costs, asset write-downs and vacant facility costs).

Restructuring expenses were $72, $75 and $132 for 2023, 2022 and 2021, respectively. The Company expects fiscal year 2024 restructuring and related costs to be approximately $160, including incremental costs related to the National Instruments acquisition.

Restructuring costs by business segment follows:

	2021	2022	2023
Measurement & Analytical	$ 58	3	9
Final Control	41	38	12
Discrete Automation	11	—	27
Safety & Productivity	4	10	—
Intelligent Devices	114	51	48
AspenTech	2	—	1
Control Systems & Software	11	11	9
Software and Control	13	11	10
Corporate	5	13	14
Total	$ 132	75	72

Actions taken in 2023 and 2022 inc luded workforce reductions of approximately 700 and 2,150 positions and the exit of ten and seven production facilities worldwide, respectively. Costs incurred in 2021 primarily relate to the Company's initiatives to improve operating margins that began in the third quarter of fiscal 2019 and were expanded in the third quarter of fiscal 2020 in response to the effects of COVID-19 on demand for the Company's products. Expenses incurred in 2021 included actions to exit five facilities and eliminate approximately 3,000 positions.

The change in the liability for restructuring costs during the years ended September 30 follows:

	2022	Expense	Utilized/Paid	2023
Severance and benefits	$ 117	42	74	85
Other	5	30	33	2
Total	$ 122	72	107	87

	2021	Expense	Utilized/Paid	2022
Severance and benefits	$ 140	44	67	117
Other	4	31	30	5
Total	$ 144	75	97	122

The tables above do not include $20, $40 and $34 of costs related to restructuring actions incurred in 2023, 2022 and 2021 respectively, that are required to be reported in cost of sales and selling, general and administrative expenses.

(8) EQUITY METHOD INVESTMENT AND NOTE RECEIVABLE

As discussed in Note 5, the Company completed the divestiture of a majority stake in Copeland on May 31, 2023, and received upfront, pre-tax cash proceeds of approximately $9.7 billion and a note receivable with a face value of $ 2.25 billion, while retaining a 40 percent non-controlling common equity interest in Copeland. As a result of the transaction, the Company deconsolidated Copeland from its financial statements, as it no longer has a controlling interest, and initially recognized its common equity investment and note receivable at fair values of $1,359 and $2,052, respectively. The fair value of the common equity investment was determined using a discounted cash flow model, which included estimating financial projections for Copeland and applying an appropriate discount rate, and an option pricing model based on various assumptions. Fair value for the note receivable was determined using a market approach primarily based on interest rates for companies with similar credit quality and the expected duration of the note.

The Company records its share of Copeland's income or loss using the equity method of accounting. For the year ended September 30, 2023 the Company recorded a loss of $177 in Other deductions to reflect its share of Copeland's reported GAAP losses and a tax benefit of $ 43 in Income taxes related to Copeland's U.S. business, which is taxed as a partnership (in total, $0.24 per share). The Company recognized non-cash interest income on the note receivable of $41, which is reported in Interest income from related party and capitalized to the carrying value of the note. Copeland's valuations of acquired assets and liabilities are in-process and subject to refinement.

As of September 30, 2023, the carrying values of the retained equity investment and note receivable were $ 1,162 and $2,093, respectively. During the year ended September 30, 2023, the Company settled a note receivable and note payable with Copeland of $918, which is reported in Investing and Financing cash flows, respectively.

Summarized financial information for Copeland as of and for the year ended September 30, 2023 is as follows. Copeland's results only reflect activity subsequent to the Company's divestiture of its majority stake.

	2023
Current assets	$ 1,737
Noncurrent assets	$ 13,818
Current liabilities	$ 1,371
Noncurrent liabilities	$ 8,007
Noncontrolling interests	$ 215

	2023
Net sales	$ 1,677
Gross profit	$ 479
Income (loss) from continuing operations	$ (442)
Net income (loss)	$ (442)
Net income (loss) attributable to shareholders	$ (442)

(9) LEASES

The components of lease expense for the years ended September 30 were as follows:

	2021	2022	2023
Operating lease expense	$ 169	160	178
Variable lease expense	$ 17	18	20

Short-term lease expense and sublease income were immaterial for the years ended September 30, 2023, 2022 and 2021. Cash paid for operating leases is classified within operating cash flows from continuing operations and was $170, $163 and $167 for the years ended September 30, 2023, 2022 and 2021, respectively. Operating lease right-of-use asset additions were $247, $94 and $162 for the years ended September 30, 2023, 2022 and 2021, respectively.

The following table summarizes the balances of the Company's operating lease right-of-use assets and operating lease liabilities as of September 30, 2022 and 2023, the vast majority of which relates to offices and manufacturing facilities:

	2022	2023
Right-of-use assets (Other assets)	$ 439	550
Current lease liabilities (Accrued expenses)	$ 128	144
Noncurrent lease liabilities (Other liabilities)	$ 312	404

The weighted-average remaining lease term for operating leases was 6.2 years and 5.7 years, and the weighted-average discount rate was 4.2 percent and 3.0 percent as of September 30, 2023 and September 30, 2022, respectively.

Future maturities of operating lease liabilities as of September 30, 2023 are summarized below:

	2023
2024	$ 160
2025	123
2026	84
2027	56
2028	41
Thereafter	150
Total lease payments	614
Less: Interest	66
Total lease liabilities	$ 548

As of September 30, 2023, the Company had one operating lease that had not yet commenced with a lease term of approximately 15 years and total undiscounted future minimum payments of approximately $80. This lease is expected to commence in 2024 and will be recorded as a right-of-use asset and lease liability.

(10) GOODWILL AND OTHER INTANGIBLES

The change in the carrying value of goodwill by business segment follows:

	Final Control	Measurement & Analytical	Discrete Automation	Safety & Productivity	Control Systems & Software	AspenTech	**Total**
Balance, September 30, 2021	$ 2,762	1,227	877	415	642	1,044	6,967
Acquisitions	—	—	—	—	40	7,289	7,329
Foreign currency translation and other	(146)	(57)	(70)	(51)	(19)	(7)	(350)
Balance, September 30, 2022	2,616	1,170	807	364	663	8,326	13,946
Acquisitions	—	374	55	—	—	—	429
Foreign currency translation and other	44	1	30	24	5	1	105
Balance, September 30, 2023	**$ 2,660**	1,545	892	388	668	8,327	14,480

The gross carrying amount and accumulated amortization of identifiable intangible assets by major class follow:

	Customer Relationships		**Intellectual Property**		**Capitalized Software**		Total	
	2022	**2023**	2022	**2023**	2022	**2023**	2022	**2023**
Gross carrying amount	$ 4,393	**4,623**	3,961	**4,118**	1,317	**1,370**	9,671	**10,111**
Less: Accumulated amortization	957	**1,270**	1,027	**1,411**	1,115	**1,167**	3,099	**3,848**
Net carrying amount	$ 3,436	**3,353**	2,934	**2,707**	202	**203**	6,572	**6,263**

Intangible asset amortization expense for the major classes included above for 2023, 2022 and 2021 was $ 764, $530 and $432, respectively. Based on intangible asset balances as of September 30, 2023, amortization expense is expected to approximate $768 in 2024, $696 in 2025, $596 in 2026, $564 in 2027 and $537 in 2028. The increase in goodwill and intangible assets in 2022 reflects the Heritage AspenTech acquisition.

(11) FINANCIAL INSTRUMENTS

Following is a discussion regarding the Company's use of financial instruments:

Hedging Activities

As of September 30, 2023, the notional amount of foreign currency hedge positions was approximately $ 2.4 billion. All derivatives receiving hedge accounting are cash flow hedges. The majority of hedging gains and losses deferred as of September 30, 2023 are expected to be recognized over the next 12 months as the underlying forecasted transactions occur. Gains and losses on foreign currency derivatives reported in Other deductions, net reflect hedges of balance sheet exposures that do not receive hedge accounting.

Net Investment Hedge

In 2019, the Company issued euro-denominated debt of € 1.5 billion. The euro notes reduce foreign currency risk associated with the Company's international subsidiaries that use the euro as their functional currency and have been designated as a hedge of a portion of the investment in these operations. Foreign currency gains or losses associated with the euro-denominated debt are deferred in accumulated other comprehensive income (loss) and will remain until the hedged investment is sold or substantially liquidated.

The following gains and losses are included in earnings and other comprehensive income (OCI):

		Gain (Loss) to Earnings			Gain (Loss) to OCI		
		2021	2022	2023	2021	2022	2023
	Location						
Commodity	Cost of sales	$ 33	12	(19)	29	(20)	6
Foreign currency	Sales	3	(2)	(3)	3	(9)	—
Foreign currency	Cost of sales	8	31	65	34	53	42
Foreign currency	Other deductions, net	53	48	(128)			
Net Investment Hedge							
Euro denominated debt				16	21	266	(128)
Total		$ 97	89	(69)	87	290	(80)

Regardless of whether derivatives and non-derivative financial instruments receive hedge accounting, the Company expects hedging gains or losses to be offset by losses or gains on the related underlying exposures. The amounts ultimately recognized will differ from those presented above for open positions, which remain subject to ongoing market price fluctuations until settlement. Derivatives receiving hedge accounting are highly effective and no amounts were excluded from the assessment of hedge effectiveness.

Equity Investment

The Company had an equity investment in National Instruments Corporation ("NI"), valued at $ 136 as of September 30, 2023 (reported in Other noncurrent assets), and recognized a mark-to-market gain of $56 in 2023. On April 12, 2023, Emerson announced an agreement to acquire NI for $ 60 per share in cash for the remaining shares not already owned by Emerson and the transaction closed on October 11, 2023. See Note 23.

Fair Value Measurement

Valuations for all derivatives, the Company's note receivable from Copeland, and the Company's long-term debt fall within Level 2 of the GAAP valuation hierarchy. The fair value of the note receivable as of September 30, 2023 was approximately $1.9 billion, which was lower than the carrying value by approximately $200. See Note 8 for further details. The fair value of long-term debt was $ 6.9 billion and $7.6 billion, respectively, as of September 30, 2023 and 2022, which was lower than the carrying value by $1,275 and $1,207, respectively. The fair values of commodity and foreign currency contracts were reported in Other current assets and Accrued expenses as summarized below:

	2022		2023	
	Assets	Liabilities	Assets	Liabilities
Commodity	$ —	25	—	—
Foreign currency	$ 51	80	30	22

Commodity contracts, which related to discontinued operations, were novated to Copeland upon the completion of the transaction and therefore no amounts are reported in the Company's balance sheet as of September 30, 2023. The fair value of the Company's equity investment in National Instruments falls within Level 1 and was based on the most recent quoted closing market price from its principal exchange for the period ended September 30, 2023.

(12) SHORT-TERM BORROWINGS AND LINES OF CREDIT

Short-term borrowings and current maturities of long-term debt are as follows:

	2022	2023
Current maturities of long-term debt	$ 516	546
Commercial paper and other short-term borrowings	1,599	1
Total	$ 2,115	547
Interest rate for weighted-average short-term borrowings at year end	2.8%	0.4%

In February 2023, the Company entered into a $3.5 billion five-year revolving backup credit facility with various banks, which replaced the May 2018 $3.5 billion facility. The credit facility is maintained to support general corporate purposes, including commercial paper borrowings. The Company has not incurred any borrowings under this or previous facilities. The credit facility contains no financial covenants and is not subject to termination based on a change of credit rating or material adverse changes. The facility is unsecured and may be accessed under various interest rate alternatives at the Company's option. Fees to maintain the facility are immaterial.

(13) LONG-TERM DEBT

The details of long-term debt follow:

	2022	2023
2.625% notes due February 2023	$ 500	—
0.375% euro notes due May 2024	490	529
3.15% notes due June 2025	500	500
1.25% euro notes due October 2025	490	529
0.875% notes due October 2026	750	750
1.8% notes due October 2027	500	500
2.0% notes due December 2028	1,000	1,000
2.0% euro notes due October 2029	490	529
1.95% notes due October 2030	500	500
2.20% notes due December 2031	1,000	1,000
6.0% notes due August 2032	250	250
6.125% notes due April 2039	250	250
5.25% notes due November 2039	300	300
2.75% notes due October 2050	500	500
2.80% notes due December 2051	1,000	1,000
Other	255	19
Long-term debt	8,775	8,156
Less: Current maturities	516	546
Total, net	$ 8,259	7,610

Long-term debt maturing during each of the four years after 2024 is $ 520, $538, $746 and $497, respectively. Total interest paid on long-term debt was approximately $200, $199 and $156 in 2023, 2022 and 2021, respectively.

During the year, the Company repaid $ 500 of 2.625% notes that matured in February 2023 and AspenTech repaid $ 264 to pay off the outstanding balance on its existing term loan facility plus accrued interest. In 2022, the Company repaid $500 of 2.625% notes that matured in December 2021. In December 2021, the Company issued $1,000 of 2.0% notes due December 2028, $ 1,000 of 2.20% notes due December 2031 and $ 1,000 of 2.80% notes due December 2051.

The Company maintains a universal shelf registration statement on file with the SEC under which it can issue debt securities, preferred stock, common stock, warrants, share purchase contracts or share purchase units without a predetermined limit. Securities can be sold in one or more separate offerings with the size, price and terms to be determined at the time of sale.

(14) PENSION AND POSTRETIREMENT PLANS

Retirement plans expense includes the following components:

	U.S. Plans			Non-U.S. Plans		
	2021	2022	2023	2021	2022	2023
Defined benefit plans:						
Service cost (benefits earned during the period)	$ 54	49	25	29	25	20
Interest cost	94	99	164	32	33	50
Expected return on plan assets	(264)	(253)	(247)	(74)	(56)	(39)
Net amortization and other	143	102	(55)	14	3	18
Net periodic pension expense (income)	27	(3)	(113)	1	5	49
Defined contribution plans	114	124	111	52	52	49
Total retirement plans expense (income)	$ 141	121	(2)	53	57	98

Net periodic pension expense decreased in 2023 primarily due to lower amortization of deferred losses partially offset by higher interest costs. Net periodic pension expense (income) includes $7, $16 and $21 and defined contribution expense includes $ 14, $32 and $30 for 2023, 2022 and 2021, respectively, related to discontinued operations. For defined contribution plans, the Company makes cash contributions based on plan requirements, which are expensed as incurred.

The Company's principal U.S. defined benefit plan is closed to employees hired after January 1, 2016 while shorter-tenured employees ceased accruing benefits effective October 1, 2016.

All of the following tables include defined benefit pension plans related to continuing and discontinued operations.

Details of the changes in the actuarial present value of the projected benefit obligation and the fair value of plan assets for defined benefit pension plans follow:

	U.S. Plans		Non-U.S. Plans	
	2022	2023	2022	2023
Projected benefit obligation, beginning	$ 4,338	3,112	1,562	965
Service cost	49	25	25	20
Interest cost	99	164	33	50
Actuarial gain	(1,170)	(75)	(404)	(25)
Curtailments	—	(31)	—	—
Benefits paid	(204)	(204)	(40)	(42)
Settlements	—	(2)	(29)	(70)
Acquisitions (Divestitures), net	—	(56)	—	(46)
Foreign currency translation and other	—	1	(182)	74
Projected benefit obligation, ending	$ 3,112	2,934	965	926
Fair value of plan assets, beginning	$ 4,844	3,625	1,474	908
Actual return on plan assets	(1,030)	230	(337)	(40)
Employer contributions	15	14	28	32
Benefits paid	(204)	(204)	(40)	(42)
Settlements	—	(2)	(29)	(70)
Acquisitions (Divestitures), net	—	(74)	—	2
Foreign currency translation and other	—	1	(188)	74
Fair value of plan assets, ending	$ 3,625	3,590	908	864
Net amount recognized in the balance sheet	$ 513	656	(57)	(62)
Location of net amount recognized in the balance sheet:				
Noncurrent asset	$ 663	815	205	180
Noncurrent asset held-for-sale	13	—	—	—
Current liability	(14)	(14)	(17)	(17)
Noncurrent liability	(149)	(145)	(203)	(225)
Net liability held-for-sale	—	—	(42)	—
Net amount recognized in the balance sheet	$ 513	656	(57)	(62)
Pretax accumulated other comprehensive loss	$ (284)	(257)	(136)	(181)

Actuarial gains in 2023 were largely due to an increase in the discount rates used to estimate the benefit obligations for the U.S. and non-U.S. plans, which were 6.03% and 5.2% at September 30, 2023 compared to 5.64% and 4.9% at September 30, 2022, respectively. Actuarial gains in 2022 were largely due to an increase in the discount rates used to estimate the benefit obligations for the U.S. and non-U.S. plans, which were 5.64% and 4.9% at September 30, 2022 compared to 2.92% and 2.2% at September 30, 2021, respectively. As of September 30, 2023, U.S. pension plans were overfunded by $656 in total, including unfunded plans totaling $159. The non-U.S. plans were underfunded by $62, including unfunded plans totaling $213.

As of the September 30, 2023 and 2022 measurement dates, the plans' total accumulated benefit obligation was $ 3,719 and $3,910, respectively. The total projected benefit obligation, accumulated benefit obligation and fair value of plan assets for individual plans with projected benefit obligations in excess of plan assets were $519, $435 and $118, respectively, for 2023, and $527, $444 and $102, respectively, for 2022. The total projected benefit obligation, accumulated benefit obligation and fair value of plan assets for individual plans with accumulated benefit obligations

in excess of plan assets were $ 469, $413 and $77, respectively, for 2023, and $ 477, $421 and $63, respectively, for 2022.

Future benefit payments by U.S. plans are estimated to be $ 222 in 2024, $226 in 2025, $229 in 2026, $230 in 2027, $231 in 2028 and $ 1,142 in total over the five years 2029 through 2033. Based on foreign currency exchange rates as of September 30, 2023, future benefit payments by non-U.S. plans are estimated to be $62 in 2024, $58 in 2025, $61 in 2026, $63 in 2027, $69 in 2028 and $ 358 in total over the five years 2029 through 2033. The Company expects to contribute approximately $45 to its retirement plans in 2024.

The weighted-average assumptions used in the valuation of pension benefits follow:

	U.S. Plans			Non-U.S. Plans		
	2021	2022	2023	2021	2022	2023
Net pension expense						
Discount rate used to determine service cost	3.16 %	3.16 %	5.66 %	1.9 %	2.2 %	4.9 %
Discount rate used to determine interest cost	2.10 %	2.31 %	5.49 %	1.9 %	2.2 %	4.9 %
Expected return on plan assets	6.50 %	6.00 %	6.00 %	5.6 %	4.4 %	4.4 %
Rate of compensation increase	3.25 %	4.00 %	4.00 %	3.6 %	3.7 %	4.0 %
Benefit obligations						
Discount rate	2.92 %	5.64 %	6.03 %	2.2 %	4.9 %	5.2 %
Rate of compensation increase	3.25 %	4.00 %	4.00 %	3.7 %	4.0 %	3.9 %

The discount rate for the U.S. retirement plans was 6.03 percent as of September 30, 2023. An actuarially developed, company-specific yield curve is used to determine the discount rate. To determine the service and interest cost components of pension expense for its U.S. retirement plans, the Company applies the specific spot rates along the yield curve, rather than the single weighted-average rate, to the projected cash flows to provide more precise measurement of these costs. The expected return on plan assets assumption is determined by reviewing the investment returns of the plans for the past 10 years plus longer-term historical returns of an asset mix approximating the Company's asset allocation targets, and periodically comparing these returns to expectations of investment advisors and actuaries to determine whether long-term future returns are expected to differ significantly from the past.

The Company's asset allocations at September 30, 2023 and 2022, and weighted-average target allocations follow:

	U.S. Plans			Non-U.S. Plans		
	2022	2023	Target	2022	2023	Target
Equity securities	39 %	39 %	35-45%	11 %	8 %	5-15%
Debt securities	54	51	50-60	73	75	70-80
Other	7	10	0-10	16	17	10-20
Total	100 %	100 %	100 %	100 %	100 %	100 %

The primary objective for the investment of pension assets is to secure participant retirement benefits by earning a reasonable rate of return. Plan assets are invested consistent with the provisions of the prudence and diversification rules of ERISA and with a long-term investment horizon. The Company continuously monitors the value of assets by class and routinely rebalances to remain within target allocations. The equity strategy is to minimize concentrations of risk by investing primarily in a mix of companies that are diversified across geographies, market capitalization, style, sectors and industries worldwide. The approach for bonds emphasizes investment-grade corporate and government debt with maturities matching a portion of the longer duration pension liabilities. The bonds strategy also includes a high-yield element which is generally shorter in duration. For diversification, a small portion of U.S. plan assets is allocated to private equity partnerships and real asset fund investments, providing opportunities for above market returns. Leveraging techniques are not used and the use of derivatives in any fund is limited and inconsequential.

The fair values of defined benefit pension assets a s of September 30, o rganized by asset class and by the fair value hierarchy of ASC 820, *Fair Value Measurement,* follow. Investments valued based on the net asset value (NAV) of fund units held, as derived from the fair value of the underlying assets, are excluded from the fair value hierarchy.

	Level 1	Level 2	Level 3	Measured at NAV	Total	%
2023						
U.S. equities	$ 396	9		619	1,024	23 %
International equities	210	14		103	327	7 %
Emerging market equities		1		118	119	3 %
Corporate bonds		1,008		843	1,851	42 %
Government bonds		505		124	629	14 %
Other	121	7	126	250	504	11 %
Total	$ 727	1,544	126	2,057	4,454	100 %
2022						
U.S. equities	$ 405	6		633	1,044	23 %
International equities	225	10		123	358	8 %
Emerging market equities		1		124	125	3 %
Corporate bonds		1,143		859	2,002	44 %
Government bonds		468		152	620	14 %
Other	(9)	7	130	256	384	8 %
Total	$ 621	1,635	130	2,147	4,533	100 %

Asset Classes

U.S. equities reflect companies domiciled in the U.S., including multinational companies. International equities are comprised of companies domiciled in developed nations outside the U.S. Emerging market equities are comprised of companies domiciled in portions of Asia, Eastern Europe and Latin America. Corporate bonds represent investment-grade debt of issuers primarily from the U.S. Government bonds include investment-grade instruments issued by federal, state and local governments, primarily in the U.S. Other includes cash, interests in mixed asset funds investing in commodities, natural resources, agriculture, real estate and infrastructure funds, life insurance contracts (U.S.), and shares in certain general investment funds of financial institutions or insurance arrangements (non-U.S.) that typically ensure no market losses or provide for a small minimum return guarantee.

Fair Value Hierarchy Categories

Valuations of Level 1 assets for all classes are based on quoted closing market prices from the principal exchanges where the individual securities are traded. Cash is valued at cost, which approximates fair value. Debt securities categorized as Level 2 assets are generally valued based on independent broker/dealer bids or by comparison to other debt securities having similar durations, yields and credit ratings. Valuation techniques and inputs for these assets include discounted cash flow analysis, earnings multiple approaches, recent transactions, transfer restrictions, prevailing discount rates, volatilities, credit ratings and other factors. In the Other class, interests in mixed asset funds are Level 2, and U.S. life insurance contracts and non-U.S. general fund investments and insurance arrangements are Level 3. Investments measured at NAV are primarily nonexchange-traded commingled or collective funds where the underlying securities have observable prices available from active markets and typically provide liquidity daily or within a few days. The NAV category also includes fund investments in private equities, real estate and infrastructure where the fair value of the underlying assets is determined by the investment manager. Total unfunded commitments for the private equity funds were approximately $115 at September 30, 2023. These investments cannot be redeemed, but instead the funds will make distributions through liquidation of the underlying assets, which is expected to occur over approximately the next 10 years. The real estate and infrastructure funds typically offer quarterly redemption.

Postretirement Plans

The Company also sponsors unfunded postretirement benefit plans (primarily health care) for certain U.S. retirees and their dependents. The Company's principal U.S. postretirement plan has been frozen to new employees since

1993. The postretirement benefit liability for all plans was $ 72 and $ 83 as of September 30, 2023 and 2022, respectively, and included deferred actuarial gains in accumulated other comprehensive income of $95 and $ 112, respectively. Service and interest costs are negligible and more than offset by the amortization of deferred actuarial gains, which resulted in net postretirement income of $19 for 2023 and $ 12 for 2022 and $ 15 for 2021. Benefits paid were $9 and $ 10 for 2023 and 2022, respectively, and the Company estimates that future health care benefit payments will be approximately $ 8 per year for 2024 through 2028, and $29 in total over the five years 2029 through 2033.

(15) CONTINGENT LIABILITIES AND COMMITMENTS

The Company is a party to a number of pending legal proceedings and claims, including those involving general and product liability (including asbestos) and other matters, several of which claim substantial amounts of damages. The Company accrues for such liabilities when it is probable that future costs (including legal fees and expenses) will be incurred and such costs can be reasonably estimated. Accruals are based on developments to date; management's estimates of the outcomes of these matters; and the Company's experience in contesting, litigating and settling similar matters. The Company engages an outside expert to develop an actuarial estimate of its expected costs to resolve all pending and future asbestos claims, including defense costs, as well as its related insurance receivables. The reserve for asbestos litigation, which is recorded on an undiscounted basis, is based on projected claims through 2065. See Note 21 for additional information about the Company's asbestos liabilities and related insurance receivables.

Although it is not possible to predict the ultimate outcome of these matters, the Company historically has been largely successful in defending itself against claims and suits that have been brought against it, and will continue to defend itself vigorously in all such matters. While the Company believes a material adverse impact is unlikely, given the inherent uncertainty of litigation, a remote possibility exists that a future development could have a material adverse impact on the Company. The Company enters into certain indemnification agreements in the ordinary course of business in which the indemnified party is held harmless and is reimbursed for losses incurred from claims by third parties, usually up to a prespecified limit. In connection with divestitures of certain assets or businesses, the Company often provides indemnities to the buyer with respect to certain matters including, for example, environmental or unidentified tax liabilities related to periods prior to the disposition. Because of the uncertain nature of the indemnities, the maximum liability cannot be quantified. As such, contingent liabilities are recorded when they are both probable and reasonably estimable. Historically, payments under indemnity arrangements have been inconsequential.

At September 30, 2023, there were no known contingent liabilities (including guarantees, pending litigation, taxes and other claims) that management believes will be material in relation to the Company's financial statements, nor were there any material commitments outside the normal course of business.

(16) INCOME TAXES

Pretax earnings from continuing operations consist of the following:

	2021	2022	2023
United States	$ 675	1,345	1,352
Non-U.S.	1,087	1,087	1,374
Total pretax earnings	$ 1,762	2,432	2,726

The principal components of income tax expense follow:

	2021	2022	2023
Current:			
U.S. federal	$ 17	315	465
State and local	14	36	47
Non-U.S.	258	306	369
Deferred:			
U.S. federal	84	(92)	(198)
State and local	(2)	(13)	(23)
Non-U.S.	(25)	(3)	(61)
Income tax expense	$ 346	549	599

Reconciliations of the U.S. federal statutory income tax rate to the Company's effective tax rate follow.

	2021	2022	2023
U.S. federal statutory rate	21.0 %	21.0 %	21.0 %
State and local taxes, net of U.S. federal tax benefit	0.5	0.7	0.7
Non-U.S. rate differential	2.6	1.2	0.8
Non-U.S. tax holidays	(1.2)	(1.1)	(0.9)
Research and development credits	(0.9)	(0.5)	(0.5)
Foreign derived intangible income	(1.6)	(2.0)	(2.8)
Subsidiary restructuring	(0.8)	0.8	—
Russia business exit	—	2.0	0.2
Other	—	0.5	3.5
Effective income tax rate	19.6 %	22.6 %	22.0 %

The 2023 increase in other was driven by a 1 percentage point impact from U.S. taxation of Non-U.S. operations and a 2 percentage point impact due to an increase in unrecognized tax benefits.

The Company has elected to recognize the tax on global intangible low-taxed income earned by certain of its non-U.S. subsidiaries as a period expense when it is incurred.

On March 27, 2020, the CARES Act was enacted in response to the COVID-19 pandemic, and among other things, provides tax relief to businesses. Tax provisions of the CARES Act included the deferral of certain payroll taxes, relief for retaining employees, and other provisions. The Company deferred $73 of certain payroll taxes through the end of calendar year 2020, of which approximately $ 37 was paid in December 2021 and the remainder paid in December 2022.

Non-U.S. tax holidays reduce tax rates in certain jurisdictions. Approximately 80 percent of the tax holidays expire over the next four years, with the remainder expiring by 2030.

Following are changes in unrecognized tax benefits before considering recoverability of any cross-jurisdictional tax credits (U.S. federal, state and non-U.S.) and temporary differences. The amount of unrecognized tax benefits is not expected to change significantly in the next 12 months.

	2022	2023
Unrecognized tax benefits, beginning	$ 209	167
Additions for current year tax positions	24	78
Additions for prior year tax positions	9	13
Reductions for prior year tax positions	(65)	(10)
Acquisitions and divestitures	1	—
Reductions for settlements with tax authorities	—	(5)
Reductions for expiration of statutes of limitations	(11)	(8)
Unrecognized tax benefits, ending	$ 167	235

If none of the unrecognized tax benefits shown is ultimately paid, the tax provision and the calculation of the effective tax rate would be favorably impacted by $196, which is net of cross-jurisdictional tax credits and temporary differences. The Company accrues interest and penalties related to income taxes in income tax expense. Total expense (income) recognized was $1, $(7) and $(6) in 2023, 2022 and 2021, respectively. As of September 30, 2023 and 2022, total accrued interest and penalties were $22 and $21, respectively.

The U.S. is the major jurisdiction for which the Company files income tax returns. Examinations for U.S. federal are complete through 2017, except for 2014. The status of state and non-U.S. tax examinations varies due to the numerous legal entities and jurisdictions in which the Company operates.

The principal items that gave rise to deferred income tax assets and liabilities follow:

	2022	2023
Deferred tax assets:		
Net operating losses, capital losses and tax credits	$ 209	253
Accrued liabilities	213	163
Postretirement and postemployment benefits	21	17
Employee compensation and benefits	122	103
Other	125	158
Total	$ 690	694
Valuation allowances	$ (171)	(164)
Deferred tax liabilities:		
Intangibles	$ (1,622)	(1,387)
Pensions	(126)	(151)
Property, plant and equipment	(207)	(148)
Undistributed non-U.S. earnings	(37)	(32)
Deferred gains	(10)	(596)
Other	(146)	(75)
Total	$ (2,148)	(2,389)
Net deferred income tax liability	$ (1,629)	(1,859)

Total income taxes paid were approximately $ 3,310, $720 and $680 in 2023, 2022 and 2021, respectively. Taxes paid in 2023 included approximately $2.3 billion related to the gains on the Copeland transaction and InSinkErator divestiture and subsidiary restructurings related to the Copeland transaction. Approximately two-thirds of the $ 253 of net operating losses can be carried forward indefinitely, while most of the remainder expire over the next 10 years.

(17) STOCK-BASED COMPENSATION

The Company's stock-based compensation plans include performance shares, restricted stock, restricted stock units, and stock options. Although the Company has discretion, shares distributed under these plans are issued from treasury stock.

In fiscal 2022, the Company changed the terms of its annual performance share awards that were issued in the first quarter. The terms meet the criteria for equity classification in accordance with ASC 718, *Compensation - Stock Compensation*, and therefore expense will be recognized on a fixed basis over the three-year performance period. The terms of the performance share awards issued in fiscal 2021 are unchanged and therefore continue to be accounted for as liability awards and marked-to-market each period based on changes in the stock price.

AspenTech also has stock-based compensation plans that are settled in its own stock. These plans consist of performance shares, restricted stock units and stock options.

Total compensation expense and income tax benefits for Emerson and AspenTech stock options and incentive shares follows.

	2021	2022	2023
Performance shares	$ 203	89	165
Restricted stock and restricted stock units	21	23	24
AspenTech stock-based compensation plans	—	32	82
Total stock compensation expense	224	144	271
Less: discontinued operations	27	19	21
Stock compensation expense from continuing operations	$ 197	125	250
Income tax benefits recognized	$ 27	19	28

As of September 30, 2023, total unrecognized compensation expense related to unvested shares awarded under Emerson plans was $ 119, which is expected to be recognized over a weighted-average period of 1.1 years, while the total future unrecognized compensation cost related to AspenTech stock options, RSUs and performance stock units was $18, $59 and $12 respectively, which is expected to be recorded over a weighted average period of 2.1 years, 3.0 years and 2.8 respectively.

Emerson Performance Shares, Restricted Stock and Restricted Stock Units

The Company's incentive shares plans include performance shares awards which distribute the value of common stock to key management employees at the conclusion of a three-year period subject to certain operating performance conditions and other terms and restrictions. The form of distribution is primarily shares of common stock, with a portion in cash in the first quarter following the end of the applicable three-year performance period. Dividend equivalents are only paid on earned awards after the performance period has concluded. Compensation expense for performance shares is recognized over the service period based on the number of shares ultimately expected to be earned.

Information related to performance share payouts for the years ended September 30, 2022 and 2023 follows (shares in thousands):

	2022	2023
Performance period	2019 - 2021	2020 - 2022
Percent payout	101 %	106 %
Total shares earned	1,341	1,557
Shares distributed in cash, primarily for tax withholding	586	684

As of September 30, 2023, approximately 1,468,000 shares awarded primarily in 2021 were outstanding, contingent on the Company achieving its performance objectives through 2023. The objectives for these shares were met at the 118 percent level and the shares will be distributed in early fiscal 2024.

Additionally, the rights to receive approximately 975,000 and 928,000 shares awarded in 2023 and 2022, respectively, are outstanding and contingent upon the Company achieving its performance objectives through 2025 and 2024, respectively.

Incentive shares plans also include restricted stock awards and restricted stock units. Restricted stock awards involve distribution of common stock to key management employees subject to cliff vesting at the end of service periods ranging from three to ten years while restricted stock units granted to employees cliff vest at the end of a three-year period. The fair value of restricted stock awards and restricted stock units is determined based on the average of the high and low market prices of the Company's common stock on the date of grant, with compensation expense recognized ratably over the applicable vesting period. In 2023, approximately 125,000 shares of restricted stock and approximately 220,000 restricted stock units vested as a result of participants fulfilling the applicable service requirements. Consequently, approximately 80,000 shares and 158,000 units were issued while 45,000 shares and 62,000 units were withheld for income taxes in accordance with minimum withholding requirements. As of September 30, 2023, there were approximately 1,065,000 shares of unvested restricted stock and restricted stock units outstanding.

In addition to the employee stock option and incentive share plans, in 2023 the Company awarded approximately 22,000 restricted stock units under the restricted stock plan for non-management directors. As of September 30, 2023, approximately 57,000 shares were available for issuance under this plan.

As of September 30, 2023, 2.7 million shares remained available for award under incentive shares plans.

Changes in shares outstanding but not yet earned under incentive shares plans during the year ended September 30, 2023 follow (shares in thousands; assumes 100 percent payout of unvested awards):

	Shares	Average Grant Date Fair Value Per Share
Beginning of year	5,280	$ 77.58
Granted	1,263	$ 87.33
Earned/vested	(1,814)	$ 71.70
Canceled	(292)	$ 89.49
End of year	4,437	$ 82.02

Information related to Emerson incentive shares plans follows:

	2021	2022	2023
Total fair value of shares earned/vested	$ 131	158	158
Share awards distributed in cash, primarily for tax withholding	$ 58	69	73

Emerson Stock Options

There were no stock option grants in 2023, 2022 and 2021. The Company's stock option plans expired in 2021. Previously awarded stock options allow key officers and employees to purchase common stock at specified prices, which are equal to 100 percent of the closing market price of the Company's stock on the date of grant. Options generally vest one-third in each of the three years subsequent to grant and expire 10 years from the date of grant. Compensation expense is recognized ratably over the vesting period based on the number of options expected to vest.

Changes in shares subject to options during the year ended September 30, 2023 follow (shares in thousands):

	Weighted- Average Exercise Price Per Share	Shares	Total Intrinsic Value of Shares	Average Remaining Life (Years)
Beginning of year	$ 58.10	1,692		
Options exercised	$ 60.66	(1,099)		
Options canceled	$ 53.05	(4)		
End of year	$ 53.35	589	$ 26	2.1
Exercisable at end of year	$ 53.35	589	$ 26	2.1

Information related to Emerson stock options follows:

	2021	2022	2023
Cash received for option exercises	$ 114	15	49
Intrinsic value of options exercised	$ 53	11	27
Tax benefits related to option exercises	$ 6	7	4

AspenTech Stock-Based Compensation

As discussed in Note 4, Emerson completed the acquisition of Heritage AspenTech in the third quarter of 2022. AspenTech, as defined in Note 4, operates as a separate publicly traded company and has various stock-based compensation plans, including stock options and restricted stock units, which are settled in their own common stock and are accounted for as equity awards. Restricted stock units generally vest over four years. Option awards have been granted with an exercise price equal to the market closing price of AspenTech's stock on the trading day prior to the grant date. These options generally vest over four years and expire within seven years or ten years of grant. AspenTech's policy is to issue new shares upon the exercise of vested stock awards.

Pursuant to the terms of the transaction agreement between Emerson and Heritage AspenTech, each outstanding option to purchase shares of Heritage AspenTech common stock, whether vested or unvested, that was unexercised as of immediately prior to the closing date was converted into an option to acquire shares of AspenTech. Each converted option is subject to the same terms and conditions as applied to the original option. In addition, each outstanding award of restricted stock units with respect to shares of Heritage AspenTech common stock that were unvested as of immediately prior to the closing date was converted into an award of restricted stock units with respect to shares of AspenTech. Each converted restricted stock unit is also subject to the same terms and conditions as applied to the original restricted stock unit.

ASC 805 required the Company to determine the fair value of the AspenTech share-based payment awards related to the replacement of the Heritage AspenTech share-based payment awards, and allocate the total fair value based on the services that are attributable to the pre- and post-combination service periods, respectively. The portion that is attributable to the pre-combination service period was considered part of the consideration transferred for Heritage AspenTech and included as part of the purchase price. The portion that is attributable to the post-combination service period is recognized as stock-based compensation expense in the post-combination consolidated financial statements over the remaining requisite service period.

AspenTech Stock Options

AspenTech utilizes the Black-Scholes option valuation model for estimating the fair value of options granted. The Black-Scholes option valuation model incorporates assumptions regarding expected stock price volatility, the expected life of the option, the risk-free interest rate, dividend yield and the market value of AspenTech's common stock. The expected stock price volatility is determined based on AspenTech's stock's historic prices over a period commensurate with the expected life of the award. The expected life of an option represents the period for which options are expected to be outstanding as determined by historic option exercises and cancellations. The risk-free interest rate is based on the U.S. Treasury yield curve for notes with terms approximating the expected life of the options granted. The expected dividend yield is zero, based on AspenTech's history and expectation of not paying dividends on common shares. Stock-based compensation expense is recognized on a straight-line basis, net of forfeitures as they occur, over the requisite service period for time-vested awards.

The weighted-average assumptions used in valuations for 2023 are: risk-free interest rate, 3.8 percent; dividend yield, none; expected volatility, 39.3 percent; and expected life, approximately 5 years.

A summary of AspenTech stock option activity in 2023 is as follows (shares in thousands):

	Weighted- Average Exercise Price Per Share	Shares	Total Intrinsic Value of Shares	Average Remaining Contractual Term (Years)
Beginning of year	$ 131.26	1,256		
Granted	$ 196.04	47		
Exercised	$ 93.04	(302)		
Canceled / Forfeited	$ 179.87	(27)		
End of year	$ 161.26	974	$ 59	6.3
Exercisable at end of year	$ 128.61	667	$ 51	5.3
Vested and expected to vest at September 30, 2023	$ 143.96	957	$ 59	6.3

The weighted average estimated fair value of option awards granted during 2023 was $ 76.99. The total intrinsic value of options exercised during 2023 was $38. Cash proceeds of $29 from issuances of shares of AspenTech common stock were received during 2023.

AspenTech Restricted Stock Units and Performance Stock Units

A summary of AspenTech restricted stock unit and performance stock unit activity in 2023 is as follows (shares in thousands):

	Weighted- Average Grant Date Fair Value	Shares
Beginning of year	$ 190.44	589
Granted	$ 192.51	367
Settled	$ 193.23	(268)
Canceled / Forfeited	$ 195.48	(35)
End of year	$ 193.17	653
Vested and expected to vest at September 30, 2023	$ 193.24	566

During 2023, AspenTech granted performance stock units with a performance condition and service condition. These performance stock units vest on a cliff basis in three years based upon the achievement of predefined performance goals, with the ability for 25 percent of granted awards to vest on an accelerated basis in each of the first two years. The performance goal relates to the sum of (i) Annual Contract Value growth and (ii) free cash flow margin over the performance period. Up to 175 percent of the performance stock units could vest upon achievement of the performance goals. Conversely, if a minimum performance goal is not met, none of the performance stock units will vest. On a quarterly basis, management evaluates the probability that the threshold performance goals will be achieved, if at all, and the anticipated level of attainment to determine the amount of compensation expense to record in the consolidated financial statements. During 2023, the total fair value of vested shares from AspenTech RSU grants amounted to $53. Withholding taxes of $ 19 were paid on vested RSUs during 2023.

At September 30, 2023, common stock reserved for future issuance under all AspenTech equity compensation plans was 3.7 million shares.

(18) COMMON AND PREFERRED STOCK

At September 30, 2023, 8.8 million shares of common stock were reserved for issuance under the Company's stock-based compensation plans. During 2023, 21.3 million common shares were purchased and 1.8 million

treasury shares were reissued. In 2022, 5.7 million common shares were purchased and 1.3 million treasury shares were reissued.

At September 30, 2023 and 2022, the Company had 5.4 million shares of $2.50 par value preferred stock authorized, with none issued.

(19) ACCUMULATED OTHER COMPREHENSIVE INCOME (LOSS)

Activity in Accumulated other comprehensive income (loss) is shown below, net of income taxes:

	2021	2022	2023
Foreign currency translation			
Beginning balance	$ (711)	(629)	(1,265)
Other comprehensive income (loss), net of tax of $(5), $(62) and $26, respectively	82	(636)	158
Reclassified to gain on sale of business	—	—	95
Ending balance	(629)	(1,265)	(1,012)
Pension and postretirement			
Beginning balance	(864)	(259)	(222)
Actuarial gains (losses) deferred during the period, net of taxes of $(150), $10 and $0, respectively	499	(33)	4
Amortization of deferred actuarial losses into earnings, net of tax of $(34), $(21) and $17, respectively	106	70	(51)
Reclassified to gain on sale of business	—	—	22
Ending balance	(259)	(222)	(247)
Cash flow hedges			
Beginning balance	(2)	16	2
Gains deferred during the period, net of taxes of $(15), $(6) and $(11), respectively	51	18	37
Reclassifications of realized (gains) losses to sales and cost of sales, net of tax of $ 11, $10 and $4, respectively	(33)	(32)	(14)
Reclassified to gain on sale of business	—	—	(19)
Ending balance	16	2	6
Accumulated other comprehensive income (loss)	$ (872)	(1,485)	(1,253)

(20) BUSINESS SEGMENTS INFORMATION

As disclosed in Note 5, the financial results of Climate Technologies, InSinkErator and Therm-O-Disc are reported as discontinued operations for all periods presented. As a result of these portfolio actions, the Company has realigned its business segments and now reports six segments and two business groups, which are highlighted in the table below. The Company also reclassified certain product sales that were previously reported in Control Systems & Software to Discrete Automation.

INTELLIGENT DEVICES	SOFTWARE AND CONTROL
• Final Control	• Control Systems & Software
• Measurement & Analytical	• AspenTech
• Discrete Automation	
• Safety & Productivity	

The new segments were previously described as follows: Final Control was the Valves, Actuators & Regulators product offering; Measurement & Analytical was the Measurement & Analytical instrumentation product offering; Discrete Automation was the Industrial Solutions product offering; Safety & Productivity was the Tools & Home Products segment, excluding the divested InSinkErator business; Control Systems & Software was the Systems &

Software product offering; and, AspenTech remains unchanged. The AspenTech segment was identified in the third quarter of fiscal 2022 as a result of the Heritage AspenTech acquisition and reflects the combined results of Heritage AspenTech and the Emerson Industrial Software Business (see Note 4 for further details). The results for this new segment include the historical results of the Emerson Industrial Software Business (which were previously reported in the Control Systems & Software segment), while results related to the Heritage AspenTech business only include periods subsequent to the close of the transaction. Prior year amounts have been reclassified to conform to the current year presentation.

The **Final Control** segment is a leading global provider of control valves, isolation valves, shutoff valves, pressure relief valves, pressure safety valves, actuators, and regulators for process and hybrid industries. These solutions respond to commands from a control system to continuously and precisely control and regulate the flow of liquids or gases to achieve safe operation along with reliability and optimized performance.

The **Measurement & Analytical** segment is a leading supplier of intelligent instrumentation measuring the physical properties of liquids or gases, such as pressure, temperature, level, flow, acoustics, corrosion, pH, conductivity, water quality, toxic gases, and flame. The instrumentation transfers data to control systems and automation software, allowing process and hybrid industry operators to make educated decisions regarding production, reliability and safety.

The **Discrete Automation** segment includes solenoid valves, pneumatic valves, valve position indicators, pneumatic cylinders and actuators, air preparation equipment, pressure and temperature switches, electric linear motion solutions, programmable automation control systems and software, electrical distribution equipment, and materials joining solutions used primarily in discrete industries.

The **Safety & Productivity** segment offers tools for professionals and homeowners that promote safety and productivity. Pipe-working tools include pipe wrenches, pipe cutters, pipe threading and roll grooving equipment, battery hydraulic tools for press connections, drain cleaners, tubing tools and diagnostic systems, including sewer inspection cameras and locating equipment. Electrical tools include conduit benders and cable pulling equipment, battery hydraulic tools for cutting and crimping electrical cable, and hole-making equipment. Other professional tools include water jetters, wet-dry vacuums, commercial vacuums and hand tools.

The **Control Systems & Software** segment provides control systems and software that control plant processes by collecting and analyzing information from measurement devices in the plant and using that information to adjust valves, pumps, motors, drives and other control hardware for maximum product quality, process efficiency and safety. These solutions include distributed control systems, safety instrumented systems, SCADA systems, application software, digital twins, asset performance management and cybersecurity. Control Systems & Software solutions are predominantly used by process and hybrid manufacturers.

AspenTech is a global leader in asset optimization software that enables industrial manufacturers to design, operate, and maintain their operations for maximum performance. AspenTech combines decades of modeling, simulation, and optimization capabilities with industrial operations expertise and applies advanced analytics to improve the profitability and sustainability of production assets. The purpose-built software drives value for customers by improving operational efficiency and maximizing productivity, reducing unplanned downtime and safety risks, and minimizing energy consumption and emissions.

The principal distribution method for each segment is direct sales forces, although the Company also uses independent sales representatives and distributors. Due to its global presence, certain of the Company's international operations are subject to risks including the stability of governments and business conditions in foreign countries which could result in adverse changes in exchange rates, changes in regulations or disruption of operations.

The primary income measure used for assessing segment performance and making operating decisions is earnings before interest and income taxes. Certain expenses are reported at Corporate, including stock compensation expense and a portion of pension and postretirement benefit costs. Corporate and other includes unallocated corporate expenses, acquisition/divestiture costs, first year acquisition accounting charges (which include fair value adjustments related to inventory, backlog and deferred revenue) and other items. Corporate assets are primarily comprised of cash and cash equivalents, investments, certain fixed assets and assets held-for-sale. Summarized below is information about the Company's operations by business segment and by geography.

Business Segments

	Sales			Earnings (Loss)			Total Assets		
	2021	2022	2023	2021	2022	2023	2021	2022	2023
Final Control	$ 3,488	3,607	3,970	$ 432	592	865	$ 5,245	4,805	5,614
Measurement & Analytical	3,078	3,215	3,595	684	785	936	4,410	4,395	3,976
Discrete Automation	2,474	2,612	2,635	457	542	509	2,405	2,284	2,493
Safety & Productivity	1,340	1,402	1,388	256	250	306	1,163	1,125	1,238
Intelligent Devices	10,380	10,836	11,588	1,829	2,169	2,616	13,223	12,609	13,321
Control Systems & Software	2,321	2,398	2,606	382	437	529	1,674	1,700	2,151
AspenTech	319	656	1,042	(7)	12	(107)	2,089	14,484	14,048
Software and Control	2,640	3,054	3,648	375	449	422	3,763	16,184	16,199
Corporate items:									
Stock compensation				(197)	(125)	(250)			
Unallocated pension and postretirement costs				94	99	171			
Corporate and other (includes assets held-for-sale)				(184)	(419)	(224)	7,729	6,879	13,226
Gain on subordinated interest				—	453	161			
Loss on Copeland equity method investment				—	—	(177)			
Eliminations/Interest	(88)	(86)	(71)	(155)	(194)	(34)			
Interest income from related party				—	—	41			
Total	$ 12,932	13,804	15,165	$ 1,762	2,432	2,726	$ 24,715	35,672	42,746

In 2023, Corporate and other includes a loss of $ 47 related to the Company's exit of business operations in Russia while 2022 includes a loss of $ 181. Corporate and other for 2023 includes acquisition/divestiture and related costs of $84 ($15 of which is reported in operating profit) while 2022 includes $91.

	Depreciation and Amortization			Capital Expenditures		
	2021	2022	2023	2021	2022	2023
Final Control	$ 210	212	170	$ 73	62	93
Measurement & Analytical	128	117	121	130	90	93
Discrete Automation	96	88	84	100	68	56
Safety & Productivity	60	57	57	59	27	35
Intelligent Devices	494	474	432	362	247	277
Control Systems & Software	103	93	90	17	27	33
AspenTech	95	242	492	6	4	6
Software and Control	198	335	582	23	31	39
Corporate and other	70	33	37	19	21	47
Total	$ 762	842	1,051	$ 404	299	363

Depreciation and amortization includes intellectual property, customer relationships and capitalized software.

Geographic Information

Sales by major geographic destination are summarized below:

	2021				2022			
	Americas	AMEA	Europe	Total	Americas	AMEA	Europe	Total
Measurement & Analytical	$ 1,338	1,181	559	3,078	$ 1,529	1,199	487	3,215
Final Control	1,504	1,385	599	3,488	1,706	1,373	528	3,607
Discrete Automation	1,084	700	690	2,474	1,217	732	663	2,612
Safety & Productivity	992	66	282	1,340	1,057	71	274	1,402
Intelligent Devices	4,918	3,332	2,130	10,380	5,509	3,375	1,952	10,836
AspenTech	200	60	59	319	362	140	154	656
Control Systems & Software	1,042	721	558	2,321	1,170	745	483	2,398
Software and Control	1,242	781	617	2,640	1,532	885	637	3,054
Total	$ 6,160	4,113	2,747	13,020	$ 7,041	4,260	2,589	13,890

	2023			
	Americas	AMEA	Europe	Total
Measurement & Analytical	$ 1,847	1,222	526	3,595
Final Control	1,949	1,481	540	3,970
Discrete Automation	1,234	720	681	2,635
Safety & Productivity	1,049	70	269	1,388
Intelligent Devices	6,079	3,493	2,016	11,588
AspenTech	470	286	286	1,042
Control Systems & Software	1,259	818	529	2,606
Software and Control	1,729	1,104	815	3,648
Total	$ 7,808	4,597	2,831	15,236

Sales in the U.S. were $6,327, $5,671 and $4,982 for 2023, 2022 and 2021, respectively, while Asia, Middle East & Africa includes sales in China of $1,804, $1,824 and $1,662 in those years.

	Property, Plant and Equipment		
	2021	2022	2023
Americas	$ 1,422	1,373	1,442
Asia, Middle East & Africa	448	398	428
Europe	574	468	493
Total	$ 2,444	2,239	2,363

Property, plant and equipment located in the U.S. w as $1,261 in 2023, $1,219 in 2022 and $ 1,273 in 2021.

(21) OTHER FINANCIAL DATA

Items reported in earnings from continuing operations during the years ended September 30 included the following:

	2021	2022	2023
Research and development expense	$ 347	385	523
Rent expense	$ 199	187	210

The components of depreciation and amortization expense reported for the years ended September 30 included the following:

	2021	2022	2023
Depreciation expense	$ 330	312	287
Amortization of intangibles (includes $57, $108 and $196 reported in Cost of Sales in 2021, 2022 and 2023, respectively) (a)	334	444	678
Amortization of capitalized software	98	86	86
Total	$ 762	842	1,051

(a) Amortization of intangibles includes $397 and $148 related to the Heritage AspenTech acquisition for 2023 and 2022, respectively, and $ 14 that is reported as a restructuring related cost in 2022. Backlog amortization of $30 related to the OSI acquisition is included in 2021.

Items reported in other noncurrent assets included the following:

	2022	2023
Pension assets	$ 868	995
Operating lease right-of-use assets	$ 439	550
Unbilled receivables (contract assets)	$ 428	559
Deferred income taxes	$ 85	100
Asbestos-related insurance receivables	$ 68	53

Items reported in accrued expenses included the following:

	2022	2023
Customer advances (contract liabilities)	$ 751	861
Employee compensation	$ 523	618
Operating lease liabilities (current)	$ 128	144
Product warranty	$ 84	84

Other liabilities are summarized as follows:

	2022	2023
Deferred income taxes	$ 1,714	1,959
Pension and postretirement liabilities	427	435
Operating lease liabilities (noncurrent)	312	404
Asbestos litigation	205	173
Other	495	535
Total	$ 3,153	3,506

(22) QUARTERLY FINANCIAL INFORMATION (UNAUDITED)

	First Quarter		Second Quarter		Third Quarter		Fourth Quarter		Full Year	
	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023
Net sales	$ 3,156	3,373	3,291	3,756	3,465	3,946	3,892	4,090	13,804	15,165
Gross profit	$ 1,415	1,620	1,476	1,801	1,586	1,994	1,829	2,012	6,306	7,427
Earnings from continuing operations common stockholders	$ 746	329	428	530	226	592	486	701	1,886	2,152
Net earnings common stockholders	$ 896	2,331	674	792	921	9,352	740	744	3,231	13,219
Earnings per common share from continuing operations:										
Basic	$ 1.25	0.56	0.72	0.93	0.38	1.04	0.82	1.23	3.17	3.74
Diluted	$ 1.25	0.56	0.72	0.92	0.38	1.03	0.82	1.22	3.16	3.72
Net earnings per common share:										
Basic	$ 1.51	3.99	1.13	1.39	1.55	16.36	1.25	1.30	5.44	23.00
Diluted	$ 1.50	3.97	1.13	1.38	1.54	16.28	1.24	1.29	5.41	22.88
Dividends per common share	$ 0.515	0.520	0.515	0.520	0.515	0.520	0.515	0.520	2.06	2.08

Earnings per share are computed independently each period; as a result, the quarterly amounts may not sum to the calculated annual figure.

Emerson Electric Co. common stock (symbol EMR) is listed on the New York Stock Exchange and NYSE Chicago.

(23) SUBSEQUENT EVENTS

On October 11, 2023, the Company completed the acquisition of National Instruments Corporation (“NI”) for $ 60 per share in cash at an equity value of $8.2 billion. The effective price per share is $59.61 considering shares previously acquired by Emerson. NI, which provides software-connected automated test and measurement systems that enable enterprises to bring products to market faster and at a lower cost, had revenues of approximately $1.7 billion and pretax earnings of approximately $ 170 for the 12 months ended September 30, 2023. NI will be referred to as Test & Measurement and reported as a new segment in the Software and Control business group in 2024. The initial accounting for this transaction is not yet complete.

Report of Independent Registered Public Accounting Firm

To the Stockholders and Board of Directors
Emerson Electric Co.:

Opinions on the Consolidated Financial Statements and Internal Control Over Financial Reporting

We have audited the accompanying consolidated balance sheets of Emerson Electric Co. and subsidiaries (the Company) as of September 30, 2023 and 2022, the related consolidated statements of earnings, comprehensive income, equity, and cash flows for each of the years in the three-year period ended September 30, 2023, and the related notes (collectively, the consolidated financial statements). We also have audited the Company's internal control over financial reporting as of September 30, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of September 30, 2023 and 2022, and the results of its operations and its cash flows for each of the years in the three-year period ended September 30, 2023, in conformity with U.S. generally accepted accounting principles. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of September 30, 2023 based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's consolidated financial statements and an opinion on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Sufficiency of Audit Evidence over Net Sales

As discussed in Notes 1, 2 and 20 to the Company's consolidated financial statements, and disclosed in the consolidated statement of earnings, the Company recorded $15.2 billion of net sales in 2023.

We identified the evaluation of the sufficiency of audit evidence over net sales as a critical audit matter. Net sales are recognized primarily from the sale of tangible products from hundreds of Company locations around the world. Evaluating the sufficiency of audit evidence obtained required especially subjective auditor judgment because of the geographical dispersion of the Company's net sales generating activities. This included determining the Company locations at which procedures were performed and the supervision and review of procedures performed at those locations.

The following are the primary procedures we performed to address this critical audit matter. We applied auditor judgment to determine the nature and extent of procedures to be performed over net sales, including the determination of the Company locations at which those procedures were to be performed. At each Company location where procedures were performed, we:

- Evaluated the design and tested the operating effectiveness of certain internal controls over the Company's net sales processes, including the Company's controls over the accurate recording of amounts.
- Assessed the recorded net sales by selecting a sample of transactions and compared the amounts recognized for consistency with underlying documentation, including contracts with customers and shipping documentation.

/s/ KPMG LLP

We or our predecessor firms have served as the Company's auditor since 193 8.

St. Louis, Missouri
November 13, 2023

ITEM 9 - CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A - CONTROLS AND PROCEDURES

The Company maintains a system of disclosure controls and procedures which is designed to ensure that information required to be disclosed by the Company in the reports filed or submitted under the Securities Exchange Act of 1934 is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and is accumulated and communicated to management, including the Company's certifying officers, as appropriate to allow timely decisions regarding required disclosure. Based on an evaluation performed, the Company's certifying officers have concluded that the disclosure controls and procedures were effective as of September 30, 2023 to provide reasonable assurance of achieving these objectives.

Notwithstanding the foregoing, there can be no assurance that the Company's disclosure controls and procedures will detect or uncover all failures of persons within the Company and its consolidated subsidiaries to report material information otherwise required to be set forth in the Company's reports. There was no change in the Company's internal control over financial reporting during the quarter ended September 30, 2023, that has materially affected, or is reasonably likely to materially affect, the Company's internal control over financial reporting.

In the fourth quarter, AspenTech implemented a new revenue management system and consequently, modified the design of certain internal controls within their revenue process.

Management's report on internal control over financial reporting, and the related report of the Company's auditor, KPMG LLP, an independent registered public accounting firm, set forth in Item 7 and Item 8, respectively, of this Annual Report on Form 10-K, are hereby incorporated by reference.

ITEM 9B - OTHER INFORMATION

During the three-month period ended September 30, 2023, none of our directors or officers adopted or terminated a Rule 10b5-1 trading arrangement or non-Rule 10b5-1 trading arrangement.

ITEM 9C - DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10 - DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

Information regarding nominees and directors appearing under "Proxy Item No. 1: Election of Directors" in the Emerson Electric Co. Notice of Annual Meeting of Shareholders and Proxy Statement for the February 2024 annual shareholders' meeting (the "2024 Proxy Statement") is hereby incorporated by reference. Information regarding executive officers is set forth in Part I of this report. Information regarding the Audit Committee and Audit Committee Financial Expert appearing under "Board and Committee Operations—Board and Corporate Governance— Committees of Our Board of Directors," "Board and Committee Operations—Corporate Governance and Nominating Committee—Nomination Process" and "— Proxy Access" in the 2024 Proxy Statement is hereby incorporated by reference.

The Company has adopted a Code of Ethics that applies to the Company's Chief Executive Officer, Chief Financial Officer and Chief Accounting Officer; has posted such Code of Ethics on its website; and intends to satisfy the disclosure requirement under Item 5.05 of Form 8-K by posting such information on its website. The Company has adopted Charters for its Audit Committee, Compensation Committee, and Corporate Governance and Nominating Committee and a Code of Business Ethics for directors, officers and employees, which are available on its website and in print to any stockholder who requests them. The Company has also adopted Corporate Governance Principles and Practices, which are available on its website and in print to any stockholder who requests them. The Corporate Governance section of the Company's website may be accessed as follows: www.Emerson.com, Investors, Corporate Governance.

ITEM 11 - EXECUTIVE COMPENSATION

Information appearing under “Executive Compensation" (including the information set forth under "Compensation Discussion and Analysis"), "Compensation Tables" (other than "Pay vs. Performance"), "Board and Committee Operations—Corporate Governance and Nominating Committee—Director Compensation," "Board and Committee Operations—Compensation Committee" (including, but not limited to, the information set forth under "Role of Executive Officers and the Compensation Consultant," "Compensation Committee Report" and "Compensation Committee Interlocks and Insider Participation") in the 2024 Proxy Statement is hereby incorporated by reference.

The information contained in the "Compensation Committee Report” shall not be deemed to be filed with the SEC or subject to the liabilities of Section 18 of the Securities Exchange Act of 1934 (the “Exchange Act”), except to the extent that the Company specifically incorporates such information into future filings under the Securities Act of 1933 or the Exchange Act.

ITEM 12 - SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information regarding beneficial ownership of shares by nominees and continuing directors, named executive officers, five percent beneficial owners, and by all directors and executive officers as a group appearing under "Ownership of Emerson Equity Securities" in the 2024 Proxy Statement is hereby incorporated by reference.

The following table sets forth aggregate information regarding the Company’s equity compensation plans as of September 30, 2023:

Plan Category	Number of Securities to be Issued upon Exercise of Outstanding Options, Warrants and Rights (a)	Weighted-Average Exercise Price of Outstanding Options, Warrants and Rights (b)	Number of Securities Remaining Available for Future Issuance under Equity Compensation Plans (Excluding Securities Reflected in Column (a)) (c)
Equity compensation plans approved by security holders (1)	6,056,000	$53.35	2,747,000
Equity compensation plans not approved by security holders	—	—	—
Total	6,056,000	$53.35	2,747,000

(1) Includes the Stock Option and Incentive Shares Plans previously approved by the Company's security holders. Shares included in column (a) assume the maximum payouts, where applicable, and are as follows: (i) 589,000 shares reserved for outstanding stock option awards, (ii) 1,414,000 shares reserved for performance share awards granted in 2023, (iii) 1,346,000 shares reserved for performance share awards granted in 2022, (iv) 2,129,000 shares reserved for performance share awards granted in 2021 and (v) 578,000 shares reserved for outstanding restricted stock unit awards. As provided by the Company’s Incentive Shares Plans, performance shares awards represent a commitment to issue such shares without cash payment by the employee, contingent upon achievement of the performance objectives and continued service by the employee.

The price in column (b) represents the weighted-average exercise price for outstanding options. Included in column (c) are shares remaining available for award under previously approved plans as follows: (i) 2,001,000 under the 2015 Incentive Shares Plan, (ii) 689,000 under the 2006 Incentive Shares Plan, and (iii) 57,000 under the Restricted Stock Plan for Non-Management Directors.

Information regarding stock option plans and incentive shares plans is set forth in Note 17.

ITEM 13 - CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

Information appearing under “Board and Committee Operations—Board and Corporate Governance—Review, Approval or Ratification of Transactions with Related Persons," "—Certain Business Relationships and Related

Party Transactions" and "—Director Independence" in the 2024 Proxy Statement is hereby incorporated by reference.

ITEM 14 - PRINCIPAL ACCOUNTANT FEES AND SERVICES

Information appearing under "Board and Committee Operations—Audit Committee—Fees Paid to KPMG LLP" in the 2024 Proxy Statement is hereby incorporated by reference.

PART IV

ITEM 15 - EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

A) Documents filed as a part of this report:

1. The consolidated financial statements and accompanying notes of the Company and subsidiaries and the report thereon of KPMG LLP set forth in Item 8 of this Annual Report on Form 10-K.

2. Financial Statement Schedules - All schedules are omitted because they are not required, not applicable or the required information is provided in the financial statements or notes thereto contained in this Annual Report on Form 10-K.

3. Exhibits (Listed by numbers corresponding to the Exhibit Table of Item 601 in Regulation S-K).

2(a)** Transaction Agreement and Plan of Merger, dated as of October 10, 2021 , among Emerson Electric Co., Aspen Technology, Inc., EMR Worldwide, Inc., Emersub CX, Inc. and Emersub CXI, Inc., incorporated by reference to the Company's Form 8-K, filed on October 12, 2021, File No. 1-278, Exhibit 2.1.

2(b) Amendment No. 1 to the Transaction Agreement and Plan of Merger , dated as of March 23, 2022, among Emerson Electric Co., Aspen Technology, Inc., EMR Worldwide Inc., Emersub CX, Inc. and Emersub CXI, Inc., incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2022, filed on May 4, 2022, File No. 1-278, Exhibit 2(b).

2(c)** Amendment No. 2 to the Transaction Agreement and Plan of Merger , dated as of May 3, 2022, among Emerson Electric Co., Aspen Technology, Inc., EMR Worldwide Inc., Emersub CX, Inc. and Emersub CXI, Inc., incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2022, filed on May 4, 2022, File No. 1-278, Exhibit 2(c).

2(d)** Transaction Agreement, dated as of October 30, 2022 , among Emerson Electric Co., BCP Emerald Aggregator L.P., Emerald Debt Merger Sub L.L.C and Emerald JV Holdings L.P, incorporated by reference to Emerson Electric Co. Form 8-K, filed on October 31, 2022, File No. 1-278, Exhibit 2.1.

2(e)** Agreement and Plan of Merger, dated as of April 12, 2023, among Emerson Electric Co., Emersub CXIV, Inc., and National Instruments Corporation*, incorporated by reference to the Company's Form 8-K, filed on April 12, 2023, File No. 1-278, Exhibit 2.1.

3(a) Restated Articles of Incorporation of Emerson Electric Co. , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2001, File No. 1-278, Exhibit 3(a); Termination of Designated Shares of Stock and Certificate of Designation, Preferences and Rights of Series B Junior Participating Preferred Stock, incorporated by reference to Emerson Electric Co. 1998 Form 10-K, File No. 1-278, Exhibit 3(a).

3(b) Bylaws of Emerson Electric Co ., as amended through May 4, 2021, incorporated by reference to the Company's Form 8-K dated May 4, 2021, filed on May 4, 2021, File No. 1-278, Exhibit 3.1.

4(a) Indenture dated as of December 10, 1998, between Emerson Electric Co. and Computershare Trust Company, N.A., as successor to Wells Fargo Bank, National Association, as successor trustee to The Bank of New York Mellon Trust Company, N.A. (successor to The Bank of New York Mellon (formerly known as the Bank of New York)), as trustee, incorporated by reference to

Emerson Electric Co. 1998 Form 10-K, File No. 1-278, Exhibit 4(b), ***Form of 2.000% Notes due 2028, incorporated by reference to Emerson Electric Co. Form 8-K, filed on December 21, 2021, File No. 1-278, Exhibit 4.2, ***Form of 2.200% Notes due 2031, incorporated by reference to Emerson Electric Co. Form 8-K, filed on December 21, 2021, File No. 1-278, Exhibit 4.3, ***Form of 2.800% Notes due 2051, incorporated by reference to Emerson Electric Co. Form 8-K, filed on December 21, 2021, File No. 1-278, Exhibit 4.4.

4(b) Agreement of Resignation, Appointment and Acceptance dated as of April 26, 2019 by and among Emerson Electric Co., Computershare Trust Company, N.A., as successor to Wells Fargo Bank, National Association, as successor trustee, and The Bank of New York Mellon Trust Company, N.A., as resigning trustee, incorporated by reference to the Company's Form 8-K dated May 15, 2019, filed on May 17, 2019, File No. 1-278, Exhibit 4.4.

4(c) Description of Capital Stock incorporated by reference to Emerson Electric Co., 2020 Form 10-K, File No. 1-278, Exhibit 4(c).

4(d) Description of 0.375% Notes due 2024, 1.250% Notes due 2025 and 2.000% Notes due 2029 , incorporated by reference to Emerson Electric Co., 2019 Form 10-K, File No. 1-278, Exhibit 4(d).

No other long-term debt instruments are filed since the total amount of securities authorized under any such instrument does not exceed 10 percent of the total assets of Emerson Electric Co. and its subsidiaries on a consolidated basis. Emerson Electric Co. agrees to furnish a copy of such instruments to the SEC upon request.

10(a)* Amended and Restated Emerson Electric Co. Continuing Compensation Plan for Non-Management Directors , incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(c).

10(b)* Amended and Restated Deferred Compensation Plan for Non-Employee Directors and Forms of Payment Election Form, Initial Notice of Election and Notice of Election Change, incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(d).

10(c)* First Amendment to the Emerson Electric Co. Supplemental Executive Retirement Plan , incorporated by reference to Emerson Electric Co. 1999 Form 10-K, File No. 1-278, Exhibit 10(h), and Form of Change of Control Election , incorporated by reference to Emerson Electric Co. Form 8-K dated October 1, 2004, Exhibit 10.9 (applicable only with respect to benefits vested as of December 31, 2004).

10(d)* Amended and Restated Emerson Electric Co. Pension Restoration Plan dated October 6, 2015 , incorporated by reference to Emerson Electric Co. 2015 Form 10-K, File No. 1-278, Exhibit 10(e); Forms of Participation Award Letter, Acceptance of Award and Benefit Election Forms (applicable only with respect to benefits after January 1, 2005), incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(f); and Lump Sum Distribution Election Forms.

10(e)* Fifth Amendment to the Supplemental Executive Savings Investment Plan , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 1999, File No. 1-278, Exhibit 10(j), and Form of Participation Agreement and Form of Annual Election, incorporated by reference to Emerson Electric Co. Form 8-K filed October 1, 2004, Exhibit 10.8 (applicable only with respect to benefits vested as of December 31, 2004).

10(f)* Amended and Restated Emerson Electric Co. Savings Investment Restoration Plan and Forms of Participation Agreement, Annual Election Form and Payment Election Form (applicable only with respect to benefits after January 1, 2005), incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(h), First Amendment to Emerson Electric Co. Savings Investment Restoration Plan, incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2008, File No. 1-278, Exhibit 10.1 and Second Amendment to the Emerson Electric Co. Savings Investment Restoration Plan , incorporated by reference to Emerson Electric Co., Form 10-Q for the quarter ended March 31, 2020, File No. 1-278, Exhibit 10.2.

10(g)* Amended and Restated Emerson Electric Co. Annual Incentive Plan and Form of Acceptance of Award , incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(i).

10(h)* Emerson Electric Co. Description of Split Dollar Life Insurance Program Transition , incorporated by reference to Emerson Electric Co. Form 8-K filed September 2, 2005, Exhibit 10.1.

10(i)* Amended and Restated Restricted Stock Plan for Non-Management Directors , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2009, File No. 1-278, Exhibit 10.1, Form of Restricted Stock Award Letter under the Emerson Electric Co. Restricted Stock Plan for Non-Management Directors, incorporated by reference to Emerson Electric Co. Form 8-K filed February 1, 2005, Exhibit 10.2, and Form of Restricted Stock Unit Award Letter under the Emerson Electric Co. Restricted Stock Plan for Non-Management Directors, incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2009, File No. 1-278, Exhibit 10.1.

10(j)* Description of Non-Management Director Compensation , incorporated by reference to Emerson Electric Co. Form 10-K filed November 20, 2017, Exhibit 10(n).

10(k)* Description of Named Executive Officer Compensation , incorporated by reference to Emerson Electric Co. Form 10-K filed November 20, 2017, Exhibit 10(o).

10(l)* Emerson Electric Co. 2006 Incentive Shares Plan , incorporated by reference to Emerson Electric Co. 2006 Proxy Statement dated December 16, 2005, Appendix C, Amendment for 409A Compliance , incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(q), Forms of Performance Shares Award Certificate and Acceptance of Award (used on or prior to September 30, 2009) and Restricted Shares Award Agreement (used on or prior to September 30, 2011), incorporated by reference to Emerson Electric Co. 2007 Form 10-K, File No. 1-278, Exhibit 10(q), Amendment to Emerson Electric Co. 2006 Incentive Shares Plan , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended June 30, 2008, File No. 1-278, Exhibit 10.1, Forms of Performance Shares Award Certificate, Acceptance of Award and 2010 Performance Shares Program Award Summary, incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2009 (used after September 30, 2009 and on or prior to September 30, 2011), File No. 1-278, Exhibit 10.2, Forms of Performance Shares Award Certificate and Acceptance of Award , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2011, File No. 1-278, Exhibit 10.3 (used after September 30, 2011), and Form of Restricted Shares Award Agreement , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2011, File No. 1-278, Exhibit 10.4 (used after September 30, 2011).

10(m) Credit Agreement dated as of February 17, 2023. , incorporated by reference to the Company's Form 8-K, filed on February 21, 2023, File No. 1-278, Exhibit 10.1.

10(n)* 2011 Stock Option Plan, incorporated by reference to Emerson Electric Co. 2011 Proxy Statement dated December 10, 2010, File No. 1-278, Appendix B, 2011 Stock Option Plan as Amended and Restated effective October 1, 2012 , incorporated by reference to Emerson Electric Co. 2012 Form 10-K, File No. 1-278, Exhibit 10(r), Forms of Notice of Grant of Stock Options, Option Agreement and Incentive Stock Option Agreement under the 2011 Stock Option Plan, incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2012, File No. 1-278, Exhibit 10.1 and Forms of Notice of Grant of Stock Options, Option Agreement and Nonqualified Stock Option Agreement under the 2011 Stock Option Plan, incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended March 31, 2012, File No. 1-278, Exhibit 10.2.

10(o)* Emerson Electric Co. 2015 Incentive Shares Plan , incorporated by reference to Emerson Electric Co. 2015 Proxy Statement dated December 12, 2014, Appendix B, Forms of Performance Shares Award Certificate and Acceptance of Award (used on or prior to November 5, 2018), Performance Shares Program Award Summary (used on or prior to November 5, 2018) and Form of Restricted Shares Award Agreement (used on or prior to November 5, 2018), incorporated by reference to Emerson Electric Co. 2015 Form 10-K, File No. 1-278, Exhibit 10(u), Form of Restricted Shares

Award Agreement (used after November 5, 2018) , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2018, Exhibit 10.1, Form of Restricted Stock Units Program Acceptance of Award (used after November 5, 2018) , incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2018, Exhibit 10.2 and Form of Performance Share Program Acceptance of Award (used after November 5, 2018), incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2018, Exhibit 10.3., Form of Emerson Electric Co. Performance Shares Program Award Agreement (used after November 1, 2021), incorporated by reference to Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2021, File No. 1-278, Exhibit 10.2

10(p) Transaction Agreement dated as of July 29, 2016 among Emerson Electric Co., Cortes NP Holdings, LLC, Cortes NP Acquisition Corporation, ASCO Power Grp, LLC and Cortes NP JV Holdings, LLC, incorporated by reference to Emerson Electric Co. 2016 Form 10-K, File No. 1-278, Exhibit 10(w).

10(q)* Emerson Electric Co. Savings Investment Restoration Plan II , incorporated by reference to the Emerson Electric Co. Form 10-Q for the quarter ended June 30, 2018, File No. 1-278, Exhibit 10.1, Second Amendment to the Emerson Electric Co. Savings Investment Restoration Plan, incorporated by reference to Emerson Electric Co., Form 10-Q for the quarter ended March 31, 2020, File No. 1-278, Exhibit 10.2 and First Amendment to the Emerson Electric Co. Savings Investment Restoration Plan II , incorporated by reference to Emerson Electric Co., Form 10-Q for the quarter ended March 31, 2020, File No. 1-278, Exhibit 10.1.

10(r)* Letter Agreement dated November 16, 2022 between Emerson Electric Co. and Mark J. Bulanda, signed November 22, 2022. , incorporated by reference to the Company's Form 8-K, filed on November 28, 2022, File No. 1-278, Exhibit 10.1.

10(s)* Amended and Restated Deferred Compensation Plan for Non-Employee Directors and Forms of Payment Election Forms , incorporated by reference to the Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2022, File No. 1-278, Exhibit 10(a).

10(t)* Amended and Restated Restricted Stock Plan for Non-Management Directors and Form of Restricted Stock Unit Award Letter under the Emerson Electric Co. Restricted Stock Plan for Non-Management Directors, incorporated by reference to the Emerson Electric Co. Form 10-Q for the quarter ended December 31, 2022, File No. 1-278, Exhibit 10(b).

10(u)* Emerson Electric Co. Annual Cash Incentive Plan and Form of Acceptance of Award, incorporated by reference to the Company's Form 10-Q, filed on February 8, 2023, File No. 1-278, Exhibit 10(c).

10(v)* Letter Agreement dated May 2, 2023 between Emerson Electric Co. and Frank J. Dellaquila. , incorporated by reference to the Company's Form 8-K, filed on May 3, 2023, File No. 1-278, Exhibit 10.1.

21 Subsidiaries of Emerson Electric Co.

23 Consent of Independent Registered Public Accounting Firm

24 Power of Attorney

31 Certifications pursuant to Exchange Act Rule 13a-14(a)

32 Certifications pursuant to Exchange Act Rule 13a-14(b) and 18 U.S.C. Section 1350

97 Incentive Compensation Recovery (Clawback) Policy

101 Attached as Exhibit 101 to this report are the following documents formatted in iXBRL (Inline Extensible Business Reporting Language): (i) Consolidated Statements of Earnings for the years ended September 30, 2021, 2022 and 2023, (ii) Consolidated Statements of Comprehensive Income for the years ended September 30, 2021, 2022, and 2023 (iii) Consolidated Balance Sheets at September 30, 2022 and 2023, (iv) Consolidated Statements of Equity for the years ended September 30, 2021, 2022 and 2023, (v) Consolidated Statements of Cash Flows for the years

ended September 30, 2021, 2022 and 2023, and (vi) Notes to Consolidated Financial Statements for the year ended September 30, 2023.

104 Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101).

* Management contract or compensatory plan.

** Certain schedules and exhibits have been omitted pursuant to Item 601(a)(5) of Regulation S-K. Emerson agrees to furnish supplementally a copy of any omitted schedule or exhibit to the SEC upon request. Portions of these exhibits have been redacted in compliance with Regulation S-K Item 601(b)(10).

*** The Company entered into two global notes for each series of notes (Notes A-1 and A-2), which are identical other than with respect to the note number

ITEM 16 - FORM 10-K SUMMARY

Not applicable.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

EMERSON ELECTRIC CO.

By /s/ M. J. Baughman

M. J. Baughman
Executive Vice President and
Chief Financial Officer
November 13, 2023

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below on November 13, 2023, by the following persons on behalf of the registrant and in the capacities indicated.

Signature	Title
/s/ S. L. Karsanbhai S. L. Karsanbhai	President and Chief Executive Officer
/s/ M. J. Baughman M. J. Baughman	Executive Vice President, Chief Financial Officer and Chief Accounting Officer
* J. S. Turley	Chair of the Board
* M. A. Blinn	Director
* J. B. Bolten	Director
* M. S. Craighead	Director
* W. H. Easter III	Director
* G. A. Flach	Director
* A. F. Golden	Director
* L. Goncalves	Director
* C. Kendle	Director
* L. M. Lee	Director
* M. S. Levatich	Director
* J. M. McKelvey	Director

* By /s/ M. J. Baughman
M. J. Baughman
Attorney-in-Fact